

MEASURE THEORY

Notes from lectures

Notes following Prof. Mariusz Mirek's lectures

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CHAPTER 1

Riemann Integrals

1.1 Partitions

1.1.1. Definition: Partition. Let $[a, b]$ be an interval. A *partition* of $[a, b]$ is a finite set

$$P = \{x_0, \dots, x_n\}, \quad a = x_0 \leq x_1 \leq \dots \leq x_{n-1} \leq x_n = b.$$

1.1.2. Example. For $[a, b] = [0, 1]$, the set $\{0, \frac{1}{2}, 1\}$ is a partition.

1.1.3. Example. For every $n \in \mathbb{N}$, the set $P_n = \{\frac{k}{n} : k = 0, 1, \dots, n\}$ is a partition of $[0, 1]$.

1.1.4. Definition: Upper and lower sums. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let $P = \{x_0, \dots, x_n\}$ be a partition. Put

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x), \quad M_i = \sup_{x \in [x_{i-1}, x_i]} f(x), \quad \Delta x_i = x_i - x_{i-1}.$$

The *upper* and *lower Riemann sums* are

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i, \quad L(P, f) = \sum_{i=1}^n m_i \Delta x_i.$$

1.1.5. Remark. For every partition P , $L(P, f) \leq U(P, f)$.

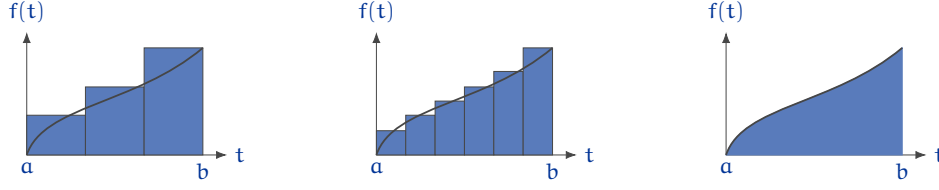
1.1.6. Example. If $f(x) = x$ and $P = \{0, \frac{1}{2}, 1\}$, then

$$U(P, f) = \frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{3}{4}, \quad L(P, f) = 0 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

1.1.7. Example. If $f(x) = x^2$ and $P_n = \{k/n : k = 0, 1, \dots, n\}$, then

$$U(P_n, f) = \sum_{i=1}^n \frac{1}{n} \left(\frac{i}{n} \right)^2, \quad L(P_n, f) = \sum_{i=1}^n \frac{1}{n} \left(\frac{i-1}{n} \right)^2.$$

1.1.8. Remark: Geometric interpretation. For an increasing nonnegative function, the upper sum is the area of the rectangles whose heights are the right-endpoint values of f ; it overshoots the region below the graph. Refining the partition narrows the rectangles and lowers this approximation.



1.2 The Integral

1.2.1. Definition: Lower and upper Riemann integrals. For a bounded function $f : [a, b] \rightarrow \mathbb{R}$, define

$$\int_a^b f(x) \, dx = \sup_P L(P, f), \quad \int_a^b f(x) \, dx = \inf_P U(P, f),$$

where P ranges over all partitions of $[a, b]$.

1.2.2. Definition: Riemann integrability. A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if

$$\int_a^b f(x) \, dx = \int_a^b f(x) \, dx = \int_a^b f(x) \, dx.$$

Then $f \in \mathcal{R}([a, b])$.

1.2.3. Proposition (Well-definedness). — The upper and lower Riemann integrals are defined for every bounded function $f : [a, b] \rightarrow \mathbb{R}$.

Proof. PICTURE. Well-definedness means that the relevant supremum and infimum exist. For this, it suffices to bound all lower and upper sums uniformly.

By hypothesis, f is bounded. Hence there exists $M \geq 0$ such that $-M \leq f(x) \leq M$ for every $x \in [a, b]$. Let $P = \{x_0, \dots, x_n\}$ be a partition. Then

$$-M \leq m_i \leq M_i \leq M$$

for every i . Since $\Delta x_i \geq 0$, the definition of lower and upper sums gives

$$-M \sum_{i=1}^n \Delta x_i \leq L(P, f) \leq U(P, f) \leq M \sum_{i=1}^n \Delta x_i.$$

By the definition of a partition, $\sum_{i=1}^n \Delta x_i = b - a$. Thus

$$-M(b - a) \leq L(P, f) \leq U(P, f) \leq M(b - a).$$

The partition $\{a, b\}$ shows that both families of sums are nonempty. The displayed inequality shows that the lower sums are bounded above and the upper sums are bounded below. Therefore, by completeness of $\mathbb{R}$,

$$\sup_P L(P, f) \quad \text{and} \quad \inf_P U(P, f)$$

exist in $\mathbb{R}$. Thus the lower and upper Riemann integrals are well-defined. $\square$

1.2.4. Example: A bounded non-integrable function. The function $f : [0, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q} \end{cases}$$

is bounded but not Riemann integrable.

1.2.5. Remark: Rationals and irrationals. Every nondegenerate interval in $\mathbb{R}$ contains both a rational and an irrational number.

1.3 Refinements

1.3.1. Definition: Refinement. A partition P^* of $[a, b]$ is a *refinement* of a partition P if $P^* \supseteq P$.

1.3.2. Definition: Common refinement. For two partitions P_1, P_2 of $[a, b]$, their *common refinement* is $P^* = P_1 \cup P_2$.

1.3.3. Example. If $P_1 = \{0, \frac{1}{2}, 1, 2\}$ and $P_2 = \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{2}, 2\}$, then their common refinement is

$$P^* = \left\{ 0, \frac{1}{4}, \frac{1}{2}, 1, \frac{3}{2}, 2 \right\}.$$

1.3.4. Theorem (Refining a partition). — If P^* is a refinement of P , then

$$L(P, f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P, f).$$

Proof. It suffices to consider $P^* = P \cup \{x^*\}$, where

$$P^* = \{x_0, \dots, x_{i-1}, \underbrace{x^*}_{\text{new point}}, x_i, \dots, x_n\}, \quad \underbrace{[x_{i-1}, x^*]}_{\text{left}} \cup \underbrace{[x^*, x_i]}_{\text{right}} = [x_{i-1}, x_i].$$

Let

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x), \quad w_1 = \inf_{x \in [x_{i-1}, x^*]} f(x), \quad w_2 = \inf_{x \in [x^*, x_i]} f(x).$$

Then $w_1, w_2 \geq m_i$, so

$$\begin{aligned} L(P^*, f) - L(P, f) &= \underbrace{w_1(x^* - x_{i-1})}_{\text{left contribution to } L(P^*, f)} + \underbrace{w_2(x_i - x^*)}_{\text{right contribution to } L(P^*, f)} - m_i(x_i - x_{i-1}) \\ &= (w_1 - m_i)(x^* - x_{i-1}) + (w_2 - m_i)(x_i - x^*) \geq 0. \end{aligned}$$

Hence $L(P, f) \leq L(P^*, f)$.

Likewise, let

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x), \quad W_1 = \sup_{x \in [x_{i-1}, x^*]} f(x), \quad W_2 = \sup_{x \in [x^*, x_i]} f(x).$$

Then $W_1, W_2 \leq M_i$, so

$$\begin{aligned} U(P, f) - U(P^*, f) &= M_i(x_i - x_{i-1}) - \underbrace{W_1(x^* - x_{i-1})}_{\text{left contribution to } U(P^*, f)} - \underbrace{W_2(x_i - x^*)}_{\text{right contribution to } U(P^*, f)} \\ &= (M_i - W_1)(x^* - x_{i-1}) + (M_i - W_2)(x_i - x^*) \geq 0. \end{aligned}$$

Thus $U(P^*, f) \leq U(P, f)$. Also, $L(P^*, f) \leq U(P^*, f)$ term-by-term.

If $P^* \setminus P$ has k points, repeat this computation k times. Hence

$$L(P, f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P, f).$$

□

1.3.5. Claim. — For partitions P_1, P_2 of $[a, b]$,

$$L(P_1, f) \leq U(P_2, f).$$

Proof. Let $P^* = P_1 \cup P_2$. Since P_1 and P_2 are finite partitions of $[a, b]$, P^* is also a partition of $[a, b]$ and is a refinement of both P_1 and P_2 . Therefore, by the refinement theorem,

$$L(P_1, f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P_2, f).$$

Hence $L(P_1, f) \leq U(P_2, f)$. □

1.3.6. Theorem (Comparing the two integrals). — For every bounded function $f : [a, b] \rightarrow \mathbb{R}$,

$$\int_a^b f(x) \, dx \leq \int_a^b f(x) \, dx.$$

Proof. By the preceding claim, for every pair of partitions P_1, P_2 of $[a, b]$,

$$L(P_1, f) \leq U(P_2, f).$$

Therefore

$$\int_a^b f(x) \, dx = \sup_{P_1} L(P_1, f) \leq \inf_{P_2} U(P_2, f) = \int_a^b f(x) \, dx.$$

□

1.4 Criterion for Riemann Integrability

1.4.1. Theorem (Condition $\mathcal{R}$). — A bounded function $f : [a, b] \rightarrow \mathbb{R}$ belongs to $\mathcal{R}([a, b])$ if and only if for every $\varepsilon > 0$ there is a partition P of $[a, b]$ such that

$$(\mathcal{R}) \quad U(P, f) - L(P, f) < \varepsilon.$$

This condition remains true for every refinement $P^* \supseteq P$.

Proof. We prove both implications.

($\implies$) Let $\varepsilon > 0$. By the defining properties of supremum and infimum, there are partitions P_1, P_2 such that

$$L(P_1, f) > \int_a^b f(x) \, dx - \frac{\varepsilon}{2}, \quad U(P_2, f) < \int_a^b f(x) \, dx + \frac{\varepsilon}{2}.$$

Let $P = P_1 \cup P_2$. By the refinement theorem,

$$\begin{aligned} U(P, f) - L(P, f) &\leq U(P_2, f) - L(P_1, f) \\ &< \left(\int_a^b f(x) \, dx + \frac{\varepsilon}{2} \right) - \left(\int_a^b f(x) \, dx - \frac{\varepsilon}{2} \right) = \varepsilon. \end{aligned}$$

($\Leftarrow$) Let $\varepsilon > 0$, and choose P satisfying Condition $\mathcal{R}$. By the definitions of the lower and upper integrals,

$$0 \leq \int_a^{\overline{b}} f(x) dx - \int_{\underline{a}}^b f(x) dx \leq U(P, f) - L(P, f) < \varepsilon.$$

Thus

$$\int_{\underline{a}}^b f(x) dx = \int_a^{\overline{b}} f(x) dx.$$

Hence $f \in \mathcal{R}([a, b])$.

If $P^* \supseteq P$, then, by the refinement theorem,

$$0 \leq U(P^*, f) - L(P^*, f) \leq U(P, f) - L(P, f) < \varepsilon.$$

□

1.4.2. Theorem (Oscillation estimate). — Let $P = \{x_0, \dots, x_n\}$ satisfy Condition $\mathcal{R}$, and choose $s_i, t_i \in [x_{i-1}, x_i]$. Then

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \Delta x_i < \varepsilon.$$

Proof. Let $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$ and $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$. By definition,

$$m_i \leq f(s_i), f(t_i) \leq M_i \implies |f(s_i) - f(t_i)| \leq M_i - m_i.$$

Therefore, by the definitions of the upper and lower sums and Condition $\mathcal{R}$,

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \Delta x_i \leq \sum_{i=1}^n (M_i - m_i) \Delta x_i = U(P, f) - L(P, f) < \varepsilon.$$

□

1.4.3. Theorem (Riemann sums approximate the integral). — Let $f \in \mathcal{R}([a, b])$, and let $P = \{x_0, \dots, x_n\}$ satisfy Condition $\mathcal{R}$. For any $t_i \in [x_{i-1}, x_i]$,

$$\left| \int_a^b f(x) dx - \sum_{i=1}^n f(t_i) \Delta x_i \right| < \varepsilon.$$

Proof. By the definitions of m_i and M_i ,

$$m_i \leq f(t_i) \leq M_i.$$

Thus

$$L(P, f) \leq \sum_{i=1}^n f(t_i) \Delta x_i \leq U(P, f).$$

Since $f \in \mathcal{R}([a, b])$, the comparing-integrals theorem gives

$$L(P, f) \leq \int_a^b f(x) dx \leq U(P, f).$$

Therefore, by Condition $\mathcal{R}$,

$$\left| \int_a^b f(x) dx - \sum_{i=1}^n f(t_i) \Delta x_i \right| \leq U(P, f) - L(P, f) < \varepsilon.$$

□

1.5 Integrability

1.5.1. Theorem (Continuous functions). — If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then $f \in \mathcal{R}([a, b])$.

Proof. PICTURE. By Condition $\mathcal{R}$, it suffices, for every $\varepsilon > 0$, to construct a partition P such that $U(P, f) - L(P, f) < \varepsilon$. For $P = \{x_0, \dots, x_n\}$,

$$U(P, f) - L(P, f) = \sum_{i=1}^n (M_i - m_i) \Delta x_i,$$

so we make $M_i - m_i$ small on every subinterval.

Let $\varepsilon > 0$. Since $[a, b]$ is compact and f is continuous, the Heine–Cantor Theorem gives $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{b - a}.$$

Choose $n > (b - a)/\delta$ and let

$$P = \left\{ a + \frac{i(b - a)}{n} : i = 0, \dots, n \right\}.$$

Then $\Delta x_i = (b - a)/n < \delta$. By the Extreme Value Theorem, there are $s_i, t_i \in [x_{i-1}, x_i]$ such that $M_i = f(s_i)$ and $m_i = f(t_i)$. Hence

$$|s_i - t_i| \leq \Delta x_i < \delta \implies M_i - m_i = |f(s_i) - f(t_i)| < \frac{\varepsilon}{b - a}.$$

Therefore

$$U(P, f) - L(P, f) = \sum_{i=1}^n (M_i - m_i) \Delta x_i < \frac{\varepsilon}{b - a} \sum_{i=1}^n \Delta x_i = \varepsilon.$$

Thus $f \in \mathcal{R}([a, b])$. □

1.5.2. PROBLEM (Monotone functions). If $f : [a, b] \rightarrow \mathbb{R}$ is monotone, prove that $f \in \mathcal{R}([a, b])$.

Proof. PICTURE. For equal partitions, the oscillations of a monotone function telescope.

For $n \in \mathbb{N}$, let

$$P_n = \{x_i = a + i(b - a)/n : i = 0, \dots, n\}.$$

Then $\Delta x_i = (b - a)/n$. By monotonicity,

$$M_i - m_i = |f(x_i) - f(x_{i-1})|, \quad \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = |f(b) - f(a)|.$$

Therefore

$$U(P_n, f) - L(P_n, f) = \frac{b - a}{n} |f(b) - f(a)|.$$

Let $\varepsilon > 0$. By the Archimedean Property, choose $n \in \mathbb{N}$ such that

$$n > \frac{(b-a)|f(b) - f(a)|}{\varepsilon}.$$

For this partition P_n ,

$$U(P_n, f) - L(P_n, f) = \frac{b-a}{n} |f(b) - f(a)| < \varepsilon.$$

Thus Condition $\mathcal{R}$ holds, so $f \in \mathcal{R}([a, b])$. $\square$

1.5.3. Theorem (Finitely many discontinuities). — *If $f : [a, b] \rightarrow \mathbb{R}$ is bounded and has only finitely many discontinuities, then $f \in \mathcal{R}([a, b])$.*

Proof. By Condition $\mathcal{R}$, it suffices to make $U(P, f) - L(P, f)$ arbitrarily small. Since f is bounded, let $M = \sup_{x \in [a, b]} |f(x)| < \infty$, and let E be the finite set of discontinuities of f .

Fix $\varepsilon > 0$. Choose pairwise disjoint intervals $[u_j, v_j] \subseteq [a, b]$ such that

$$E \cap (a, b) \subseteq \bigcup_j (u_j, v_j), \quad \sum_j (v_j - u_j) < \varepsilon.$$

Set $K = [a, b] \setminus \bigcup_j (u_j, v_j)$. Then K is compact and $f|_K$ is continuous. By Heine–Cantor, there is $\delta > 0$ such that

$$s, t \in K, \quad |s - t| < \delta \implies |f(s) - f(t)| < \varepsilon.$$

Choose $P = \{x_0, \dots, x_n\}$ containing all u_j, v_j , with no points in (u_j, v_j) , and such that

$$x_{i-1} \notin \{u_j\}_j \implies \Delta x_i < \delta.$$

Then $M_i - m_i \leq 2M$ for every i , while $x_{i-1} \notin \{u_j\}_j$ implies $M_i - m_i < \varepsilon$. Hence

$$(*) \quad U(P, f) - L(P, f) \leq \varepsilon \sum_{\substack{i \\ x_{i-1} \notin \{u_j\}_j}} \Delta x_i + 2M \sum_j (v_j - u_j) \leq \varepsilon((b-a) + 2M).$$

Now let $\varepsilon > 0$. Choose $\varepsilon > 0$ such that

$$\varepsilon((b-a) + 2M) < \varepsilon.$$

Applying the preceding construction with this ε , the estimate $(*)$ gives

$$U(P, f) - L(P, f) < \varepsilon.$$

Thus $f \in \mathcal{R}([a, b])$. $\square$

1.6 Properties of Riemann Integrals

1.6.1. Theorem (Linearity). — If $f_1, f_2 \in \mathcal{R}([a, b])$ and $c \in \mathbb{R}$, then $f_1 + f_2, cf_1 \in \mathcal{R}([a, b])$, and

- $\int_a^b (f_1(x) + f_2(x)) dx = \int_a^b f_1(x) dx + \int_a^b f_2(x) dx;$
- $\int_a^b cf_1(x) dx = c \int_a^b f_1(x) dx.$

Proof. PICTURE. We use Condition $\mathcal{R}$: the oscillation of $f_1 + f_2$ is at most the sum of the oscillations of f_1 and f_2 , while scaling multiplies oscillation by $|c|$.

Let $\varepsilon > 0$. Since $f_1, f_2 \in \mathcal{R}([a, b])$, by Condition $\mathcal{R}$ there are partitions P_1, P_2 such that

$$U(P_j, f_j) - L(P_j, f_j) < \frac{\varepsilon}{2} \quad (j = 1, 2).$$

Let P be a common refinement of P_1 and P_2 . By the refinement theorem,

$$U(P, f_j) - L(P, f_j) < \frac{\varepsilon}{2} \quad (j = 1, 2).$$

For $I_i = [x_{i-1}, x_i]$, let $M_i^{(j)}$ and $m_i^{(j)}$ be the supremum and infimum of f_j on I_i . By the definitions of supremum and infimum,

$$\sup_{I_i} (f_1 + f_2) - \inf_{I_i} (f_1 + f_2) \leq (M_i^{(1)} - m_i^{(1)}) + (M_i^{(2)} - m_i^{(2)}).$$

Therefore, by the definitions of upper and lower sums,

$$\begin{aligned} U(P, f_1 + f_2) - L(P, f_1 + f_2) &\leq U(P, f_1) - L(P, f_1) + U(P, f_2) - L(P, f_2) \\ &< \varepsilon. \end{aligned}$$

Thus $f_1 + f_2 \in \mathcal{R}([a, b])$.

If $c = 0$, then $cf_1 = 0$ is integrable. Assume $c \neq 0$. For every partition Q , the definitions of supremum and infimum give

$$U(Q, cf_1) - L(Q, cf_1) = |c|(U(Q, f_1) - L(Q, f_1)).$$

Since $f_1 \in \mathcal{R}([a, b])$, Condition $\mathcal{R}$ gives a partition Q with $U(Q, f_1) - L(Q, f_1) < \varepsilon/|c|$. Hence

$$U(Q, cf_1) - L(Q, cf_1) < \varepsilon,$$

so $cf_1 \in \mathcal{R}([a, b])$.

It remains to prove the formulas. Let $\varepsilon > 0$. By Condition $\mathcal{R}$ and the refinement theorem, choose one partition $P = \{x_0, \dots, x_n\}$ satisfying Condition $\mathcal{R}$ for $f_1, f_2, f_1 + f_2$, each with $\varepsilon/3$. Choose $t_i \in [x_{i-1}, x_i]$. By the Riemann-sum approximation theorem,

$$\begin{aligned} \left| \int_a^b (f_1 + f_2) - \int_a^b f_1 - \int_a^b f_2 \right| &\leq \left| \int_a^b (f_1 + f_2) - \sum_{i=1}^n (f_1(t_i) + f_2(t_i)) \Delta x_i \right| \\ &\quad + \left| \sum_{i=1}^n f_1(t_i) \Delta x_i - \int_a^b f_1 \right| + \left| \sum_{i=1}^n f_2(t_i) \Delta x_i - \int_a^b f_2 \right| \\ &< \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\int_a^b (f_1(x) + f_2(x)) dx = \int_a^b f_1(x) dx + \int_a^b f_2(x) dx.$$

Similarly, choose P satisfying Condition $\mathcal{R}$ for f_1 and cf_1 , each with $\varepsilon/2$. By the Riemann-sum approximation theorem,

$$\left| \int_a^b cf_1 - c \int_a^b f_1 \right| \leq \left| \int_a^b cf_1 - \sum_{i=1}^n cf_1(t_i) \Delta x_i \right| + |c| \left| \sum_{i=1}^n f_1(t_i) \Delta x_i - \int_a^b f_1 \right|.$$

Applying the Riemann-sum approximation theorem with sufficiently small errors makes the right-hand side smaller than ε . Hence

$$\int_a^b cf_1(x) dx = c \int_a^b f_1(x) dx.$$

□

1.6.2. PROBLEM (Order and additivity). Assume $f_1, f_2 \in \mathcal{R}([a, b])$.

- (a) If $f_1 \leq f_2$, prove $\int_a^b f_1(x) dx \leq \int_a^b f_2(x) dx$.
 (b) If $a < c < b$, $f_1 \in \mathcal{R}([a, c])$, and $f_1 \in \mathcal{R}([c, b])$, prove

$$\int_a^b f_1(x) dx = \int_a^c f_1(x) dx + \int_c^b f_1(x) dx.$$

Solution. (a) Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. By hypothesis, $f_1 \leq f_2$. Hence, by the definitions of m_i and M_i ,

$$m_i(f_1) \leq m_i(f_2), \quad M_i(f_1) \leq M_i(f_2)$$

for every i . By the definitions of lower and upper sums,

$$L(P, f_1) \leq L(P, f_2), \quad U(P, f_1) \leq U(P, f_2).$$

Since $L(P, f_2) \leq \int_a^b f_2(x) dx$, by the definition of the lower integral, we obtain

$$L(P, f_1) \leq \int_a^b f_2(x) dx$$

for every partition P . Taking the supremum over P and using $f_1 \in \mathcal{R}([a, b])$,

$$\int_a^b f_1(x) dx = \sup_P L(P, f_1) \leq \int_a^b f_2(x) dx.$$

- (b) **PICTURE.** We choose one partition containing c , so that its Riemann sum splits into its parts on $[a, c]$ and $[c, b]$.

Let $\varepsilon > 0$. Since $f_1 \in \mathcal{R}([a, b])$, by Condition $\mathcal{R}$ there is a partition P of $[a, b]$ such that

$$U(P, f_1) - L(P, f_1) < \frac{\varepsilon}{3}.$$

By the refinement theorem, after adjoining c to P , the same inequality remains true. Write

$$P = \{a = x_0 < \dots < x_k = c < \dots < x_n = b\},$$

and choose $t_i \in [x_{i-1}, x_i]$. By the definitions of the restrictions of P ,

$$P_{[a,c]} = \{x_0, \dots, x_k\}, \quad P_{[c,b]} = \{x_k, \dots, x_n\}.$$

Since

$U(P, f_1) - L(P, f_1) = (U(P_{[a,c]}, f_1) - L(P_{[a,c]}, f_1)) + (U(P_{[c,b]}, f_1) - L(P_{[c,b]}, f_1))$,
each restricted partition satisfies Condition $\mathcal{R}$ with $\varepsilon/3$. Therefore, by the Riemann-sum approximation theorem,

$$\begin{aligned} \left| \int_a^b f_1(x) dx - \int_a^c f_1(x) dx - \int_c^b f_1(x) dx \right| &\leq \left| \int_a^b f_1(x) dx - \sum_{i=1}^n f_1(t_i) \Delta x_i \right| \\ &\quad + \left| \sum_{i=1}^k f_1(t_i) \Delta x_i - \int_a^c f_1(x) dx \right| \\ &\quad + \left| \sum_{i=k+1}^n f_1(t_i) \Delta x_i - \int_c^b f_1(x) dx \right| \\ &< \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\int_a^b f_1(x) dx = \int_a^c f_1(x) dx + \int_c^b f_1(x) dx.$$

□

1.6.3. PROBLEM (Composition). Suppose $f \in \mathcal{R}([a, b])$, $m \leq f \leq M$, $\varphi : [m, M] \rightarrow \mathbb{R}$ is continuous, and $h(x) = \varphi(f(x))$. Prove that $h \in \mathcal{R}([a, b])$.

Solution. PICTURE. The oscillation of $\varphi \circ f$ is small where the oscillation of f is small. The remaining intervals have small total length because f is Riemann integrable.

Since $[m, M]$ is compact and φ is continuous, the Extreme Value Theorem gives

$$B = \max_{t \in [m, M]} |\varphi(t)| < \infty.$$

For a partition $P = \{x_0, \dots, x_n\}$, write

$$\omega_i(f) = \sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x), \quad \omega_i(h) = \sup_{x \in [x_{i-1}, x_i]} h(x) - \inf_{x \in [x_{i-1}, x_i]} h(x).$$

By the definitions of upper and lower sums,

$$U(P, h) - L(P, h) = \sum_{i=1}^n \omega_i(h) \Delta x_i.$$

Let $\varepsilon > 0$. We first make the contribution from intervals on which f has small oscillation less than $\varepsilon/2$. Choose $\eta > 0$ such that

$$\eta(b-a) < \frac{\varepsilon}{2}.$$

Since φ is continuous on the compact interval $[m, M]$, the Heine–Cantor Theorem gives $\delta > 0$ such that

$$|s - t| < \delta \implies |\varphi(s) - \varphi(t)| < \eta \quad (s, t \in [m, M]).$$

Let $\mathbf{P}$ be any partition, and separate its subintervals according to whether $\omega_i(f) < \delta$ or $\omega_i(f) \geq \delta$. If $\omega_i(f) < \delta$, then, by the definition of $\omega_i(f)$,

$$|f(x) - f(y)| \leq \omega_i(f) < \delta \implies |h(x) - h(y)| < \eta,$$

so $\omega_i(h) \leq \eta$. If $\omega_i(f) \geq \delta$, then $|h| \leq B$, so $\omega_i(h) \leq 2B$. Therefore

$$\begin{aligned} U(\mathbf{P}, h) - L(\mathbf{P}, h) &= \sum_{i=1}^n \omega_i(h) \Delta x_i \\ &\leq \eta \sum_{\omega_i(f) < \delta} \Delta x_i + 2B \sum_{\omega_i(f) \geq \delta} \Delta x_i \\ &\leq \eta(b-a) + \frac{2B}{\delta} \sum_{\omega_i(f) \geq \delta} \omega_i(f) \Delta x_i \\ &\leq \eta(b-a) + \frac{2B}{\delta} (U(\mathbf{P}, f) - L(\mathbf{P}, f)). \end{aligned}$$

We now make the second term less than $\varepsilon/2$. Since $f \in \mathcal{R}([a, b])$, Condition $\mathcal{R}$ gives a partition $\mathbf{P}$ such that

$$U(\mathbf{P}, f) - L(\mathbf{P}, f) < \frac{\delta \varepsilon}{2(1+2B)}.$$

For this partition,

$$U(\mathbf{P}, h) - L(\mathbf{P}, h) < \frac{\varepsilon}{2} + \frac{2B}{\delta} \cdot \frac{\delta \varepsilon}{2(1+2B)} < \varepsilon.$$

Thus Condition $\mathcal{R}$ holds for h , so

$$h \in \mathcal{R}([a, b]).$$

□

1.6.4. Proposition (Products and absolute values). — *If $f, g \in \mathcal{R}([a, b])$, then*

- (a) $fg \in \mathcal{R}([a, b])$;
- (b) $|f| \in \mathcal{R}([a, b])$ and

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

Proof. (a) **PRODUCT IDENTITY.** By linearity,

$$f + g, f - g \in \mathcal{R}([a, b]).$$

Since every Riemann integrable function is bounded, the composition theorem applied to $\varphi(t) = t^2$ gives

$$(f + g)^2, (f - g)^2 \in \mathcal{R}([a, b]).$$

By the algebraic identity

$$fg = \frac{1}{4}((f + g)^2 - (f - g)^2),$$

linearity gives

$$fg \in \mathcal{R}([a, b]).$$

(b) Since $f \in \mathcal{R}([a, b])$, it is bounded. The function $\varphi(t) = |t|$ is continuous, so the composition theorem gives

$$|f| \in \mathcal{R}([a, b]).$$

By the elementary inequalities

$$-|f(x)| \leq f(x) \leq |f(x)| \quad (x \in [a, b]),$$

the order theorem and linearity give

$$-\int_a^b |f(x)| \, dx = \int_a^b -|f(x)| \, dx \leq \int_a^b f(x) \, dx \leq \int_a^b |f(x)| \, dx.$$

Hence

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

□

1.7 Substitution

1.7.1. Theorem (Change of variables). — Let $\varphi : [A, B] \rightarrow [a, b]$ be strictly increasing and continuous, with $\varphi' \in \mathcal{R}([A, B])$. If $f \in \mathcal{R}([a, b])$, then $(f \circ \varphi)\varphi' \in \mathcal{R}([A, B])$ and

$$\int_a^b f(x) \, dx = \int_A^B f(\varphi(x))\varphi'(x) \, dx.$$

Proof. PICTURE. We pull a partition of $[a, b]$ back through φ . On each resulting interval, $f \circ \varphi$ lies between the corresponding lower and upper values of f ; multiplying by $\varphi' \geq 0$ turns the interval lengths into the increments of φ .

Since $f \in \mathcal{R}([a, b])$ and $\varphi' \in \mathcal{R}([A, B])$, both functions are bounded. Hence

$$(f \circ \varphi)\varphi'$$

is bounded. By hypothesis, φ is increasing and differentiable, so

$$\varphi'(t) \geq 0 \quad (t \in [A, B]).$$

Let $\varepsilon > 0$. By Condition $\mathcal{R}$, choose a partition

$$Q = \{a = y_0 < \cdots < y_n = b\}$$

such that

$$U(Q, f) - L(Q, f) < \varepsilon.$$

Since φ is continuous and strictly increasing, it is a bijection $[A, B] \rightarrow [a, b]$. Define

$$x_i = \varphi^{-1}(y_i), \quad P = \{A = x_0 < \cdots < x_n = B\}.$$

Let

$$m_i = \inf_{y \in [y_{i-1}, y_i]} f(y), \quad M_i = \sup_{y \in [y_{i-1}, y_i]} f(y).$$

Define step functions on $[A, B]$ by

$$\ell_P(t) = m_i \varphi'(t), \quad u_P(t) = M_i \varphi'(t) \quad (t \in [x_{i-1}, x_i]).$$

Their values at the finitely many partition points may be chosen arbitrarily. Since the step functions and φ' are Riemann integrable, the product theorem gives

$$\ell_P, u_P \in \mathcal{R}([A, B]).$$

For $t \in [x_{i-1}, x_i]$, monotonicity of φ gives

$$\varphi(t) \in [y_{i-1}, y_i], \quad m_i \leq f(\varphi(t)) \leq M_i.$$

Since $\varphi'(t) \geq 0$,

$$\ell_P(t) \leq f(\varphi(t))\varphi'(t) \leq u_P(t).$$

By the Fundamental Theorem of Calculus for a function with Riemann integrable derivative,

$$\begin{aligned} \int_A^B \ell_P(t) dt &= \sum_{i=1}^n m_i \int_{x_{i-1}}^{x_i} \varphi'(t) dt = \sum_{i=1}^n m_i (\varphi(x_i) - \varphi(x_{i-1})) \\ &= \sum_{i=1}^n m_i (y_i - y_{i-1}) = L(Q, f), \end{aligned}$$

and similarly

$$\int_A^B u_P(t) dt = U(Q, f).$$

By the definitions of the lower and upper integrals and the preceding point-wise inequalities,

$$\begin{aligned} \overline{\int}_A^B (f \circ \varphi) \varphi' - \underline{\int}_A^B (f \circ \varphi) \varphi' &\leq \int_A^B u_P(t) dt - \int_A^B \ell_P(t) dt \\ &= U(Q, f) - L(Q, f) < \varepsilon. \end{aligned}$$

Thus $(f \circ \varphi) \varphi' \in \mathcal{R}([A, B])$.

Finally, by the order theorem,

$$L(Q, f) = \int_A^B \ell_P(t) dt \leq \int_A^B f(\varphi(t)) \varphi'(t) dt \leq \int_A^B u_P(t) dt = U(Q, f).$$

Also, by the definition of the Riemann integral,

$$L(Q, f) \leq \int_a^b f(y) dy \leq U(Q, f).$$

Hence

$$\left| \int_a^b f(y) dy - \int_A^B f(\varphi(t)) \varphi'(t) dt \right| \leq U(Q, f) - L(Q, f) < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\int_a^b f(x) dx = \int_A^B f(\varphi(x)) \varphi'(x) dx.$$

□

1.7.2. Example: A substitution. For $\varphi(x) = x^2$ on $[0, 2]$ and $f(x) = e^{-x}$,

$$\int_0^2 2xe^{-x^2} dx = \int_0^4 e^{-x} dx.$$

1.8 Integration and Differentiation

1.8.1. Theorem. — Let $f \in \mathcal{R}([a, b])$ and set

$$F(x) = \int_a^x f(y) dy \quad (x \in [a, b]).$$

Then F is continuous on $[a, b]$. If f is continuous at $x_0 \in [a, b]$, then F is differentiable at x_0 and

$$F'(x_0) = f(x_0).$$

Proof. Since $f \in \mathcal{R}([a, b])$, f is bounded. Choose $M > 0$ such that

$$|f(y)| \leq M \quad (y \in [a, b]).$$

Let $x, x_0 \in [a, b]$. By additivity of the integral and the integral inequality,

$$\begin{aligned} |F(x) - F(x_0)| &= \left| \int_{x_0}^x f(y) dy \right| \\ &\leq \int_{\min\{x, x_0\}}^{\max\{x, x_0\}} |f(y)| dy \\ &\leq M|x - x_0|. \end{aligned}$$

Thus $x \rightarrow x_0$ implies $F(x) \rightarrow F(x_0)$, so F is continuous on $[a, b]$.

Now suppose that f is continuous at x_0 . For $x \neq x_0$, by additivity and linearity,

$$\begin{aligned} \left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| &= \left| \frac{1}{x - x_0} \int_{x_0}^x (f(y) - f(x_0)) dy \right| \\ &\leq \frac{1}{|x - x_0|} \int_{\min\{x, x_0\}}^{\max\{x, x_0\}} |f(y) - f(x_0)| dy. \end{aligned}$$

Let $\varepsilon > 0$. By continuity of f at x_0 , there is $\delta > 0$ such that

$$|y - x_0| < \delta \implies |f(y) - f(x_0)| < \varepsilon.$$

If $0 < |x - x_0| < \delta$, then every y between x and x_0 satisfies $|y - x_0| < \delta$. Therefore

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| < \frac{1}{|x - x_0|} \varepsilon |x - x_0| = \varepsilon.$$

Hence

$$F'(x_0) = f(x_0).$$

At $x_0 = a$ or $x_0 = b$, this is the corresponding one-sided derivative. $\square$

1.8.2. Theorem (Fundamental theorem of calculus). — Let $f \in \mathcal{R}([a, b])$. If a differentiable function F on $[a, b]$ satisfies $F' = f$, then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

Proof. Let $\varepsilon > 0$. Since $f \in \mathcal{R}([a, b])$, by Condition $\mathcal{R}$ choose a partition

$$P = \{a = x_0 < \cdots < x_n = b\}.$$

By the Riemann-sum approximation theorem, for every $t_i \in [x_{i-1}, x_i]$,

$$\left| \int_a^b f(x) \, dx - \sum_{i=1}^n f(t_i) \Delta x_i \right| < \varepsilon.$$

For every i , F is continuous on $[x_{i-1}, x_i]$ and differentiable on (x_{i-1}, x_i) . Thus the hypotheses of the Mean Value Theorem are satisfied, so there is $t_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = F'(t_i)(x_i - x_{i-1}) = f(t_i) \Delta x_i.$$

Summing and telescoping,

$$\sum_{i=1}^n f(t_i) \Delta x_i = \sum_{i=1}^n (F(x_i) - F(x_{i-1})) = F(b) - F(a).$$

Therefore

$$\left| \int_a^b f(x) \, dx - (F(b) - F(a)) \right| < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

□

1.8.3. Example: Differentiating an integral. Let

$$h(x) = \int_0^{x^2} \frac{1}{y^2 + e^{y^6}} \, dy.$$

Then

$$h'(x) = \frac{2x}{x^4 + e^{x^{12}}}.$$

1.8.4. Theorem (Integration by parts). — Suppose F, G are differentiable on $[a, b]$, $F' = f \in \mathcal{R}([a, b])$, and $G' = g \in \mathcal{R}([a, b])$. Then

$$\int_a^b F(x)g(x) \, dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x) \, dx.$$

Proof. Since F and G are differentiable, they are continuous. Hence, by the theorem on continuous functions,

$$F, G \in \mathcal{R}([a, b]).$$

By the product theorem,

$$Fg, fG \in \mathcal{R}([a, b]).$$

By the product rule and the hypotheses $F' = f$, $G' = g$,

$$(FG)' = F'G + FG' = fG + Fg.$$

Therefore $(FG)' \in \mathcal{R}([a, b])$. Applying the Fundamental Theorem of Calculus and then linearity,

$$\begin{aligned} F(b)G(b) - F(a)G(a) &= \int_a^b (FG)'(x) \, dx \\ &= \int_a^b (f(x)G(x) + F(x)g(x)) \, dx \\ &= \int_a^b f(x)G(x) \, dx + \int_a^b F(x)g(x) \, dx. \end{aligned}$$

Rearranging gives

$$\int_a^b F(x)g(x) \, dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x) \, dx.$$

□

Set Algebras, Borel Sets, and Generating Classes

2.1 Algebras and σ -Algebras

Let X be a nonempty set, and let $\mathcal{P}(X)$ denote its power set.

2.1.1. Definition: Algebra of sets. An algebra of sets on X is a collection $\mathcal{A} \subseteq \mathcal{P}(X)$ such that

- (a) $\mathcal{A} \neq \emptyset$;
- (b) $E \in \mathcal{A} \Rightarrow E^c \in \mathcal{A}$;
- (c) $E_1, \dots, E_N \in \mathcal{A} \Rightarrow \bigcup_{j=1}^N E_j \in \mathcal{A}$.

2.1.2. Example. For $E \subseteq X$, the collection $\{\emptyset, X, E, E^c\}$ is an algebra on X .

2.1.3. Definition: σ -algebra and measurable space. A σ -algebra on X is a collection $\mathcal{A} \subseteq \mathcal{P}(X)$ such that

- (a) $\mathcal{A} \neq \emptyset$;
- (b) $E \in \mathcal{A} \Rightarrow E^c \in \mathcal{A}$;
- (c) $(E_j)_{j \in \mathbb{N}} \subseteq \mathcal{A} \Rightarrow \bigcup_{j=1}^{\infty} E_j \in \mathcal{A}$.

The pair $(X, \mathcal{A})$ is then a measurable space.

2.1.4. Example. For every set X , both $\{\emptyset, X\}$ and $\mathcal{P}(X)$ are σ -algebras on X .

2.1.5. Example: Countable and co-countable sets. If X is uncountable, then

$$\mathcal{A} = \{E \subseteq X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ -algebra on X .

2.1.6. Remark. An algebra is closed under finite intersections, and a σ -algebra is closed under countable intersections. In the countable case this follows from De Morgan's identity

$$\left(\bigcup_{j=1}^{\infty} E_j \right)^c = \bigcap_{j=1}^{\infty} E_j^c.$$

2.1.7. Remark. If $\mathcal{A}$ is an algebra, then $\emptyset, X \in \mathcal{A}$.

2.1.8. Proposition. — An algebra $\mathcal{A}$ is a σ -algebra if and only if it is closed under countable disjoint unions.

Proof. We prove both implications.

($\Rightarrow$) If $\mathcal{A}$ is a σ -algebra, then it is closed under countable disjoint unions by definition.

($\Leftarrow$) Suppose that $\mathcal{A}$ is an algebra closed under countable disjoint unions. Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$. We must prove that $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

By disjointification, define

$$B_1 = A_1, \quad B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k \quad (n \geq 2).$$

Since $\mathcal{A}$ is an algebra, finite unions and differences of its sets belong to $\mathcal{A}$. Hence $B_n \in \mathcal{A}$ for every n . By construction, the sets B_n are pairwise disjoint.

We prove both inclusions.

($\subseteq$) Since $B_n \subseteq A_n$ for every n ,

$$\bigcup_{n=1}^{\infty} B_n \subseteq \bigcup_{n=1}^{\infty} A_n.$$

($\supseteq$) Let $x \in \bigcup_{n=1}^{\infty} A_n$. By the well-ordering of $\mathbb{N}$, choose the smallest n such that $x \in A_n$. Then $x \notin A_k$ for $k < n$, so

$$x \in A_n \setminus \bigcup_{k=1}^{n-1} A_k = B_n.$$

Therefore

$$\bigcup_{n=1}^{\infty} A_n \subseteq \bigcup_{n=1}^{\infty} B_n.$$

Hence

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n.$$

By the assumed closure under countable disjoint unions,

$$\bigcup_{n=1}^{\infty} B_n \in \mathcal{A}.$$

Therefore

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ -algebra. □

2.1.9. Proposition. — *The intersection of any family of σ -algebras on X is a σ -algebra on X .*

Proof. By the intersection construction, define

$$\mathcal{A} = \bigcap_{\alpha \in I} \mathcal{A}_{\alpha}.$$

If $I = \emptyset$, then $\mathcal{A} = \mathcal{P}(X)$, which is a σ -algebra. Assume $I \neq \emptyset$.

We verify the defining properties of a σ -algebra.

Since each $\mathcal{A}_{\alpha}$ is a σ -algebra,

$$X \in \mathcal{A}_{\alpha} \quad (\alpha \in I) \implies X \in \bigcap_{\alpha \in I} \mathcal{A}_{\alpha} = \mathcal{A}.$$

Let $E \in \mathcal{A}$. By the definition of intersection,

$$E \in \mathcal{A} \implies E \in \mathcal{A}_{\alpha} \quad (\alpha \in I).$$

Since each $\mathcal{A}_\alpha$ is closed under complements,

$$E \in \mathcal{A}_\alpha \implies E^c \in \mathcal{A}_\alpha \quad (\alpha \in I) \implies E^c \in \mathcal{A}.$$

Finally, let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$. By the definition of intersection,

$$E_n \in \mathcal{A} \implies E_n \in \mathcal{A}_\alpha \quad (n \in \mathbb{N}, \alpha \in I).$$

Since each $\mathcal{A}_\alpha$ is closed under countable unions,

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}_\alpha \quad (\alpha \in I) \implies \bigcup_{n=1}^{\infty} E_n \in \mathcal{A}.$$

Thus $\mathcal{A}$ contains X and is closed under complements and countable unions. Therefore $\mathcal{A}$ is a σ -algebra on X . $\square$

2.1.10. Definition: Generated σ -algebra. For $\mathcal{E} \subseteq \mathcal{P}(X)$, define

$$\sigma(\mathcal{E}) = \bigcap_{\substack{\mathcal{A} \supseteq \mathcal{E} \\ \mathcal{A} \text{ a } \sigma\text{-algebra}}} \mathcal{A}.$$

This is the unique smallest σ -algebra containing $\mathcal{E}$, called the σ -algebra generated by $\mathcal{E}$.

2.1.11. Example. The family $\mathcal{E} = \{\{x\} : x \in X\}$ generates the σ -algebra of countable and co-countable subsets of X .

2.1.12. Lemma (Monotonicity of $\sigma(\cdot)$). — If $\mathcal{E} \subseteq \sigma(\mathcal{F})$, then $\sigma(\mathcal{E}) \subseteq \sigma(\mathcal{F})$. In particular, $\sigma(\sigma(\mathcal{E})) = \sigma(\mathcal{E})$.

Proof. Suppose

$$\mathcal{E} \subseteq \sigma(\mathcal{F}).$$

By definition, $\sigma(\mathcal{F})$ is a σ -algebra containing $\mathcal{E}$. Since $\sigma(\mathcal{E})$ is the smallest σ -algebra containing $\mathcal{E}$, minimality gives

$$\sigma(\mathcal{E}) \subseteq \sigma(\mathcal{F}).$$

For the particular statement, we prove both inclusions.

($\subseteq$) By definition,

$$\sigma(\mathcal{E}) \subseteq \sigma(\mathcal{E}).$$

Applying the first implication with $\mathcal{E}$ replaced by $\sigma(\mathcal{E})$, we obtain

$$\sigma(\sigma(\mathcal{E})) \subseteq \sigma(\mathcal{E}).$$

($\supseteq$) By definition, every family is contained in the σ -algebra it generates:

$$\mathcal{E} \subseteq \sigma(\mathcal{E}).$$

Applying the first implication gives

$$\sigma(\mathcal{E}) \subseteq \sigma(\sigma(\mathcal{E})).$$

Hence

$$\sigma(\sigma(\mathcal{E})) = \sigma(\mathcal{E}).$$

$\square$

2.2 Borel Sets

2.2.1. Definition: Borel σ -algebra. Let X be a topological space. The *Borel σ -algebra* of X is

$$\text{Bor}(X) = \sigma(\{\mathcal{U} \subseteq X : \mathcal{U} \text{ is open}\}).$$

Equivalently, it is the σ -algebra generated by the closed subsets of X . Its members are the *Borel sets*.

2.2.2. Definition: F_σ , G_δ , and related classes. A countable intersection of open sets is a G_δ set, and a countable union of closed sets is an F_σ set. A countable union of G_δ sets is a $G_{\delta\sigma}$ set, while a countable intersection of F_σ sets is an $F_{\sigma\delta}$ set.

2.2.3. PROBLEM (Generating $\text{Bor}(\mathbb{R})$). Prove that $\text{Bor}(\mathbb{R})$ is generated by each of the following families:

- (a) $\mathcal{E}_1 = \{(a, b) : a < b\}$;
- (b) $\mathcal{E}_2 = \{[a, b] : a < b\}$;
- (c) $\mathcal{E}_3 = \{(a, b] : a < b\}$ or $\mathcal{E}_4 = \{[a, b) : a < b\}$;
- (d) $\mathcal{E}_5 = \{(a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) : a \in \mathbb{R}\}$;
- (e) $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a] : a \in \mathbb{R}\}$.

Solution. REDUCTION. We first establish

$$\text{Bor}(\mathbb{R}) = \sigma(\mathcal{E}_1), \quad \mathcal{E}_1 = \{(a, b) : a < b\}.$$

Since every open interval is Borel,

$$\sigma(\mathcal{E}_1) \subseteq \text{Bor}(\mathbb{R}).$$

Conversely, if $\mathcal{U} \subseteq \mathbb{R}$ is open, then by the countable-base property of $\mathbb{R}$,

$$\mathcal{U} = \bigcup_{\substack{p, q \in \mathbb{Q} \\ (p, q) \subseteq \mathcal{U}}} (p, q),$$

so $\mathcal{U} \in \sigma(\mathcal{E}_1)$. Hence

$$\text{Bor}(\mathbb{R}) = \sigma(\mathcal{E}_1).$$

We will repeatedly use the following consequence of monotonicity:

$$\mathcal{G} \subseteq \text{Bor}(\mathbb{R}) \quad \text{and} \quad \mathcal{E}_1 \subseteq \sigma(\mathcal{G}) \quad \implies \quad \sigma(\mathcal{G}) = \text{Bor}(\mathbb{R}).$$

Indeed,

$$\sigma(\mathcal{G}) \subseteq \text{Bor}(\mathbb{R})$$

by monotonicity, while

$$\text{Bor}(\mathbb{R}) = \sigma(\mathcal{E}_1) \subseteq \sigma(\sigma(\mathcal{G})) = \sigma(\mathcal{G}).$$

(a) By the reduction,

$$\sigma(\mathcal{E}_1) = \text{Bor}(\mathbb{R}).$$

- (b) **ENDPOINT TRIMMING.** Every closed interval is Borel. Fix $a < b$. Since $b - a > 0$, the Archimedean property gives $N \in \mathbb{N}$ such that

$$N > \frac{2}{b - a}.$$

Taking reciprocals gives

$$\frac{2}{N} < b - a.$$

Therefore

$$(a, b) = \bigcup_{n=N}^{\infty} \left[a + \frac{1}{n}, b - \frac{1}{n} \right].$$

Hence $\mathcal{E}_1 \subseteq \sigma(\mathcal{E}_2)$, and therefore

$$\sigma(\mathcal{E}_2) = \text{Bor}(\mathbb{R}).$$

- (c) **ENDPOINT APPROXIMATION.** Every interval in $\mathcal{E}_3$ is Borel, and

$$(a, b] = \bigcap_{n=1}^{\infty} \left(a, b + \frac{1}{n} \right).$$

Thus $\mathcal{E}_3 \subseteq \text{Bor}(\mathbb{R})$. Since $b - a > 0$, the Archimedean property gives $N \in \mathbb{N}$ such that

$$N > \frac{1}{b - a}, \quad \text{hence} \quad \frac{1}{N} < b - a.$$

Then

$$(a, b) = \bigcup_{n=N}^{\infty} \left(a, b - \frac{1}{n} \right).$$

Hence

$$\sigma(\mathcal{E}_3) = \text{Bor}(\mathbb{R}).$$

Similarly, every interval in $\mathcal{E}_4$ is Borel, and

$$(a, b) = \bigcup_{n=N}^{\infty} \left[a + \frac{1}{n}, b \right),$$

where N is chosen by the Archimedean property so that $1/N < b - a$. Therefore

$$\sigma(\mathcal{E}_4) = \text{Bor}(\mathbb{R}).$$

- (d) **SET DIFFERENCE OF RAYS.** Since the rays in $\mathcal{E}_5$ are Borel,

$$(a, b) = (a, \infty) \setminus (b, \infty) \in \sigma(\mathcal{E}_5).$$

Hence

$$\sigma(\mathcal{E}_5) = \text{Bor}(\mathbb{R}).$$

For $\mathcal{E}_6$, choose $N \in \mathbb{N}$ such that $1/N < b - a$. Then

$$\left[a + \frac{1}{n}, b \right) = (-\infty, b) \setminus \left(-\infty, a + \frac{1}{n} \right),$$

and therefore

$$(a, b) = \bigcup_{n=N}^{\infty} \left((-\infty, b) \setminus \left(-\infty, a + \frac{1}{n} \right) \right) \in \sigma(\mathcal{E}_6).$$

Thus

$$\sigma(\mathcal{E}_6) = \text{Bor}(\mathbb{R}).$$

(e) APPROXIMATION BY CLOSED RAYS. Every member of $\mathcal{E}_7$ is Borel, and

$$(a, \infty) = \bigcup_{n=1}^{\infty} \left[a + \frac{1}{n}, \infty \right).$$

Consequently,

$$(a, b) = (a, \infty) \setminus [b, \infty) \in \sigma(\mathcal{E}_7).$$

Hence

$$\sigma(\mathcal{E}_7) = \text{Bor}(\mathbb{R}).$$

Finally, for $\mathcal{E}_8$,

$$(-\infty, b) = \bigcup_{n=1}^{\infty} \left(-\infty, b - \frac{1}{n} \right],$$

so

$$(a, b) = (-\infty, b) \setminus (-\infty, a] \in \sigma(\mathcal{E}_8).$$

Therefore

$$\sigma(\mathcal{E}_8) = \text{Bor}(\mathbb{R}).$$

□

2.3 Product σ -Algebras

2.3.1. Definition: Product σ -algebra. Let $(X_\alpha)_{\alpha \in A}$ be nonempty sets, $X = \prod_{\alpha \in A} X_\alpha$, and $\pi_\alpha : X \rightarrow X_\alpha$ the coordinate maps. If $\mathcal{A}_\alpha$ is a σ -algebra on X_α for every $\alpha \in A$, the *product σ -algebra* is

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha = \sigma(\{\pi_\alpha^{-1}[E_\alpha] : E_\alpha \in \mathcal{A}_\alpha, \alpha \in A\}).$$

For $A = \{1, \dots, n\}$, write $\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$.

2.3.2. Proposition (Countable index sets). — If A is countable, then $\bigotimes_{\alpha \in A} \mathcal{A}_\alpha$ is generated by

$$\left\{ \prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{A}_\alpha \right\}.$$

Proof. Let

$$X = \prod_{\alpha \in A} X_\alpha, \quad \pi_\alpha : X \rightarrow X_\alpha$$

be the coordinate projections. Write

$$\mathcal{R} = \left\{ \prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{A}_\alpha \right\}.$$

TECHNIQUE. Because A is countable, an arbitrary rectangle can be written as a countable intersection of cylinders.

Let $\mathcal{C}$ be the family of finite-coordinate cylinders

$$\mathcal{C} = \left\{ \bigcap_{j=1}^n \pi_{\alpha_j}^{-1}(E_{\alpha_j}) : E_{\alpha_j} \in \mathcal{A}_{\alpha_j} \right\}.$$

By definition,

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha = \sigma(\mathcal{C}).$$

We prove both inclusions.

($\subseteq$) Choose an enumeration $A = \{\alpha_1, \alpha_2, \dots\}$. Let $R = \prod_{\alpha \in A} E_\alpha \in \mathcal{R}$. By the definition of a Cartesian product,

$$R = \bigcap_{n=1}^{\infty} \pi_{\alpha_n}^{-1}(E_{\alpha_n}).$$

Each set $\pi_{\alpha_n}^{-1}(E_{\alpha_n})$ is a cylinder and hence belongs to $\bigotimes_{\alpha \in A} \mathcal{A}_\alpha$. Since this is a σ -algebra, it is closed under countable intersections. Therefore

$$R \in \bigotimes_{\alpha \in A} \mathcal{A}_\alpha.$$

Thus

$$\mathcal{R} \subseteq \bigotimes_{\alpha \in A} \mathcal{A}_\alpha,$$

and monotonicity of generated σ -algebras gives

$$\sigma(\mathcal{R}) \subseteq \bigotimes_{\alpha \in A} \mathcal{A}_\alpha.$$

($\supseteq$) Let

$$C = \bigcap_{j=1}^n \pi_{\alpha_j}^{-1}(E_{\alpha_j}) \in \mathcal{C}.$$

Define

$$F_\alpha = \begin{cases} E_\alpha, & \alpha \in \{\alpha_1, \dots, \alpha_n\}, \\ X_\alpha, & \text{otherwise.} \end{cases}$$

Since $X_\alpha \in \mathcal{A}_\alpha$, we have

$$C = \prod_{\alpha \in A} F_\alpha \in \mathcal{R}.$$

Hence $\mathcal{C} \subseteq \mathcal{R}$. By monotonicity,

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha = \sigma(\mathcal{C}) \subseteq \sigma(\mathcal{R}).$$

Therefore

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha = \sigma\left(\left\{\prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{A}_\alpha\right\}\right).$$

□

2.3.3. Remark: Role of countability. The countability of A is essential in the preceding proposition: a general product set is an intersection indexed by A of coordinate preimages, whereas a σ -algebra need only be closed under countable intersections.

2.3.4. Proposition (Generators of a product σ -algebra). — Suppose $\mathcal{A}_\alpha = \sigma(\mathcal{E}_\alpha)$ for every $\alpha \in A$. Then $\bigotimes_{\alpha \in A} \mathcal{A}_\alpha$ is generated by

$$\mathcal{F}_1 = \{\pi_\alpha^{-1}[E_\alpha] : E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\}.$$

If A is countable and $X_\alpha \in \mathcal{E}_\alpha$ for every $\alpha \in A$, it is also generated by

$$\mathcal{F}_2 = \left\{\prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{E}_\alpha\right\}.$$

Proof. Let

$$X = \prod_{\alpha \in A} X_\alpha, \quad \pi_\alpha : X \rightarrow X_\alpha$$

denote the coordinate projections.

GENERATED- σ -ALGEBRA ARGUMENT. Set

$$\Sigma = \sigma(\mathcal{F}_1).$$

Since $\mathcal{E}_\alpha \subseteq \mathcal{A}_\alpha$, every generator $\pi_\alpha^{-1}[E_\alpha] \in \mathcal{F}_1$ is measurable in the product σ -algebra. Hence

$$\Sigma \subseteq \bigotimes_{\alpha \in A} \mathcal{A}_\alpha.$$

For the reverse inclusion, fix $\alpha \in A$ and define

$$\mathcal{G}_\alpha = \{E \subseteq X_\alpha : \pi_\alpha^{-1}[E] \in \Sigma\}.$$

Since inverse images preserve complements and countable unions, $\mathcal{G}_\alpha$ is a σ -algebra on X_α . Moreover,

$$\mathcal{E}_\alpha \subseteq \mathcal{G}_\alpha$$

by the definition of $\mathcal{F}_1$. Since $\mathcal{A}_\alpha = \sigma(\mathcal{E}_\alpha)$, minimality gives

$$\mathcal{A}_\alpha \subseteq \mathcal{G}_\alpha.$$

Thus

$$\pi_\alpha^{-1}[E] \in \Sigma \quad (E \in \mathcal{A}_\alpha).$$

Therefore all measurable cylinders belong to Σ , and hence

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha \subseteq \Sigma.$$

Consequently,

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha = \sigma(\mathcal{F}_1).$$

Now suppose that A is countable and $X_\alpha \in \mathcal{E}_\alpha$ for every $\alpha \in A$. Define

$$\mathcal{F}_2 = \left\{ \prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{E}_\alpha \right\}.$$

COUNTABLE-COORDINATE ARGUMENT. By the preceding proposition, every element of $\mathcal{F}_2$ belongs to the product σ -algebra. Hence

$$\sigma(\mathcal{F}_2) \subseteq \bigotimes_{\alpha \in A} \mathcal{A}_\alpha.$$

Conversely, let $E_\alpha \in \mathcal{E}_\alpha$. Since $X_\beta \in \mathcal{E}_\beta$ for every $\beta \in A$,

$$\pi_\alpha^{-1}[E_\alpha] = \prod_{\beta \in A} F_\beta, \quad F_\beta = \begin{cases} E_\alpha, & \beta = \alpha, \\ X_\beta, & \beta \neq \alpha. \end{cases}$$

Thus

$$\mathcal{F}_1 \subseteq \sigma(\mathcal{F}_2),$$

and monotonicity gives

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha = \sigma(\mathcal{F}_1) \subseteq \sigma(\mathcal{F}_2).$$

Therefore

$$\bigotimes_{\alpha \in A} \mathcal{A}_\alpha = \sigma(\mathcal{F}_2).$$

□

2.3.5. PROBLEM (Borel sets on finite products). Let $N \in \mathbb{N}$, let $X_1, \dots, X_N$ be metric spaces, and equip $X = \prod_{j=1}^N X_j$ with a product metric. Show that

$$\bigotimes_{j=1}^N \text{Bor}(X_j) \subseteq \text{Bor}(X).$$

If the X_j are separable, prove equality. What can be said for $N = \infty$?

Solution. Let

$$X = \prod_{j=1}^N X_j, \quad \pi_j : X \rightarrow X_j$$

be the coordinate projections.

CONTINUOUS-PREIMAGE ARGUMENT. Since X is equipped with a product metric, each coordinate projection π_j is continuous. Hence, for every $E_j \in \text{Bor}(X_j)$,

$$\pi_j^{-1}(E_j) \in \text{Bor}(X).$$

The product σ -algebra is generated by these coordinate cylinders. Therefore

$$\bigotimes_{j=1}^N \text{Bor}(X_j) \subseteq \text{Bor}(X).$$

Now assume that every X_j is separable. Since each X_j is a separable metric space, it has a countable basis $\mathcal{B}_j$. The family

$$\mathcal{B} = \{U_1 \times \dots \times U_N : U_j \in \mathcal{B}_j\}$$

is a countable basis for the product topology on X .

COUNTABLE-BASE ARGUMENT. Let $U \subseteq X$ be open. By the basis property,

$$U = \bigcup_{\substack{B \in \mathcal{B} \\ B \subseteq U}} B.$$

Since $\mathcal{B}$ is countable, this is a countable union. Moreover, for

$$B = U_1 \times \dots \times U_N \in \mathcal{B},$$

we have

$$B = \bigcap_{j=1}^N \pi_j^{-1}(U_j).$$

Each U_j is open and therefore Borel in X_j . Hence

$$B \in \bigotimes_{j=1}^N \text{Bor}(X_j).$$

Closure under countable unions gives

$$U \in \bigotimes_{j=1}^N \text{Bor}(X_j).$$

Thus every open subset of X belongs to the product σ -algebra, and therefore

$$\text{Bor}(X) \subseteq \bigotimes_{j=1}^N \text{Bor}(X_j).$$

Combining both inclusions,

$$\bigotimes_{j=1}^{\mathbb{N}} \text{Bor}(X_j) = \text{Bor}(X).$$

THE COUNTABLE PRODUCT. For

$$X = \prod_{j=1}^{\infty} X_j$$

with the product topology, the coordinate projections are still continuous. Hence, without additional assumptions,

$$\bigotimes_{j=1}^{\infty} \text{Bor}(X_j) \subseteq \text{Bor}(X).$$

If every X_j is separable, choose countable bases $\mathcal{B}_j$. The product topology has the countable basis consisting of the cylinders

$$\bigcap_{j \in F} \pi_j^{-1}(U_j), \quad F \subseteq \mathbb{N} \text{ finite}, \quad U_j \in \mathcal{B}_j.$$

Each such cylinder belongs to the product σ -algebra, and every open set is a countable union of them. Consequently,

$$\bigotimes_{j=1}^{\infty} \text{Bor}(X_j) = \text{Bor}\left(\prod_{j=1}^{\infty} X_j\right).$$

Without separability, the inclusion remains valid, but equality need not hold in general. $\square$

2.4 Semi-Algebras

2.4.1. Definition: Semi-algebra. A semi-algebra of sets on X is a collection $\mathcal{I} \subseteq \mathcal{P}(X)$ such that

- (a) $\emptyset \in \mathcal{I}$;
- (b) $E, F \in \mathcal{I} \Rightarrow E \cap F \in \mathcal{I}$;
- (c) $E \in \mathcal{I} \Rightarrow E^c$ is a finite disjoint union of members of $\mathcal{I}$.

2.4.2. Example. The collections consisting of $\emptyset, \mathbb{R}$, and all intervals $[a, b]$, respectively $(a, b]$, for $-\infty \leq a < b \leq \infty$, are semi-algebras on $\mathbb{R}$.

2.4.3. Example. If $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ are measurable spaces, then $\mathcal{A} \times \mathcal{B} = \{A \times B : A \in \mathcal{A}, B \in \mathcal{B}\}$ is a semi-algebra on $X \times Y$.

2.4.4. Proposition (From semi-algebras to algebras). — If $\mathcal{I}$ is a semi-algebra, then the collection of finite disjoint unions of members of $\mathcal{I}$ is an algebra.

Proof. Let

$$\mathcal{A} = \left\{ \bigcup_{i=1}^n I_i : n \in \mathbb{N}, I_i \in \mathcal{I} \right\} \cup \{\emptyset\}.$$

UNIT AND EMPTY SET. By the definition of a semi-algebra,

$$X \in \mathcal{I}, \quad \emptyset \in \mathcal{I}.$$

Hence

$$X \in \mathcal{A}, \quad \emptyset \in \mathcal{A}.$$

AUXILIARY TECHNIQUE: ITERATED DISJOINTIFICATION. We first show that

$$I \in \mathcal{I}, B \in \mathcal{A} \implies I \setminus B \in \mathcal{A}.$$

Write $B = \bigsqcup_{j=1}^n J_j$, where $J_j \in \mathcal{I}$. We use induction on n . For $n = 1$, the conclusion is part of the definition of a semi-algebra.

Assume the statement for $n - 1$. By the induction hypothesis,

$$I \setminus \bigcup_{j=1}^{n-1} J_j = \bigsqcup_{k=1}^m K_k$$

for some $K_k \in \mathcal{I}$. Therefore

$$I \setminus \bigcup_{j=1}^n J_j = \bigsqcup_{k=1}^m (K_k \setminus J_n).$$

By the definition of a semi-algebra, every $K_k \setminus J_n$ is a finite disjoint union of members of $\mathcal{I}$. Since the K_k 's are pairwise disjoint,

$$I \setminus B \in \mathcal{A}.$$

CLOSURE UNDER COMPLEMENTS. Let

$$A = \bigsqcup_{i=1}^m I_i \in \mathcal{A}.$$

By the auxiliary result,

$$X \setminus I_i \in \mathcal{A}$$

for every i . More directly, applying the same result successively to the disjoint union A ,

$$A^c = X \setminus A \in \mathcal{A}.$$

CLOSURE UNDER FINITE UNIONS. Let $A, B \in \mathcal{A}$. By the auxiliary result,

$$B \setminus A \in \mathcal{A}.$$

By the disjointification identity,

$$A \cup B = A \bigsqcup (B \setminus A).$$

Thus $A \cup B \in \mathcal{A}$.

Therefore $\mathcal{A}$ contains X , is closed under complements, and is closed under finite unions. Hence $\mathcal{A}$ is an algebra. $\square$

2.4.5. Definition: Intervals in $\mathbb{R}$. Put

$$\mathcal{J}_o = \{(a, b) : -\infty \leq a < b \leq \infty\} \cup \{\emptyset\}, \quad \mathcal{J}_{co} = \{[a, b) : -\infty \leq a < b \leq \infty\} \cup \{\emptyset\},$$

$$\mathcal{J}_{oc} = \{(a, b] : -\infty \leq a < b \leq \infty\} \cup \{\emptyset\}, \quad \mathcal{J}_c = \{[a, b] : -\infty \leq a < b \leq \infty\} \cup \{\emptyset\},$$

and $\mathcal{J} = \mathcal{J}_o \cup \mathcal{J}_{oc} \cup \mathcal{J}_{co} \cup \mathcal{J}_c$.

2.4.6. Definition: Rectangles. For $\ell \in \{o, oc, co, c\}$, let

$$\mathcal{J}_\ell^d = \{E_1 \times \cdots \times E_d : E_1, \dots, E_d \in \mathcal{J}_\ell\}.$$

The family of all rectangles in $\mathbb{R}^d$ is

$$\mathcal{J}^d = \mathcal{J}_o^d \cup \mathcal{J}_{oc}^d \cup \mathcal{J}_{co}^d \cup \mathcal{J}_c^d.$$

Thus $\mathcal{J}_o^d, \mathcal{J}_{oc}^d, \mathcal{J}_{co}^d, \mathcal{J}_c^d$ are respectively the families of open, open–closed, closed–open, and closed rectangles in $\mathbb{R}^d$.

2.4.7. PROBLEM. Show that $\mathcal{J}_{oc}^d$ and $\mathcal{J}_{co}^d$ are semi-algebras.

Solution. Recall that a family $\mathcal{S}$ is a semi-algebra if $\emptyset \in \mathcal{S}$, it is closed under finite intersections, and the difference of two members is a finite disjoint union of members of $\mathcal{S}$.

For $\tau \in \{oc, co\}$, write

$$I_{oc}(a, b) = (a, b], \quad I_{co}(a, b) = [a, b),$$

so that

$$\mathcal{J}_\tau^d = \left\{ \prod_{j=1}^d I_\tau(a_j, b_j) : a_j < b_j \right\} \cup \{\emptyset\}.$$

INTERSECTION. Let

$$R = \prod_{j=1}^d I_\tau(a_j, b_j), \quad S = \prod_{j=1}^d I_\tau(c_j, d_j).$$

By the definition of Cartesian products,

$$R \cap S = \prod_{j=1}^d (I_\tau(a_j, b_j) \cap I_\tau(c_j, d_j)).$$

For each j ,

$$I_{oc}(a_j, b_j) \cap I_{oc}(c_j, d_j) = (\max\{a_j, c_j\}, \min\{b_j, d_j\}],$$

and

$$I_{co}(a_j, b_j) \cap I_{co}(c_j, d_j) = [\max\{a_j, c_j\}, \min\{b_j, d_j\}).$$

If the left endpoint is at least the right endpoint, the intersection is empty; otherwise it is an interval of the same type. Hence

$$R \cap S \in \mathcal{J}_\tau^d.$$

DISJOINTIFICATION BY AN ENDPOINT GRID. We prove that $R \setminus S$ is a finite disjoint union of members of $\mathcal{J}_\tau^d$. Fix j , collect the four endpoints

$$\{a_j, b_j, c_j, d_j\},$$

remove repetitions, and arrange the remaining points as

$$t_{j,0} < t_{j,1} < \cdots < t_{j,m_j}.$$

For $\tau = \text{oc}$, consider the cells

$$(t_{j,r-1}, t_{j,r}],$$

and for $\tau = \text{co}$, consider the cells

$$[t_{j,r-1}, t_{j,r}).$$

Retain only those cells contained in $I_\tau(a_j, b_j)$. These cells form a finite pairwise disjoint partition of $I_\tau(a_j, b_j)$.

Because all endpoints of $I_\tau(c_j, d_j)$ occur in the grid, each retained cell is either contained in $I_\tau(c_j, d_j)$ or disjoint from it. Therefore, the products of the retained cells form a finite pairwise disjoint partition of $\mathbb{R}$, and

$$\mathbb{R} \setminus S = \bigcup_{\substack{\text{grid cells } C \subseteq \mathbb{R} \\ C \cap S = \emptyset}} C.$$

Each C is a product of intervals of the same type, so $C \in \mathcal{J}_\tau^d$. Thus $\mathbb{R} \setminus S$ is a finite disjoint union of members of $\mathcal{J}_\tau^d$.

Therefore both

$$\mathcal{J}_{\text{oc}}^d \quad \text{and} \quad \mathcal{J}_{\text{co}}^d$$

are semi-algebras. □

2.5 Preimages and σ -Algebras

2.5.1. PROBLEM. Let $f : X \rightarrow Y$.

- (a) If Σ is a σ -algebra on X , show that $\{F \subseteq Y : f^{-1}[F] \in \Sigma\}$ is a σ -algebra on Y .
- (b) If Σ is a σ -algebra on Y , show that $\{f^{-1}[F] : F \in \Sigma\}$ is a σ -algebra on X .

Solution. (a) Set

$$\mathcal{A} = \{F \subseteq Y : f^{-1}[F] \in \Sigma\}.$$

WHOLE SPACE. By the properties of preimages,

$$f^{-1}[Y] = X.$$

Since $X \in \Sigma$,

$$Y \in \mathcal{A}.$$

COMPLEMENTS. Let $F \in \mathcal{A}$. By the definition of $\mathcal{A}$, $f^{-1}[F] \in \Sigma$. By the properties of preimages,

$$f^{-1}[Y \setminus F] = X \setminus f^{-1}[F].$$

Since Σ is closed under complements,

$$f^{-1}[Y \setminus F] \in \Sigma.$$

Hence

$$Y \setminus F \in \mathcal{A}.$$

COUNTABLE UNIONS. Let $(F_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$. By the definition of $\mathcal{A}$,

$$f^{-1}[F_n] \in \Sigma \quad (n \in \mathbb{N}).$$

By the properties of preimages,

$$f^{-1}\left[\bigcup_{n=1}^{\infty} F_n\right] = \bigcup_{n=1}^{\infty} f^{-1}[F_n].$$

Since Σ is closed under countable unions,

$$f^{-1}\left[\bigcup_{n=1}^{\infty} F_n\right] \in \Sigma.$$

Therefore

$$\bigcup_{n=1}^{\infty} F_n \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ -algebra on Y .

(b) Set

$$\mathcal{B} = \{f^{-1}[F] : F \in \Sigma\}.$$

WHOLE SPACE. Since $Y \in \Sigma$,

$$X = f^{-1}[Y] \in \mathcal{B}.$$

COMPLEMENTS. Let $f^{-1}[F] \in \mathcal{B}$, where $F \in \Sigma$. By the properties of preimages,

$$X \setminus f^{-1}[F] = f^{-1}[Y \setminus F].$$

Since Σ is closed under complements, $Y \setminus F \in \Sigma$. Hence

$$X \setminus f^{-1}[F] \in \mathcal{B}.$$

COUNTABLE UNIONS. Let $(f^{-1}[F_n])_{n \in \mathbb{N}} \subseteq \mathcal{B}$, where $F_n \in \Sigma$. Since Σ is closed under countable unions,

$$\bigcup_{n=1}^{\infty} F_n \in \Sigma.$$

By the properties of preimages,

$$\bigcup_{n=1}^{\infty} f^{-1}[F_n] = f^{-1}\left[\bigcup_{n=1}^{\infty} F_n\right] \in \mathcal{B}.$$

Thus $\mathcal{B}$ is a σ -algebra on X .

□

2.5.2. PROBLEM. Let $\mathcal{E} \subseteq \mathcal{P}(X)$ and $f : Y \rightarrow X$. Show that $\{f^{-1}[E] : E \in \sigma(\mathcal{E})\}$ is the σ -algebra on Y generated by $\{f^{-1}[E] : E \in \mathcal{E}\}$.

Solution. Set

$$\mathcal{A} = \{f^{-1}[E] : E \in \sigma(\mathcal{E})\}, \quad \mathcal{B} = \sigma(\{f^{-1}[E] : E \in \mathcal{E}\}).$$

We prove both inclusions.

($\subseteq$) **PULLBACK σ -ALGEBRA TECHNIQUE.** Define

$$\mathcal{G} = \{E \subseteq X : f^{-1}[E] \in \mathcal{B}\}.$$

Since $\mathcal{B}$ is a σ -algebra on Y , the preceding problem gives that $\mathcal{G}$ is a σ -algebra on X .

For every $E \in \mathcal{E}$, the definition of $\mathcal{B}$ gives

$$f^{-1}[E] \in \mathcal{B}.$$

By the definition of $\mathcal{G}$,

$$E \in \mathcal{G}.$$

Thus

$$\mathcal{E} \subseteq \mathcal{G}.$$

Since $\sigma(\mathcal{E})$ is the smallest σ -algebra containing $\mathcal{E}$, minimality gives

$$\sigma(\mathcal{E}) \subseteq \mathcal{G}.$$

Therefore, for every $E \in \sigma(\mathcal{E})$,

$$f^{-1}[E] \in \mathcal{B}.$$

Hence

$$\mathcal{A} \subseteq \mathcal{B}.$$

($\supseteq$) By the preceding problem,

$$\mathcal{A} = \{f^{-1}[E] : E \in \sigma(\mathcal{E})\}$$

is a σ -algebra on Y . Since

$$\mathcal{E} \subseteq \sigma(\mathcal{E}),$$

we have

$$\{f^{-1}[E] : E \in \mathcal{E}\} \subseteq \mathcal{A}.$$

By the minimality of the generated σ -algebra,

$$\mathcal{B} = \sigma(\{f^{-1}[E] : E \in \mathcal{E}\}) \subseteq \mathcal{A}.$$

Hence

$$\{f^{-1}[E] : E \in \sigma(\mathcal{E})\} = \sigma(\{f^{-1}[E] : E \in \mathcal{E}\}).$$

□

2.5.3. PROBLEM (Borel σ -algebra of a metric space). Show that $\text{Bor}(X)$ is the smallest family $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfying each of the following respective conditions:

- (a) it contains every open subset of X and is closed under countable intersections and countable unions;
- (b) it contains every closed subset of X and is closed under countable intersections and countable unions;
- (c) it contains every open subset of X and is closed under countable intersections and countable disjoint unions.

Solution. METRIC APPROXIMATION. We use the following two consequences of the distance function theorem.

If $F \subseteq X$ is closed and nonempty, then

$$F = \bigcap_{n=1}^{\infty} \left\{ x \in X : d(x, F) < \frac{1}{n} \right\}.$$

Indeed, the sets on the right are open, and $d(x, F) = 0 \iff x \in F$ because F is closed.

If $U \subseteq X$ is open and $U \neq X$, then

$$U = \bigcup_{n=1}^{\infty} \left\{ x \in X : d(x, U^c) \geq \frac{1}{n} \right\}.$$

The sets on the right are closed. If $x \in U$, then, by openness, $B(x, r) \subseteq U$ for some $r > 0$, so $d(x, U^c) \geq r$. The reverse inclusion is immediate. The case $U = X$ is clear.

- (a) Let $\mathcal{C} \subseteq \mathcal{P}(X)$ contain every open subset of X and be closed under countable intersections and countable unions.

By the preceding metric approximation, every closed subset of X belongs to $\mathcal{C}$. Define

$$\mathcal{D} = \{E \subseteq X : E \in \mathcal{C}, E^c \in \mathcal{C}\}.$$

COMPLEMENT-PAIR TECHNIQUE. We show that $\mathcal{D}$ is a σ -algebra. Since every open set and every closed set belong to $\mathcal{C}$,

$$U \text{ open} \implies U, U^c \in \mathcal{C} \implies U \in \mathcal{D}.$$

If $E \in \mathcal{D}$, then $E, E^c \in \mathcal{C}$, so $E^c \in \mathcal{D}$. Finally, if $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{D}$, then

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{C}, \quad \left(\bigcup_{n=1}^{\infty} E_n \right)^c = \bigcap_{n=1}^{\infty} E_n^c \in \mathcal{C}.$$

Hence

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{D}.$$

Thus $\mathcal{D}$ is a σ -algebra containing every open set, so

$$\text{Bor}(X) \subseteq \mathcal{D} \subseteq \mathcal{C}.$$

Since $\text{Bor}(X)$ itself has the required properties, it is the smallest such family.

- (b) Let $\mathcal{C} \subseteq \mathcal{P}(X)$ contain every closed subset of X and be closed under countable intersections and countable unions.

By the preceding metric approximation of open sets, every open subset of X belongs to $\mathcal{C}$. Define

$$\mathcal{D} = \{E \subseteq X : E \in \mathcal{C}, E^c \in \mathcal{C}\}.$$

The complement-pair argument from part 1 shows that $\mathcal{D}$ is a σ -algebra containing every open set. Therefore

$$\text{Bor}(X) \subseteq \mathcal{D} \subseteq \mathcal{C}.$$

Since $\text{Bor}(X)$ contains every closed set and is closed under countable intersections and countable unions, it is the smallest such family.

- (c) Let $\mathcal{C} \subseteq \mathcal{P}(X)$ contain every open subset of X and be closed under countable intersections and countable disjoint unions.

By the closed-set approximation above, every closed subset of X belongs to $\mathcal{C}$. Again define

$$\mathcal{D} = \{E \subseteq X : E \in \mathcal{C}, E^c \in \mathcal{C}\}.$$

As before, $\mathcal{D}$ contains every open set and is closed under complements.

DISJOINTIFICATION. Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{D}$, and define

$$D_1 = E_1, \quad D_n = E_n \cap \bigcap_{k=1}^{n-1} E_k^c \quad (n \geq 2).$$

Since $E_n, E_k^c \in \mathcal{C}$ and $\mathcal{C}$ is closed under countable intersections,

$$D_n \in \mathcal{C} \quad (n \in \mathbb{N}).$$

By construction, the D_n 's are pairwise disjoint, and

$$\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} D_n.$$

Hence, by closure under countable disjoint unions,

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{C}.$$

Also, by De Morgan's law and closure under countable intersections,

$$\left(\bigcup_{n=1}^{\infty} E_n \right)^c = \bigcap_{n=1}^{\infty} E_n^c \in \mathcal{C}.$$

Therefore

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{D}.$$

Thus $\mathcal{D}$ is a σ -algebra containing every open set, so

$$\text{Bor}(X) \subseteq \mathcal{D} \subseteq \mathcal{C}.$$

Since $\text{Bor}(X)$ itself has the stated properties, it is the smallest such family. $\square$

2.6 Monotone Classes

2.6.1. Definition: Monotone class. A collection $\mathcal{M} \subseteq \mathcal{P}(X)$ is a *monotone class* if

- (a) $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$ and $E_1 \supseteq E_2 \supseteq \cdots \Rightarrow \bigcap_{n=1}^{\infty} E_n \in \mathcal{M}$;
- (b) $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$ and $E_1 \subseteq E_2 \subseteq \cdots \Rightarrow \bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$.

2.6.2. PROBLEM. Show that the intersection of all monotone classes containing $\mathcal{E} \subseteq \mathcal{P}(X)$ is the unique smallest monotone class containing $\mathcal{E}$.

Solution. Let

$$\mathfrak{C} = \{\mathcal{M} \subseteq \mathcal{P}(X) : \mathcal{M} \text{ is a monotone class and } \mathcal{E} \subseteq \mathcal{M}\},$$

and define

$$\mathcal{M}_{\mathcal{E}} = \bigcap_{\mathcal{M} \in \mathfrak{C}} \mathcal{M}.$$

INTERSECTION CONSTRUCTION. The family $\mathfrak{C}$ is nonempty: $\mathcal{P}(X)$ is a monotone class containing $\mathcal{E}$. Since every $\mathcal{M} \in \mathfrak{C}$ contains $\mathcal{E}$,

$$\mathcal{E} \subseteq \mathcal{M}_{\mathcal{E}}.$$

We verify that $\mathcal{M}_{\mathcal{E}}$ is a monotone class. Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}_{\mathcal{E}}$ be increasing. By the definition of intersection,

$$E_n \in \mathcal{M} \quad (n \in \mathbb{N}, \mathcal{M} \in \mathfrak{C}).$$

Since every $\mathcal{M} \in \mathfrak{C}$ is a monotone class,

$$E_n \uparrow \implies \bigcup_{n=1}^{\infty} E_n \in \mathcal{M} \quad (\mathcal{M} \in \mathfrak{C}).$$

Therefore

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}_{\mathcal{E}}.$$

Similarly, if $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}_{\mathcal{E}}$ is decreasing, then

$$E_n \downarrow \implies \bigcap_{n=1}^{\infty} E_n \in \mathcal{M} \quad (\mathcal{M} \in \mathfrak{C}),$$

and hence

$$\bigcap_{n=1}^{\infty} E_n \in \mathcal{M}_{\mathcal{E}}.$$

Thus $\mathcal{M}_{\mathcal{E}}$ is a monotone class containing $\mathcal{E}$.

MINIMALITY. Let $\mathcal{N}$ be any monotone class containing $\mathcal{E}$. Then $\mathcal{N} \in \mathfrak{C}$. By the definition of $\mathcal{M}_{\mathcal{E}}$,

$$\mathcal{M}_{\mathcal{E}} = \bigcap_{\mathcal{M} \in \mathfrak{C}} \mathcal{M} \subseteq \mathcal{N}.$$

Hence $\mathcal{M}_{\mathcal{E}}$ is the smallest monotone class containing $\mathcal{E}$.

UNIQUENESS. Let $\mathcal{N}$ be another smallest monotone class containing $\mathcal{E}$. By the minimality of $\mathcal{M}_{\mathcal{E}}$,

$$\mathcal{M}_{\mathcal{E}} \subseteq \mathcal{N}.$$

By the minimality of $\mathcal{N}$,

$$\mathcal{N} \subseteq \mathcal{M}_{\mathcal{E}}.$$

Therefore

$$\mathcal{N} = \mathcal{M}_{\mathcal{E}}.$$

□

2.6.3. PROBLEM (Monotone class theorem). Show that if $\mathcal{A}$ is an algebra on X , then the smallest monotone class containing $\mathcal{A}$ equals $\sigma(\mathcal{A})$.

Solution. Let $\mathcal{M}$ denote the smallest monotone class containing $\mathcal{A}$. We prove both inclusions.

($\subseteq$) Since $\sigma(\mathcal{A})$ is a σ -algebra, it is closed under increasing unions and decreasing intersections. Hence $\sigma(\mathcal{A})$ is a monotone class containing $\mathcal{A}$. By the minimality of $\mathcal{M}$,

$$\mathcal{M} \subseteq \sigma(\mathcal{A}).$$

($\supseteq$) **TWO-STAGE CLOSURE BOOTSTRAP.** We prove that $\mathcal{M}$ is a σ -algebra. Fix $A_0 \in \mathcal{A}$, and define

$$\mathcal{M}_{A_0} = \{B \in \mathcal{M} : A_0 \setminus B \in \mathcal{M}\}.$$

We verify that $\mathcal{M}_{A_0}$ is a monotone class. If $B_n \uparrow B$ with $B_n \in \mathcal{M}_{A_0}$, then, by the monotone-class property of $\mathcal{M}$,

$$B = \bigcup_{n=1}^{\infty} B_n \in \mathcal{M}, \quad A_0 \setminus B = \bigcap_{n=1}^{\infty} (A_0 \setminus B_n) \in \mathcal{M}.$$

Thus $B \in \mathcal{M}_{A_0}$. Similarly, if $B_n \downarrow B$, then

$$B = \bigcap_{n=1}^{\infty} B_n \in \mathcal{M}, \quad A_0 \setminus B = \bigcup_{n=1}^{\infty} (A_0 \setminus B_n) \in \mathcal{M},$$

so $B \in \mathcal{M}_{A_0}$.

Since $\mathcal{A}$ is an algebra,

$$C \in \mathcal{A} \implies A_0 \setminus C \in \mathcal{A} \subseteq \mathcal{M}.$$

Hence

$$\mathcal{A} \subseteq \mathcal{M}_{A_0}.$$

By the minimality of $\mathcal{M}$,

$$\mathcal{M} = \mathcal{M}_{A_0}.$$

Therefore

$$(*) \quad A_0 \in \mathcal{A}, B \in \mathcal{M} \implies A_0 \setminus B \in \mathcal{M}.$$

Now fix $B \in \mathcal{M}$, and define

$$\mathcal{N}_B = \{C \in \mathcal{M} : C \setminus B \in \mathcal{M}\}.$$

The same increasing-union and decreasing-intersection argument shows that $\mathcal{N}_B$ is a monotone class. By (*),

$$A_0 \in \mathcal{A} \implies A_0 \setminus B \in \mathcal{M},$$

so

$$\mathcal{A} \subseteq \mathcal{N}_B.$$

By the minimality of $\mathcal{M}$,

$$\mathcal{M} = \mathcal{N}_B.$$

Hence

$$(**) \quad C, B \in \mathcal{M} \implies C \setminus B \in \mathcal{M}.$$

ALGEBRA AXIOMS. Since $X \in \mathcal{A} \subseteq \mathcal{M}$, $(**)$ gives

$$B \in \mathcal{M} \implies B^c = X \setminus B \in \mathcal{M}.$$

Also, for $B, C \in \mathcal{M}$,

$$B \cap C = B \setminus (B \setminus C) \in \mathcal{M}$$

by $(**)$. Therefore, by complements and intersections,

$$B \cup C = (B^c \cap C^c)^c \in \mathcal{M}.$$

Finally, let $(B_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$. By finite-union closure,

$$C_n = \bigcup_{k=1}^n B_k \in \mathcal{M}.$$

Since $C_n \uparrow \bigcup_{n=1}^{\infty} B_n$, the monotone-class property gives

$$\bigcup_{n=1}^{\infty} B_n \in \mathcal{M}.$$

Thus $\mathcal{M}$ is a σ -algebra containing $\mathcal{A}$. By the minimality of $\sigma(\mathcal{A})$,

$$\sigma(\mathcal{A}) \subseteq \mathcal{M}.$$

Hence

$$\mathcal{M} = \sigma(\mathcal{A}).$$

□

2.6.4. PROBLEM (σ -algebra generated by a finite family). Let $\mathcal{E} = \{E_1, \dots, E_n\}$ be a finite collection of distinct, not necessarily disjoint subsets of X . Let

$$\mathcal{D} = \{F_1 \cap \dots \cap F_n : F_i \in \{E_i, E_i^c\}\},$$

and let $\mathcal{U}$ be the collection of finite unions of members of $\mathcal{D}$, including $\emptyset$ as the empty union. Show that

- (a) $\mathcal{D}$ has at most 2^n distinct members;
- (b) $\sigma(\mathcal{E}) = \mathcal{U}$;
- (c) $\mathcal{U}$ has at most 2^{2^n} distinct members.

Solution. BOOLEAN ATOMS. For $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{0, 1\}^n$, write

$$E_i^{\varepsilon_i} = \begin{cases} E_i, & \varepsilon_i = 1, \\ E_i^c, & \varepsilon_i = 0, \end{cases} \quad D_\varepsilon = \bigcap_{i=1}^n E_i^{\varepsilon_i}.$$

The family $\mathcal{D}$ consists precisely of the sets D_ε .

- (a) There are 2^n possible choices of $\varepsilon \in \{0, 1\}^n$. Hence there are at most 2^n sets of the form D_ε , so

$$|\mathcal{D}| \leq 2^n.$$

- (b) ATOMIC PARTITION. The nonempty members of $\mathcal{D}$ are pairwise disjoint. Indeed, if $\varepsilon \neq \delta$, choose i with $\varepsilon_i \neq \delta_i$. Then one of D_ε and D_δ is contained in E_i , while the other is contained in E_i^c . Therefore

$$D_\varepsilon \cap D_\delta = \emptyset.$$

Also,

$$X = \bigcup_{D \in \mathcal{D}} D.$$

Indeed, for $x \in X$, choose $\varepsilon_i = 1$ when $x \in E_i$ and $\varepsilon_i = 0$ when $x \notin E_i$. Then $x \in D_\varepsilon$.

We show that $\mathcal{U}$ is a σ -algebra. By definition,

$$\emptyset \in \mathcal{U}, \quad X = \bigcup_{D \in \mathcal{D}} D \in \mathcal{U}.$$

COMPLEMENTS BY ATOMS. If

$$A = \bigcup_{D \in \mathcal{S}} D, \quad \mathcal{S} \subseteq \mathcal{D},$$

then, because the members of $\mathcal{D}$ form a partition of X ,

$$A^c = \bigcup_{D \in \mathcal{D} \setminus \mathcal{S}} D \in \mathcal{U}.$$

COUNTABLE UNIONS COLLAPSE TO FINITE UNIONS. Let $(A_k)_{k \in \mathbb{N}} \subseteq \mathcal{U}$. For every k , choose $\mathcal{S}_k \subseteq \mathcal{D}$ such that

$$A_k = \bigcup_{D \in \mathcal{S}_k} D.$$

Since $\mathcal{D}$ is finite,

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{D \in \bigcup_{k=1}^{\infty} \mathcal{S}_k} D \in \mathcal{U}.$$

Thus $\mathcal{U}$ is a σ -algebra.

For every i ,

$$E_i = \bigcup_{\varepsilon_i=1} D_\varepsilon \in \mathcal{U}.$$

Hence

$$\mathcal{E} \subseteq \mathcal{U}.$$

By the minimality of $\sigma(\mathcal{E})$,

$$\sigma(\mathcal{E}) \subseteq \mathcal{U}.$$

We prove the reverse inclusion. Every $D \in \mathcal{D}$ is a finite intersection of sets E_i and E_i^c . Since $\sigma(\mathcal{E})$ is a σ -algebra containing $\mathcal{E}$,

$$\mathcal{D} \subseteq \sigma(\mathcal{E}).$$

By closure under finite unions,

$$\mathcal{U} \subseteq \sigma(\mathcal{E}).$$

Therefore

$$\sigma(\mathcal{E}) = \mathcal{U}.$$

- (c) Every element of $\mathcal{U}$ is determined by a subfamily of $\mathcal{D}$. Since $\mathcal{D}$ has at most 2^n members, it has at most

$$2^{|\mathcal{D}|} \leq 2^{2^n}$$

subfamilies. Hence

$$|\mathcal{U}| \leq 2^{2^n}.$$

□

2.6.5. PROBLEM. Let $\mathcal{M}$ be an infinite σ -algebra. Show that

- (a) $\mathcal{M}$ contains an infinite sequence of nonempty pairwise disjoint sets;
 (b) $|\mathcal{M}| \geq |\mathbb{R}|$.

Solution. (a) **NESTED-SET CONSTRUCTION.** For $A \in \mathcal{M}$, define

$$\mathcal{M}_A = \{E \in \mathcal{M} : E \subseteq A\}.$$

We construct a decreasing sequence

$$X = A_0 \supsetneq A_1 \supsetneq A_2 \supsetneq \cdots$$

such that $\mathcal{M}_{A_n}$ is infinite for every n .

Since

$$\mathcal{M}_{A_0} = \mathcal{M}$$

is infinite, the construction begins. Suppose that A_n has been chosen and that $\mathcal{M}_{A_n}$ is infinite. Then there is

$$C \in \mathcal{M}, \quad \emptyset \subsetneq C \subsetneq A_n;$$

otherwise $\mathcal{M}_{A_n} = \{\emptyset, A_n\}$ would be finite.

At least one of

$$\mathcal{M}_C, \quad \mathcal{M}_{A_n \setminus C}$$

is infinite. Indeed, suppose both were finite. Then every $E \in \mathcal{M}_{A_n}$ is determined by

$$E = (E \cap C) \cup (E \cap (A_n \setminus C)),$$

where

$$E \cap C \in \mathcal{M}_C, \quad E \cap (A_n \setminus C) \in \mathcal{M}_{A_n \setminus C}.$$

Thus $\mathcal{M}_{A_n}$ injects into the finite set

$$\mathcal{M}_C \times \mathcal{M}_{A_n \setminus C},$$

contradicting the hypothesis that $\mathcal{M}_{A_n}$ is infinite.

Choose A_{n+1} to be either C or $A_n \setminus C$ so that $\mathcal{M}_{A_{n+1}}$ is infinite. In either case,

$$\emptyset \subsetneq A_{n+1} \subsetneq A_n.$$

Define

$$D_n = A_{n-1} \setminus A_n \quad (n \in \mathbb{N}).$$

Since $A_n \subsetneq A_{n-1}$,

$$D_n \neq \emptyset.$$

Also $D_n \in \mathcal{M}$ because $\mathcal{M}$ is closed under differences. Finally, if $m < n$, then

$$D_m \subseteq X \setminus A_m, \quad D_n \subseteq A_{n-1} \subseteq A_m,$$

so

$$D_m \cap D_n = \emptyset.$$

Hence $(D_n)_{n \in \mathbb{N}}$ is an infinite sequence of nonempty pairwise disjoint members of $\mathcal{M}$.

(b) CODING SUBSETS OF $\mathbb{N}$. Define

$$\Phi : \mathcal{P}(\mathbb{N}) \longrightarrow \mathcal{M}, \quad \Phi(S) = \bigcup_{n \in S} D_n.$$

Since $\mathcal{M}$ is closed under countable unions,

$$\Phi(S) \in \mathcal{M} \quad (S \subseteq \mathbb{N}).$$

We prove that Φ is injective. Suppose

$$\Phi(S) = \Phi(T).$$

If $S \neq T$, choose $n \in S \Delta T$. Without loss of generality, $n \in S \setminus T$. Since $D_n \neq \emptyset$, choose $x \in D_n$. By pairwise disjointness,

$$x \in \Phi(S), \quad x \notin \Phi(T),$$

contradicting $\Phi(S) = \Phi(T)$. Hence

$$S = T.$$

Therefore

$$|\mathcal{M}| \geq |\mathcal{P}(\mathbb{N})| = |\mathbb{R}|.$$

□

2.6.6. PROBLEM. Let $\mathcal{A}$ be a σ -algebra on X , and let $S \subseteq X$ with $S \notin \mathcal{A}$. Show that

$$\sigma(\mathcal{A} \cup \{S\}) = \{(A \cap S) \cup (B \cap S^c) : A, B \in \mathcal{A}\}.$$

Solution. Set

$$\mathcal{C} = \{(A \cap S) \cup (B \cap S^c) : A, B \in \mathcal{A}\}.$$

SIGMA-ALGEBRA VERIFICATION. Since $X, \emptyset \in \mathcal{A}$,

$$X = (X \cap S) \cup (X \cap S^c) \in \mathcal{C}.$$

Let

$$E = (A \cap S) \cup (B \cap S^c) \in \mathcal{C}.$$

Since $S \cap S^c = \emptyset$, by taking complements,

$$E^c = (A^c \cap S) \cup (B^c \cap S^c).$$

Since $A^c, B^c \in \mathcal{A}$,

$$E^c \in \mathcal{C}.$$

Now let

$$E_n = (A_n \cap S) \cup (B_n \cap S^c) \in \mathcal{C} \quad (n \in \mathbb{N}).$$

By distributivity of unions over intersections,

$$\bigcup_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} A_n \cap S \right) \cup \left(\bigcup_{n=1}^{\infty} B_n \cap S^c \right).$$

Since $\mathcal{A}$ is closed under countable unions,

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}, \quad \bigcup_{n=1}^{\infty} B_n \in \mathcal{A}.$$

Therefore

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{C}.$$

Thus $\mathcal{C}$ is a σ -algebra on X .

We prove both inclusions.

($\subseteq$) For every $A \in \mathcal{A}$,

$$A = (A \cap S) \cup (A \cap S^c) \in \mathcal{C}.$$

Also,

$$S = (X \cap S) \cup (\emptyset \cap S^c) \in \mathcal{C}.$$

Hence

$$\mathcal{A} \cup \{S\} \subseteq \mathcal{C}.$$

Since $\mathcal{C}$ is a σ -algebra, the minimality of $\sigma(\mathcal{A} \cup \{S\})$ gives

$$\sigma(\mathcal{A} \cup \{S\}) \subseteq \mathcal{C}.$$

($\supseteq$) Let

$$E = (A \cap S) \cup (B \cap S^c) \in \mathcal{C}.$$

Since

$$A, B, S \in \sigma(\mathcal{A} \cup \{S\}),$$

closure under complements, intersections, and unions gives

$$A \cap S \in \sigma(\mathcal{A} \cup \{S\}), \quad B \cap S^c \in \sigma(\mathcal{A} \cup \{S\}),$$

and hence

$$E \in \sigma(\mathcal{A} \cup \{S\}).$$

Therefore

$$\mathcal{C} \subseteq \sigma(\mathcal{A} \cup \{S\}).$$

Hence

$$\sigma(\mathcal{A} \cup \{S\}) = \{(A \cap S) \cup (B \cap S^c) : A, B \in \mathcal{A}\}.$$

□

2.6.7. PROBLEM. Let $\mathcal{E} \subseteq \mathcal{P}(X)$. Show that

$$\sigma(\mathcal{E}) = \bigcup_{\substack{\mathcal{F} \subseteq \mathcal{E} \\ \mathcal{F} \text{ countable}}} \sigma(\mathcal{F}).$$

Solution. Set

$$\mathcal{C} = \bigcup_{\substack{\mathcal{F} \subseteq \mathcal{E} \\ \mathcal{F} \text{ countable}}} \sigma(\mathcal{F}).$$

COUNTABLE-SUPPORT TECHNIQUE. We first show that $\mathcal{C}$ is a σ -algebra. Since $\emptyset \subseteq \mathcal{E}$ is countable,

$$X \in \sigma(\emptyset) \subseteq \mathcal{C}.$$

Let $A \in \mathcal{C}$. By the definition of $\mathcal{C}$, there is a countable $\mathcal{F} \subseteq \mathcal{E}$ such that

$$A \in \sigma(\mathcal{F}).$$

Since $\sigma(\mathcal{F})$ is a σ -algebra,

$$A^c \in \sigma(\mathcal{F}) \subseteq \mathcal{C}.$$

Now let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{C}$. For every n , choose a countable $\mathcal{F}_n \subseteq \mathcal{E}$ such that

$$A_n \in \sigma(\mathcal{F}_n).$$

Define

$$\mathcal{F} = \bigcup_{n=1}^{\infty} \mathcal{F}_n.$$

Since $\mathcal{F}$ is a countable union of countable sets,

$$\mathcal{F} \subseteq \mathcal{E} \quad \text{and} \quad \mathcal{F} \text{ is countable.}$$

For every n ,

$$\mathcal{F}_n \subseteq \mathcal{F} \implies \sigma(\mathcal{F}_n) \subseteq \sigma(\mathcal{F})$$

by monotonicity of generated σ -algebras. Hence

$$A_n \in \sigma(\mathcal{F}) \quad (n \in \mathbb{N}).$$

Since $\sigma(\mathcal{F})$ is closed under countable unions,

$$\bigcup_{n=1}^{\infty} A_n \in \sigma(\mathcal{F}) \subseteq \mathcal{C}.$$

Thus $\mathcal{C}$ is a σ -algebra.

For every $E \in \mathcal{E}$, the family $\{E\}$ is a countable subfamily of $\mathcal{E}$, and

$$E \in \sigma(\{E\}) \subseteq \mathcal{C}.$$

Therefore

$$\mathcal{E} \subseteq \mathcal{C}.$$

We prove both inclusions.

($\subseteq$) Since $\mathcal{C}$ is a σ -algebra containing $\mathcal{E}$, the minimality of $\sigma(\mathcal{E})$ gives

$$\sigma(\mathcal{E}) \subseteq \mathcal{C}.$$

($\supseteq$) Let

$$A \in \mathcal{C}.$$

By the definition of $\mathcal{C}$, there is a countable $\mathcal{F} \subseteq \mathcal{E}$ such that

$$A \in \sigma(\mathcal{F}).$$

Since $\mathcal{F} \subseteq \mathcal{E}$, monotonicity gives

$$\sigma(\mathcal{F}) \subseteq \sigma(\mathcal{E}).$$

Hence

$$A \in \sigma(\mathcal{E}).$$

Therefore

$$\mathcal{C} \subseteq \sigma(\mathcal{E}).$$

Hence

$$\sigma(\mathcal{E}) = \bigcup_{\substack{\mathcal{F} \subseteq \mathcal{E} \\ \mathcal{F} \text{ countable}}} \sigma(\mathcal{F}).$$

□

2.7 π -Systems and λ -Systems

2.7.1. Definition: π -system. A nonempty collection $\Pi \subseteq \mathcal{P}(X)$ is a π -system if $A, B \in \Pi \Rightarrow A \cap B \in \Pi$.

2.7.2. Definition: λ -system. A collection $\Lambda \subseteq \mathcal{P}(X)$ is a λ -system if

- (a) $X \in \Lambda$;
- (b) $A, B \in \Lambda$ and $A \subseteq B \Rightarrow B \setminus A \in \Lambda$;
- (c) $(A_n)_{n \in \mathbb{N}} \subseteq \Lambda$ and $A_1 \subseteq A_2 \subseteq \dots \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \Lambda$.

2.7.3. PROBLEM. Show that if $\mathcal{D}$ is a λ -system containing a π -system $\mathcal{A}$, then $\sigma(\mathcal{A}) \subseteq \mathcal{D}$.

Solution. MINIMAL λ -SYSTEM AND CLOSURE BOOTSTRAP. Let $\mathcal{L}$ be the smallest λ -system containing $\mathcal{A}$. Equivalently,

$$\mathcal{L} = \bigcap \{ \mathcal{H} : \mathcal{H} \text{ is a } \lambda\text{-system and } \mathcal{A} \subseteq \mathcal{H} \}.$$

By the intersection construction for λ -systems, $\mathcal{L}$ is a λ -system containing $\mathcal{A}$. Since $\mathcal{D}$ is a λ -system containing $\mathcal{A}$, minimality gives

$$(*) \quad \mathcal{L} \subseteq \mathcal{D}.$$

It remains to prove that $\mathcal{L}$ is a σ -algebra.

Fix $A_0 \in \mathcal{A}$, and define

$$\mathcal{L}_{A_0} = \{ B \in \mathcal{L} : A_0 \cap B \in \mathcal{L} \}.$$

FIRST BOOTSTRAP: INTERSECT WITH GENERATORS. We verify that $\mathcal{L}_{A_0}$ is a λ -system. Since $A_0 \in \mathcal{A} \subseteq \mathcal{L}$,

$$A_0 \cap X = A_0 \in \mathcal{L},$$

so $X \in \mathcal{L}_{A_0}$. If $B \in \mathcal{L}_{A_0}$, then $A_0 \cap B \subseteq A_0$, and the difference property of a λ -system gives

$$A_0 \cap B^c = A_0 \setminus (A_0 \cap B) \in \mathcal{L}.$$

Hence $B^c \in \mathcal{L}_{A_0}$.

If $(B_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}_{A_0}$ are pairwise disjoint, then $(A_0 \cap B_n)_{n \in \mathbb{N}}$ are pairwise disjoint. Therefore, by closure under countable disjoint unions,

$$A_0 \cap \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A_0 \cap B_n) \in \mathcal{L}.$$

Thus $\bigcup_n B_n \in \mathcal{L}_{A_0}$.

Finally, if $C \in \mathcal{A}$, then, since $\mathcal{A}$ is a π -system,

$$A_0 \cap C \in \mathcal{A} \subseteq \mathcal{L}.$$

Hence

$$\mathcal{A} \subseteq \mathcal{L}_{A_0}.$$

By the minimality of $\mathcal{L}$,

$$\mathcal{L} = \mathcal{L}_{A_0}.$$

Therefore

$$(**) \quad A_0 \in \mathcal{A}, B \in \mathcal{L} \implies A_0 \cap B \in \mathcal{L}.$$

SECOND BOOTSTRAP: INTERSECT ARBITRARY MEMBERS. Fix $B \in \mathcal{L}$, and define

$$\mathcal{L}^B = \{C \in \mathcal{L} : B \cap C \in \mathcal{L}\}.$$

The preceding complement and disjoint-union argument shows that $\mathcal{L}^B$ is a λ -system. By $(**)$,

$$A_0 \in \mathcal{A} \implies B \cap A_0 \in \mathcal{L},$$

so

$$\mathcal{A} \subseteq \mathcal{L}^B.$$

By the minimality of $\mathcal{L}$,

$$\mathcal{L} = \mathcal{L}^B.$$

Hence

$$(\dagger) \quad B, C \in \mathcal{L} \implies B \cap C \in \mathcal{L}.$$

FROM A λ -SYSTEM TO A σ -ALGEBRA. Since $\mathcal{L}$ is a λ -system, it contains X and is closed under complements. By $(\dagger)$,

$$B, C \in \mathcal{L} \implies B \cup C = (B^c \cap C^c)^c \in \mathcal{L}.$$

Thus $\mathcal{L}$ is closed under finite unions.

Let $(B_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}$. By disjointification, define

$$D_1 = B_1, \quad D_n = B_n \setminus \bigcup_{k=1}^{n-1} B_k \quad (n \geq 2).$$

By finite-union closure and the difference property of a λ -system,

$$D_n \in \mathcal{L} \quad (n \in \mathbb{N}).$$

The D_n 's are pairwise disjoint and

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} D_n.$$

By closure under countable disjoint unions,

$$\bigcup_{n=1}^{\infty} B_n \in \mathcal{L}.$$

Thus $\mathcal{L}$ is a σ -algebra containing $\mathcal{A}$. By the minimality of $\sigma(\mathcal{A})$,

$$\sigma(\mathcal{A}) \subseteq \mathcal{L}.$$

Combining this with (*),

$$\sigma(\mathcal{A}) \subseteq \mathcal{D}.$$

□

2.8 The Borel Hierarchy on $\mathbb{R}$ (optional)

We defined $\text{Bor}(\mathbb{R})$ from above, as an intersection, which does not display its members. From below, one begins with the open sets, repeatedly closes under complements and countable unions, and obtains the Borel hierarchy. Neither finitely many nor ω many stages suffice. The appropriate length is ω_1 , the least uncountable ordinal; it satisfies

$$\sup S < \omega_1 \quad \text{whenever } S \subseteq \omega_1 \text{ is countable.}$$

2.8.1. Definition: Borel hierarchy. Let Σ_1^0 be the family of open subsets of $\mathbb{R}$. For $1 \leq \alpha < \omega_1$, define

$$\Pi_\alpha^0 = \{A^c : A \in \Sigma_\alpha^0\}, \quad \Sigma_\alpha^0 = \left\{ \bigcup_{n \in \mathbb{N}} A_n : (A_n)_{n \in \mathbb{N}} \subseteq \bigcup_{\beta < \alpha} \Pi_\beta^0 \right\} \quad (\alpha > 1).$$

Here ω_1 is the least uncountable ordinal. Thus $\Sigma_2^0, \Pi_2^0, \Sigma_3^0, \Pi_3^0$ are respectively the $F_\sigma, G_\delta, G_{\delta\sigma}, F_{\sigma\delta}$ sets.

2.8.2. PROBLEM. Show that

- (a) $\Sigma_\beta^0 \cup \Pi_\beta^0 \subseteq \Sigma_\alpha^0 \cap \Pi_\alpha^0$ whenever $1 \leq \beta < \alpha < \omega_1$;
- (b) $\text{Bor}(\mathbb{R}) = \bigcup_{\alpha < \omega_1} \Sigma_\alpha^0 = \bigcup_{\alpha < \omega_1} \Pi_\alpha^0$;
- (c) $|\text{Bor}(\mathbb{R})| = |\mathbb{R}|$, hence $\text{Bor}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R})$.

Hint. For (a), every open subset of $\mathbb{R}$ is an F_σ set. For (b), the inclusion $\supseteq$ uses transfinite induction; for $\subseteq$, check that the right-hand side is a σ -algebra. Only countable unions require an argument, and this is where the displayed property of ω_1 enters. For (c), use transfinite induction.

Solution. We use the recursive definition of the Borel hierarchy:

$$\Sigma_1^0 = \{U \subseteq \mathbb{R} : U \text{ is open}\}, \quad \Pi_1^0 = \{F \subseteq \mathbb{R} : F \text{ is closed}\},$$

and, for $1 < \alpha < \omega_1$,

$$\Sigma_\alpha^0 = \left\{ \bigcup_{n=1}^{\infty} E_n : E_n \in \bigcup_{\beta < \alpha} \Pi_\beta^0 \right\}, \quad \Pi_\alpha^0 = \{E^c : E \in \Sigma_\alpha^0\}.$$

(a) TRANSFINITE INDUCTION. We prove, by transfinite induction on α , that

$$(*) \quad 1 \leq \beta < \alpha \implies \Sigma_\beta^0 \cup \Pi_\beta^0 \subseteq \Sigma_\alpha^0 \cap \Pi_\alpha^0.$$

For $\alpha = 2$, every closed set belongs to Σ_2^0 as a one-term union, and every open set belongs to Π_2^0 as a one-term intersection. By the hint, every open subset of $\mathbb{R}$ is an F_σ set, so

$$\Sigma_1^0 \subseteq \Sigma_2^0.$$

Taking complements gives

$$\Pi_1^0 \subseteq \Pi_2^0.$$

Hence $(*)$ holds for $\alpha = 2$.

Suppose first that $\alpha = \delta + 1$. If $\beta < \delta$, the induction hypothesis gives

$$\Sigma_\beta^0 \cup \Pi_\beta^0 \subseteq \Sigma_\delta^0 \cap \Pi_\delta^0.$$

Since a member of Π_δ^0 is a one-term union and a member of Σ_δ^0 is a one-term intersection,

$$\Sigma_\delta^0 \cap \Pi_\delta^0 \subseteq \Sigma_{\delta+1}^0 \cap \Pi_{\delta+1}^0.$$

If $\beta = \delta$, then

$$\Pi_\delta^0 \subseteq \Sigma_{\delta+1}^0, \quad \Sigma_\delta^0 \subseteq \Pi_{\delta+1}^0$$

by one-term unions and intersections. Also, for $\delta \geq 2$, the recursive definition gives

$$\Sigma_\delta^0 \subseteq \Sigma_{\delta+1}^0;$$

taking complements gives

$$\Pi_\delta^0 \subseteq \Pi_{\delta+1}^0.$$

Thus $(*)$ holds at successors.

Now let α be a limit ordinal and let $\beta < \alpha$. Choose γ such that

$$\beta < \gamma < \alpha.$$

By the induction hypothesis,

$$\Sigma_\beta^0 \cup \Pi_\beta^0 \subseteq \Sigma_\gamma^0 \cap \Pi_\gamma^0.$$

A member of Π_γ^0 is a one-term union in Σ_α^0 , and a member of Σ_γ^0 is a one-term intersection in Π_α^0 . Therefore

$$\Sigma_\beta^0 \cup \Pi_\beta^0 \subseteq \Sigma_\alpha^0 \cap \Pi_\alpha^0.$$

This completes the transfinite induction.

(b) Set

$$\mathcal{H} = \bigcup_{\alpha < \omega_1} \Sigma_\alpha^0.$$

By part 1,

$$\Sigma_\alpha^0 \subseteq \Pi_{\alpha+1}^0, \quad \Pi_\alpha^0 \subseteq \Sigma_{\alpha+1}^0.$$

Since $\alpha + 1 < \omega_1$,

$$(**) \quad \mathcal{H} = \bigcup_{\alpha < \omega_1} \Pi_\alpha^0.$$

TRANSFINITE INDUCTION: EVERY LEVEL IS BOREL. By transfinite induction on α , every member of $\Sigma_\alpha^0 \cup \Pi_\alpha^0$ is Borel. At level 1, this is the definition of $\text{Bor}(\mathbb{R})$. At higher levels, the conclusion follows from closure of $\text{Bor}(\mathbb{R})$ under complements and countable unions. Hence

$$\mathcal{H} \subseteq \text{Bor}(\mathbb{R}).$$

COUNTABLE-UNION ARGUMENT. We show that $\mathcal{H}$ is a σ -algebra. Since all open sets belong to Σ_1^0 ,

$$\Sigma_1^0 \subseteq \mathcal{H}.$$

If $E \in \Sigma_\alpha^0$, then

$$E^c \in \Pi_\alpha^0 \subseteq \Sigma_{\alpha+1}^0 \subseteq \mathcal{H}.$$

Thus $\mathcal{H}$ is closed under complements.

Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{H}$. Choose $\alpha_n < \omega_1$ such that

$$E_n \in \Sigma_{\alpha_n}^0.$$

Since the supremum of countably many countable ordinals is countable,

$$\gamma = \sup_{n \in \mathbb{N}} (\alpha_n + 1) < \omega_1.$$

Then $\alpha_n < \gamma$, so part 1 gives

$$E_n \in \Pi_\gamma^0 \quad (n \in \mathbb{N}).$$

By the definition of $\Sigma_{\gamma+1}^0$,

$$\bigcup_{n=1}^{\infty} E_n \in \Sigma_{\gamma+1}^0 \subseteq \mathcal{H}.$$

Thus $\mathcal{H}$ is a σ -algebra containing every open set. By the minimality of the Borel σ -algebra,

$$\text{Bor}(\mathbb{R}) \subseteq \mathcal{H}.$$

Combining this with the preceding inclusion and $(**)$,

$$\text{Bor}(\mathbb{R}) = \bigcup_{\alpha < \omega_1} \Sigma_\alpha^0 = \bigcup_{\alpha < \omega_1} \Pi_\alpha^0.$$

(c) Write

$$\mathfrak{c} = |\mathbb{R}|.$$

CARDINALITY INDUCTION. We prove by transfinite induction that

$$|\Sigma_\alpha^0| \leq \mathfrak{c}, \quad |\Pi_\alpha^0| \leq \mathfrak{c} \quad (\alpha < \omega_1).$$

At level 1, every open subset of $\mathbb{R}$ is a union of intervals with rational end-points. Since there are countably many such intervals,

$$|\Sigma_1^0| \leq |\mathcal{P}(\mathbb{N})| = \mathfrak{c}.$$

Taking complements gives

$$|\Pi_1^0| \leq \mathfrak{c}.$$

Assume the conclusion for every $\beta < \alpha$. Since $\alpha < \omega_1$ is countable,

$$\left| \bigcup_{\beta < \alpha} \Pi_\beta^0 \right| \leq \mathfrak{c}.$$

Every member of Σ_α^0 is determined by a sequence from this family. Therefore

$$|\Sigma_\alpha^0| \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c}.$$

Taking complements gives

$$|\Pi_\alpha^0| \leq \mathfrak{c}.$$

By part 2,

$$|\text{Bor}(\mathbb{R})| \leq |\omega_1| \cdot \mathfrak{c} = \mathfrak{c},$$

because $|\omega_1| \leq \mathfrak{c}$. On the other hand,

$$\mathbb{R} \longrightarrow \text{Bor}(\mathbb{R}), \quad x \longmapsto \{x\},$$

is injective. Hence

$$|\text{Bor}(\mathbb{R})| \geq \mathfrak{c}.$$

Therefore

$$|\text{Bor}(\mathbb{R})| = |\mathbb{R}|.$$

Finally, by Cantor's Theorem,

$$|\mathcal{P}(\mathbb{R})| > |\mathbb{R}| = |\text{Bor}(\mathbb{R})|.$$

Hence

$$\text{Bor}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R}).$$

□

2.9 The Hierarchy Does Not Close Up Below ω_1 (optional)

One must ask whether the length ω_1 is genuinely needed. Equivalently, does $\Sigma_\gamma^0 = \Sigma_{\gamma+1}^0$ hold for some $\gamma < \omega_1$? Counting does not decide this: every level has cardinality $\mathfrak{c}$, as does $\text{Bor}(\mathbb{R})$. The levels are separated by a diagonal argument, which requires a single set parametrizing an entire level.

The hierarchy makes sense in every metric space Z ; write $\Sigma_\alpha^0(Z)$ for its levels. Let $\mathcal{C} = \{0, 1\}^{\mathbb{N}}$ carry the product topology.

2.9.1. Definition: Universal set. Let $\mathcal{C} = \{0, 1\}^{\mathbb{N}}$ carry the product topology, and let Z be a metric space. A set $U \subseteq \mathcal{C} \times Z$ is *universal* for $\Sigma_\alpha^0(Z)$ if $U \in \Sigma_\alpha^0(\mathcal{C} \times Z)$ and

$$\Sigma_\alpha^0(Z) = \{U_x : x \in \mathcal{C}\}, \quad U_x = \{z \in Z : (x, z) \in U\}.$$

2.9.2. PROBLEM. For $Z \in \{\mathbb{R}, \mathcal{C}\}$ and $1 \leq \alpha < \omega_1$,

- (a) construct a set $U^\alpha \subseteq \mathcal{C} \times Z$ universal for $\Sigma_\alpha^0(Z)$;
- (b) prove $\Sigma_\alpha^0 \neq \Pi_\alpha^0$, and deduce $\Sigma_\alpha^0 \subsetneq \Sigma_{\alpha+1}^0$ and $\Pi_\alpha^0 \subsetneq \Pi_{\alpha+1}^0$;
- (c) conclude that $\bigcup_{\alpha < \gamma} \Sigma_\alpha^0 \subsetneq \text{Bor}(\mathbb{R})$ for every $\gamma < \omega_1$.

Hint. In (a), fix a countable base $(B_n)_{n \in \mathbb{N}}$ of Z and put

$$U^1 = \{(x, z) : z \in B_n \text{ for some } n \text{ with } x(n) = 1\}.$$

For $\alpha > 1$, choose $\alpha_n < \alpha$ with $\sup_n (\alpha_n + 1) = \alpha$, fix a homeomorphism $x \mapsto (x_n)_{n \in \mathbb{N}}$ from $\mathcal{C}$ onto $\mathcal{C}^{\mathbb{N}}$, and put

$$U^\alpha = \{(x, z) : (x_n, z) \notin U^{\alpha_n} \text{ for some } n\}.$$

In (b), take $Z = \mathcal{C}$ and $D = \{x : (x, x) \in U^\alpha\}$; if D^c were Σ_α^0 , then $D^c = U_{x_0}^\alpha$ for some x_0 , forcing $x_0 \in D$ if and only if $x_0 \in D^c$. Transfer the conclusion to $\mathbb{R}$ via the Cantor set. In (c), use part (a) of the preceding problem.

Set Functions and Measures

3.1 Additivity and Subadditivity

3.1.1. Definition: Additive set functions. Let $\mathcal{U} \subseteq \mathcal{P}(X)$ contain $\emptyset$, and let $\nu : \mathcal{U} \rightarrow [0, \infty]$ satisfy $\nu(\emptyset) = 0$.

- (a) ν is *finitely additive* on $\mathcal{U}$ if, whenever $A_1, \dots, A_n \in \mathcal{U}$ are pairwise disjoint and $\bigcup_{j=1}^n A_j \in \mathcal{U}$,

$$\nu\left(\bigcup_{j=1}^n A_j\right) = \sum_{j=1}^n \nu(A_j).$$

- (b) ν is *countably additive* on $\mathcal{U}$ if, whenever $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{U}$ is pairwise disjoint and $\bigcup_{n=1}^{\infty} A_n \in \mathcal{U}$,

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \nu(A_n).$$

3.1.2. Definition: Subadditive set functions. Let $\mathcal{U} \subseteq \mathcal{P}(X)$ contain $\emptyset$, and let $\nu : \mathcal{U} \rightarrow [0, \infty]$ satisfy $\nu(\emptyset) = 0$.

- (a) ν is *finitely subadditive* on $\mathcal{U}$ if

$$\nu\left(\bigcup_{j=1}^n A_j\right) \leq \sum_{j=1}^n \nu(A_j)$$

whenever $A_1, \dots, A_n \in \mathcal{U}$ and $\bigcup_{j=1}^n A_j \in \mathcal{U}$.

- (b) ν is *countably subadditive* on $\mathcal{U}$ if

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \nu(A_n)$$

whenever $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{U}$ and $\bigcup_{n=1}^{\infty} A_n \in \mathcal{U}$.

3.1.3. Remark. Countable additivity implies finite additivity, and countable subadditivity implies finite subadditivity.

3.1.4. PROBLEM (Continuity at zero). Let μ be finitely additive on an algebra $\mathcal{A}$, with $\mu(A) < \infty$ for every $A \in \mathcal{A}$. Prove that the following are equivalent:

- (a) μ is countably additive;
 (b) whenever $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$ is decreasing and $\bigcap_{n \in \mathbb{N}} A_n = \emptyset$,

$$\lim_{n \rightarrow \infty} \mu(A_n) = 0.$$

Solution. We prove both implications.

($\implies$) Suppose that μ is countably additive. Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$ satisfy

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_{n=1}^{\infty} A_n = \emptyset.$$

TAIL DISJOINTIFICATION. Define

$$D_n = A_n \setminus A_{n+1} \quad (n \in \mathbb{N}).$$

Since $\mathcal{A}$ is an algebra,

$$D_n \in \mathcal{A}.$$

The sets D_n are pairwise disjoint. Moreover, for every n ,

$$A_n = \bigsqcup_{k=n}^{\infty} D_k.$$

We prove both inclusions.

($\subseteq$) Let $x \in A_n$. Since $\bigcap_{k=1}^{\infty} A_k = \emptyset$, there is $r \geq n$ such that $x \notin A_{r+1}$. Choose the smallest such r . Then

$$x \in A_r \setminus A_{r+1} = D_r,$$

so

$$x \in \bigsqcup_{k=n}^{\infty} D_k.$$

($\supseteq$) If $k \geq n$, then

$$D_k \subseteq A_k \subseteq A_n.$$

Hence

$$\bigsqcup_{k=n}^{\infty} D_k \subseteq A_n.$$

By countable additivity,

$$\mu(A_n) = \sum_{k=n}^{\infty} \mu(D_k) = \mu(A_1) - \sum_{k=1}^{n-1} \mu(D_k).$$

Since

$$\mu(A_1) < \infty,$$

the partial sums on the right converge to $\mu(A_1)$. Therefore

$$\lim_{n \rightarrow \infty} \mu(A_n) = 0.$$

($\impliedby$) Suppose that every decreasing sequence in $\mathcal{A}$ with empty intersection has measure converging to 0.

Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint, and suppose that

$$E = \bigcup_{n=1}^{\infty} E_n \in \mathcal{A}.$$

We must prove that

$$\mu(E) = \sum_{n=1}^{\infty} \mu(E_n).$$

Define

$$S_n = \bigcup_{k=1}^n E_k, \quad R_n = E \setminus S_n.$$

Since $\mathcal{A}$ is an algebra,

$$S_n, R_n \in \mathcal{A}.$$

Also,

$$R_1 \supseteq R_2 \supseteq \cdots.$$

If $x \in E$, then, by the definition of E , there is $m \in \mathbb{N}$ such that $x \in E_m$. Hence

$$x \in S_n \quad (n \geq m),$$

so $x \notin R_n$ for $n \geq m$. Therefore

$$\bigcap_{n=1}^{\infty} R_n = \emptyset.$$

By the assumed continuity at zero,

$$\lim_{n \rightarrow \infty} \mu(R_n) = 0.$$

Since

$$E = S_n \bigsqcup R_n,$$

finite additivity gives

$$\mu(E) = \mu(S_n) + \mu(R_n) = \sum_{k=1}^n \mu(E_k) + \mu(R_n).$$

Letting $n \rightarrow \infty$ and using $\mu(R_n) \rightarrow 0$, we obtain

$$\mu(E) = \sum_{k=1}^{\infty} \mu(E_k).$$

Thus μ is countably additive. $\square$

3.1.5. PROBLEM (Extension from a semi-algebra). Let $\rho : \mathcal{E} \rightarrow [0, \infty]$ be a finitely additive set function on a semi-algebra $\mathcal{E}$ of subsets of X , with $\rho(\emptyset) = 0$. Let $\mathcal{A}$ be the algebra of finite disjoint unions of members of $\mathcal{E}$. For $A = \bigcup_{i=1}^n E_i$, where the $E_i \in \mathcal{E}$ are pairwise disjoint, define

$$\mu_0(A) = \sum_{i=1}^n \rho(E_i).$$

- (a) Prove that μ_0 is well defined and additive on $\mathcal{A}$, with $\mu_0(\emptyset) = 0$ and $\mu_0 = \rho$ on $\mathcal{E}$.
- (b) If ρ is countably subadditive on $\mathcal{E}$, prove that μ_0 is countably additive on $\mathcal{A}$.

Solution. (a) **COMMON-REFINEMENT TECHNIQUE.** Suppose that $A \in \mathcal{A}$ has two disjoint representations

$$A = \bigsqcup_{i=1}^m E_i = \bigsqcup_{j=1}^n F_j, \quad E_i, F_j \in \mathcal{E}.$$

Since $\mathcal{E}$ is a semi-algebra,

$$E_i \cap F_j \in \mathcal{E} \quad (1 \leq i \leq m, 1 \leq j \leq n).$$

For each i , the sets $E_i \cap F_j$ are pairwise disjoint and

$$E_i = \bigsqcup_{j=1}^n (E_i \cap F_j).$$

Hence, by finite additivity of ρ ,

$$\rho(E_i) = \sum_{j=1}^n \rho(E_i \cap F_j).$$

Similarly,

$$\rho(F_j) = \sum_{i=1}^m \rho(E_i \cap F_j).$$

Therefore

$$\begin{aligned} \sum_{i=1}^m \rho(E_i) &= \sum_{i=1}^m \sum_{j=1}^n \rho(E_i \cap F_j) \\ &= \sum_{j=1}^n \sum_{i=1}^m \rho(E_i \cap F_j) \\ &= \sum_{j=1}^n \rho(F_j). \end{aligned}$$

Thus μ_0 is well defined.

FINITE ADDITIVITY. Let

$$A = \bigsqcup_{i=1}^m E_i, \quad B = \bigsqcup_{j=1}^n F_j$$

belong to $\mathcal{A}$, and suppose that $A \cap B = \emptyset$. Then

$$A \cup B = \left(\bigsqcup_{i=1}^m E_i \right) \sqcup \left(\bigsqcup_{j=1}^n F_j \right)$$

is a disjoint representation by members of $\mathcal{E}$. Therefore, by the definition of μ_0 ,

$$\begin{aligned} \mu_0(A \cup B) &= \sum_{i=1}^m \rho(E_i) + \sum_{j=1}^n \rho(F_j) \\ &= \mu_0(A) + \mu_0(B). \end{aligned}$$

Also,

$$\mu_0(\emptyset) = 0$$

by the empty sum, and, for $E \in \mathcal{E}$,

$$\mu_0(E) = \rho(E)$$

by the one-term representation $E = E$.

- (b) Suppose that ρ is countably subadditive on $\mathcal{E}$. Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint, and suppose that

$$A = \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

We must prove that

$$\mu_0(A) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

Write

$$A = \bigsqcup_{i=1}^m E_i, \quad A_n = \bigsqcup_{j=1}^{m_n} E_{n,j},$$

where every displayed set belongs to $\mathcal{E}$.

COUNTABLE COMMON REFINEMENT. Since $A = \bigcup_n A_n$, for every i ,

$$E_i = \bigcup_{n=1}^{\infty} \bigcup_{j=1}^{m_n} (E_i \cap E_{n,j}).$$

All sets $E_i \cap E_{n,j}$ belong to $\mathcal{E}$. By countable subadditivity of ρ ,

$$\rho(E_i) \leq \sum_{n=1}^{\infty} \sum_{j=1}^{m_n} \rho(E_i \cap E_{n,j}).$$

Summing over i gives

$$\begin{aligned} \mu_0(A) &= \sum_{i=1}^m \rho(E_i) \\ &\leq \sum_{n=1}^{\infty} \sum_{j=1}^{m_n} \sum_{i=1}^m \rho(E_i \cap E_{n,j}). \end{aligned}$$

Since $E_{n,j} \subseteq A = \bigsqcup_{i=1}^m E_i$,

$$E_{n,j} = \bigsqcup_{i=1}^m (E_i \cap E_{n,j}).$$

By finite additivity of ρ ,

$$\sum_{i=1}^m \rho(E_i \cap E_{n,j}) = \rho(E_{n,j}).$$

Therefore

$$\begin{aligned} (*) \quad \mu_0(A) &\leq \sum_{n=1}^{\infty} \sum_{j=1}^{m_n} \rho(E_{n,j}) \\ &= \sum_{n=1}^{\infty} \mu_0(A_n). \end{aligned}$$

On the other hand, for every $N \in \mathbb{N}$,

$$A = \left(\bigsqcup_{n=1}^N A_n \right) \sqcup \left(A \setminus \bigcup_{n=1}^N A_n \right).$$

By finite additivity and nonnegativity of μ_0 ,

$$\sum_{n=1}^N \mu_0(A_n) \leq \mu_0(A).$$

Taking the supremum over N gives

$$(**) \quad \sum_{n=1}^{\infty} \mu_0(A_n) \leq \mu_0(A).$$

Combining (*) and (**),

$$\mu_0(A) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

Thus μ_0 is countably additive on $\mathcal{A}$.

□

3.2 Volume on $\mathbb{R}^d$

3.2.1. Definition: Volume function. Let $\mathcal{J}^d$ be the family of rectangles in $\mathbb{R}^d$. Define $v_d : \mathcal{J}^d \rightarrow [0, \infty]$ by

$$v_d(I) = \begin{cases} 0, & I = \emptyset, \\ \prod_{j=1}^d (b_j - a_j), & \bar{I} = \prod_{j=1}^d [a_j, b_j], \\ \infty, & \text{otherwise.} \end{cases}$$

For $d = 1$, v_1 is the length function on $\mathbb{R}$.

3.2.2. Definition: Almost disjoint rectangles. A finite family of rectangles is *almost disjoint* if their interiors are pairwise disjoint.

3.2.3. Lemma (Finite additivity of volume). — Let $R_1, \dots, R_n \in \mathcal{J}^d$ be almost disjoint and suppose $R = \bigcup_{j=1}^n R_j \in \mathcal{J}^d$. Then

$$v_d(R) = \sum_{j=1}^n v_d(R_j).$$

Proof. PICTURE. Put all coordinate endpoints of $R_1, \dots, R_n$ into one grid. Each R_j is then a union of grid boxes, and almost disjointness says that no grid box is used twice.

Write $R = \prod_{\ell=1}^d (a_\ell, b_\ell]$. For each ℓ , let

$$a_\ell = t_{\ell,0} < t_{\ell,1} < \dots < t_{\ell,N_\ell} = b_\ell$$

be the ordered list of all ℓ -th coordinate endpoints of $R_1, \dots, R_n$. The grid boxes are

$$Q_k = \prod_{\ell=1}^d (t_{\ell, k_\ell - 1}, t_{\ell, k_\ell}], \quad 1 \leq k_\ell \leq N_\ell.$$

For every $S \in \{R, R_1, \dots, R_n\}$, let $\mathcal{G}(S)$ be the set of grid boxes contained in S . By the construction of the grid,

$$\begin{aligned} v_d(S) &= \prod_{\ell=1}^d (b_\ell - a_\ell) = \prod_{\ell=1}^d \sum_{k_\ell=1}^{N_\ell} (t_{\ell, k_\ell} - t_{\ell, k_\ell - 1}) \\ &= \sum_{Q \in \mathcal{G}(S)} v_d(Q). \end{aligned}$$

Since $R = \bigcup_{j=1}^n R_j$ and the R_j 's are almost disjoint,

$$\mathcal{G}(R) = \bigsqcup_{j=1}^n \mathcal{G}(R_j).$$

Therefore

$$\begin{aligned} v_d(R) &= \sum_{Q \in \mathcal{G}(R)} v_d(Q) = \sum_{j=1}^n \sum_{Q \in \mathcal{G}(R_j)} v_d(Q) \\ &= \sum_{j=1}^n v_d(R_j). \end{aligned}$$

□

3.2.4. PROBLEM. If $R = \bigcup_{j=1}^n R_j \in \mathcal{J}^d$ and $R_1, \dots, R_n \in \mathcal{J}^d$ are not necessarily almost disjoint, prove

$$v_d(R) \leq \sum_{j=1}^n v_d(R_j).$$

Solution. FIRST-OCCURRENCE DISJOINTIFICATION. Define

$$S_1 = R_1, \quad S_j = R_j \setminus \bigcup_{i=1}^{j-1} R_i \quad (2 \leq j \leq n).$$

By the disjointification identity,

$$R = \bigsqcup_{j=1}^n S_j.$$

Since $\mathcal{J}^d$ is a semi-algebra, each S_j is a finite disjoint union of members of $\mathcal{J}^d$. Write

$$S_j = \bigsqcup_{k=1}^{m_j} Q_{j,k}, \quad Q_{j,k} \in \mathcal{J}^d.$$

Then all $Q_{j,k}$ are pairwise almost disjoint and

$$R = \bigsqcup_{j=1}^n \bigsqcup_{k=1}^{m_j} Q_{j,k}.$$

By finite additivity of volume on almost disjoint rectangles,

$$(*) \quad v_d(R) = \sum_{j=1}^n \sum_{k=1}^{m_j} v_d(Q_{j,k}).$$

For every j ,

$$\bigsqcup_{k=1}^{m_j} Q_{j,k} = S_j \subseteq R_j.$$

By the semi-algebra difference property, the remaining set

$$R_j \setminus S_j$$

is a finite disjoint union of members of $\mathcal{J}^d$. Thus we may write

$$R_j = \left(\bigsqcup_{k=1}^{m_j} Q_{j,k} \right) \sqcup \left(\bigsqcup_{\ell=1}^{r_j} T_{j,\ell} \right), \quad T_{j,\ell} \in \mathcal{J}^d.$$

Applying finite additivity of volume again,

$$v_d(R_j) = \sum_{k=1}^{m_j} v_d(Q_{j,k}) + \sum_{\ell=1}^{r_j} v_d(T_{j,\ell}) \geq \sum_{k=1}^{m_j} v_d(Q_{j,k}).$$

Summing over j and using $(*)$,

$$v_d(R) = \sum_{j=1}^n \sum_{k=1}^{m_j} v_d(Q_{j,k}) \leq \sum_{j=1}^n v_d(R_j).$$

□

3.3 Measures

3.3.1. Definition: Measure and measure space. Let $(X, \mathcal{A})$ be a measurable space. A *measure* on $\mathcal{A}$ is a countably additive function $\mu : \mathcal{A} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$. The triple $(X, \mathcal{A}, \mu)$ is a *measure space*.

3.3.2. Definition: Finite, probability, σ -finite, and semifinite measures. Let $(X, \mathcal{A}, \mu)$ be a measure space.

- (a) μ is *finite* if $\mu(X) < \infty$.
- (b) μ is a *probability measure* if $\mu(X) = 1$.
- (c) μ is *σ -finite* if $X = \bigcup_{j \in \mathbb{N}} E_j$ for measurable E_j satisfying $\mu(E_j) < \infty$.
- (d) μ is *semifinite* if, for every $E \in \mathcal{A}$ with $\mu(E) = \infty$, there exists $F \in \mathcal{A}$ such that $F \subseteq E$ and $0 < \mu(F) < \infty$.

3.3.3. Remark. If μ is finite, then every measurable set has finite measure.

3.3.4. Example: Dirac measure. Let $X \neq \emptyset$ and fix $x_0 \in X$. The *Dirac measure* on $(X, \mathcal{P}(X))$ is

$$\delta_{x_0}(E) = \begin{cases} 0, & x_0 \notin E, \\ 1, & x_0 \in E. \end{cases}$$

3.3.5. Example: Counting measure. If $X \neq \emptyset$ is countable, the *counting measure* on $(X, \mathcal{P}(X))$ is

$$\mu(E) = \sum_{x \in X} \delta_x(E), \quad E \subseteq X.$$

3.3.6. Example. If X is infinite, the function $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ given by

$$\mu(E) = \begin{cases} \infty, & E \text{ is infinite,} \\ 0, & E \text{ is finite} \end{cases}$$

is finitely additive but is not a measure.

3.3.7. Example: Normalized counting measure. If $X \neq \emptyset$ is finite, then

$$\mu(E) = \frac{1}{|X|} \sum_{x \in X} \delta_x(E)$$

is the normalized counting measure on $(X, \mathcal{P}(X))$. It satisfies $\mu(X) = 1$, hence is a probability measure.

3.3.8. PROBLEM. Let X be uncountable and let $\mathcal{A}$ be the σ -algebra of countable or co-countable subsets of X . Define

$$\mu(E) = \begin{cases} 1, & E \text{ is uncountable,} \\ 0, & E \text{ is countable.} \end{cases} \quad (E \in \mathcal{A}).$$

Show that μ is a measure on $(X, \mathcal{A})$.

Solution. Since $\emptyset$ is countable,

$$\mu(\emptyset) = 0.$$

Also,

$$\mu(E) \in \{0, 1\} \subseteq [0, \infty] \quad (E \in \mathcal{A}).$$

It remains to prove countable additivity. Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint.

CO-COUNTABLE SETS CANNOT BE DISJOINT. There is at most one uncountable member of the sequence. Indeed, if E_m and E_n were both uncountable, then, by the definition of $\mathcal{A}$,

$$E_m^c, E_n^c$$

would be countable. Hence

$$(E_m \cap E_n)^c = E_m^c \cup E_n^c$$

would be countable. Therefore $E_m \cap E_n$ would be co-countable, and thus nonempty because X is uncountable. This contradicts pairwise disjointness.

We consider the two possible cases.

(all E_n are countable) By closure of countable sets under countable unions,

$$\bigcup_{n=1}^{\infty} E_n$$

is countable. Therefore

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = 0 = \sum_{n=1}^{\infty} \mu(E_n).$$

(some E_m is uncountable) By the preceding argument, E_m is the unique uncountable member of the sequence. Hence

$$\mu(E_m) = 1, \quad \mu(E_n) = 0 \quad (n \neq m).$$

Since

$$E_m \subseteq \bigcup_{n=1}^{\infty} E_n,$$

the union is uncountable. Therefore

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = 1 = \sum_{n=1}^{\infty} \mu(E_n).$$

Thus μ is countably additive on $\mathcal{A}$. Hence μ is a measure on $(X, \mathcal{A})$. $\square$

3.3.9. Theorem (Basic properties of measures). — *Let $(X, \mathcal{A}, \mu)$ be a measure space.*

- (a) Monotonicity: $E, F \in \mathcal{A}$ and $E \subseteq F \Rightarrow \mu(E) \leq \mu(F)$.
- (b) Countable subadditivity: $(E_j)_{j \in \mathbb{N}} \subseteq \mathcal{A} \Rightarrow \mu(\bigcup_{j=1}^{\infty} E_j) \leq \sum_{j=1}^{\infty} \mu(E_j)$.
- (c) Continuity from below: if $E_1 \subseteq E_2 \subseteq \cdots$, then

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

- (d) Continuity from above: if $E_1 \supseteq E_2 \supseteq \cdots$ and $\mu(E_1) < \infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

Proof. (a) DECOMPOSITION INTO A SET AND ITS REMAINDER. Since $E \subseteq F$,

$$F = E \bigsqcup (F \setminus E).$$

By finite additivity,

$$\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E).$$

(b) FIRST-OCCURRENCE DISJOINTIFICATION. Define

$$D_1 = E_1, \quad D_j = E_j \setminus \bigcup_{k=1}^{j-1} E_k \quad (j \geq 2).$$

Then $D_j \in \mathcal{A}$, the sets D_j are pairwise disjoint, and

$$\bigcup_{j=1}^{\infty} E_j = \bigsqcup_{j=1}^{\infty} D_j.$$

Since $D_j \subseteq E_j$, monotonicity gives

$$\mu(D_j) \leq \mu(E_j).$$

Therefore, by countable additivity,

$$\begin{aligned}\mu\left(\bigcup_{j=1}^{\infty} E_j\right) &= \sum_{j=1}^{\infty} \mu(D_j) \\ &\leq \sum_{j=1}^{\infty} \mu(E_j).\end{aligned}$$

(c) **SUCCESSIVE DIFFERENCES.** Define

$$D_1 = E_1, \quad D_n = E_n \setminus E_{n-1} \quad (n \geq 2).$$

Since $E_1 \subseteq E_2 \subseteq \dots$, the sets D_n are pairwise disjoint, and

$$E_N = \bigsqcup_{n=1}^N D_n, \quad \bigcup_{n=1}^{\infty} E_n = \bigsqcup_{n=1}^{\infty} D_n.$$

Thus, by finite and countable additivity,

$$\begin{aligned}\mu\left(\bigcup_{n=1}^{\infty} E_n\right) &= \sum_{n=1}^{\infty} \mu(D_n) \\ &= \lim_{N \rightarrow \infty} \sum_{n=1}^N \mu(D_n) \\ &= \lim_{N \rightarrow \infty} \mu(E_N).\end{aligned}$$

(d) **TAKING COMPLEMENTS INSIDE A FINITE-MEASURE SET.** Define

$$F_n = E_1 \setminus E_n.$$

Since (E_n) is decreasing,

$$F_1 \subseteq F_2 \subseteq \dots.$$

By De Morgan's law,

$$\bigcup_{n=1}^{\infty} F_n = E_1 \setminus \bigcap_{n=1}^{\infty} E_n.$$

By continuity from below,

$$\mu\left(E_1 \setminus \bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_1 \setminus E_n).$$

Since $E_n \subseteq E_1$ and $\mu(E_1) < \infty$, finite additivity gives

$$\mu(E_1 \setminus E_n) = \mu(E_1) - \mu(E_n),$$

and similarly

$$\mu\left(E_1 \setminus \bigcap_{n=1}^{\infty} E_n\right) = \mu(E_1) - \mu\left(\bigcap_{n=1}^{\infty} E_n\right).$$

Therefore

$$\begin{aligned}\mu(E_1) - \mu\left(\bigcap_{n=1}^{\infty} E_n\right) &= \lim_{n \rightarrow \infty} (\mu(E_1) - \mu(E_n)) \\ &= \mu(E_1) - \lim_{n \rightarrow \infty} \mu(E_n).\end{aligned}$$

Hence

$$\mu \left(\bigcap_{n=1}^{\infty} E_n \right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

□

3.3.10. PROBLEM. Give an example showing that the hypothesis $\mu(E_1) < \infty$ in continuity from above cannot be removed.

Solution. INFINITE-MEASURE OBSTRUCTION. Let $X = \mathbb{N}$, let $\mathcal{A} = \mathcal{P}(\mathbb{N})$, and let μ be counting measure. Define

$$E_n = \{n, n+1, n+2, \dots\}.$$

Then $E_1 \supseteq E_2 \supseteq \dots$, and

$$\bigcap_{n=1}^{\infty} E_n = \emptyset.$$

However, by the definition of counting measure,

$$\mu(E_n) = \infty \quad (n \in \mathbb{N}).$$

Hence

$$\mu \left(\bigcap_{n=1}^{\infty} E_n \right) = \mu(\emptyset) = 0 \neq \infty = \lim_{n \rightarrow \infty} \mu(E_n).$$

Thus continuity from above can fail when $\mu(E_1) = \infty$. □

3.3.11. PROBLEM. Let $(X, \mathcal{M}, \mu)$ be a measure space. If $(E_j)_{j \in \mathbb{N}} \subseteq \mathcal{M}$ and $\mu(E_m \cap E_n) = 0$ for $m \neq n$, prove

$$\mu \left(\bigcup_{j \in \mathbb{N}} E_j \right) = \sum_{j \in \mathbb{N}} \mu(E_j).$$

Solution. DISJOINTIFICATION MODULO NULL SETS. Define

$$D_1 = E_1, \quad D_j = E_j \setminus \bigcup_{k=1}^{j-1} E_k \quad (j \geq 2).$$

By construction, the sets $(D_j)_{j \in \mathbb{N}}$ are pairwise disjoint and

$$\bigcup_{j \in \mathbb{N}} D_j = \bigcup_{j \in \mathbb{N}} E_j.$$

For $j \geq 2$, let

$$N_j = E_j \setminus D_j = E_j \cap \bigcup_{k=1}^{j-1} E_k \subseteq \bigcup_{k=1}^{j-1} (E_j \cap E_k).$$

By finite subadditivity and the hypothesis $\mu(E_j \cap E_k) = 0$ for $j \neq k$,

$$\mu(N_j) \leq \sum_{k=1}^{j-1} \mu(E_j \cap E_k) = 0.$$

Since $E_j = D_j \sqcup N_j$, finite additivity gives

$$\mu(E_j) = \mu(D_j) + \mu(N_j) = \mu(D_j).$$

Therefore, by countable additivity applied to the pairwise disjoint sets D_j ,

$$\mu\left(\bigcup_{j \in \mathbb{N}} E_j\right) = \mu\left(\bigcup_{j \in \mathbb{N}} D_j\right) = \sum_{j \in \mathbb{N}} \mu(D_j) = \sum_{j \in \mathbb{N}} \mu(E_j).$$

□

3.3.12. PROBLEM (Uniqueness from an algebra). Let $\mathcal{A}$ be an algebra with $\sigma(\mathcal{A}) = \mathcal{M}$. If μ_1, μ_2 are finite measures on $(X, \mathcal{M})$ such that $\mu_1(A) = \mu_2(A)$ for all $A \in \mathcal{A}$, prove that $\mu_1 = \mu_2$ on $\mathcal{M}$.

Solution. π - λ TECHNIQUE. We show that the measures agree on a λ -system containing $\mathcal{A}$. Define

$$\mathcal{D} = \{E \in \mathcal{M} : \mu_1(E) = \mu_2(E)\}.$$

By hypothesis, $\mathcal{A} \subseteq \mathcal{D}$.

THE WHOLE SPACE. Since $X \in \mathcal{A}$, the hypothesis gives

$$\mu_1(X) = \mu_2(X),$$

so $X \in \mathcal{D}$.

COMPLEMENTS. Let $E \in \mathcal{D}$. Since μ_1 and μ_2 are finite measures, $\mu_1(X), \mu_2(X) < \infty$. By finite additivity,

$$\mu_1(E^c) = \mu_1(X) - \mu_1(E) = \mu_2(X) - \mu_2(E) = \mu_2(E^c).$$

Hence $E^c \in \mathcal{D}$.

COUNTABLE DISJOINT UNIONS. Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{D}$ be pairwise disjoint. By countable additivity,

$$\mu_1\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \sum_{n \in \mathbb{N}} \mu_1(E_n) = \sum_{n \in \mathbb{N}} \mu_2(E_n) = \mu_2\left(\bigcup_{n \in \mathbb{N}} E_n\right).$$

Thus $\bigcup_{n \in \mathbb{N}} E_n \in \mathcal{D}$.

Therefore $\mathcal{D}$ is a λ -system. Since $\mathcal{A}$ is an algebra, it is a π -system. By the π - λ Theorem,

$$\mathcal{M} = \sigma(\mathcal{A}) \subseteq \mathcal{D}.$$

Since $\mathcal{D} \subseteq \mathcal{M}$ by definition, $\mathcal{D} = \mathcal{M}$. Hence

$$\mu_1(E) = \mu_2(E) \quad (E \in \mathcal{M}),$$

so $\mu_1 = \mu_2$. □

3.4 Null Sets and Completion

3.4.1. Definition: Null sets and almost everywhere. In a measure space $(X, \mathcal{A}, \mu)$, a measurable set E satisfying $\mu(E) = 0$ is a *null set*, also called a *negligible set* or a *set of measure zero*. A property that holds outside a null set holds μ -almost everywhere, abbreviated μ -a.e., or for μ -almost every x .

3.4.2. Remark. Countable unions of null sets are null. If $E \in \mathcal{A}$ is null and $F \subseteq E$ is measurable, then F is null; without measurability, F need not lie in $\mathcal{A}$.

3.4.3. Definition: Complete measure. A measure is *complete* if its domain contains every subset of every null set.

3.4.4. Theorem (Completion). — Let $(X, \mathcal{A}, \mu)$ be a measure space, and put

$$\mathcal{N} = \{N \in \mathcal{A} : \mu(N) = 0\}, \quad \overline{\mathcal{A}} = \{E \cup F : E \in \mathcal{A}, F \subseteq N \text{ for some } N \in \mathcal{N}\}.$$

Then $\overline{\mathcal{A}}$ is a σ -algebra, and μ has a unique extension $\overline{\mu}$ to a complete measure on $\overline{\mathcal{A}}$, given by

$$\overline{\mu}(E \cup F) = \mu(E), \quad E \in \mathcal{A}, \quad F \subseteq N \in \mathcal{N}.$$

Proof. NULL-SET ENLARGEMENT. We first show that $\overline{\mathcal{A}}$ is a σ -algebra.

THE WHOLE SPACE. Since $X \in \mathcal{A}$,

$$X = X \cup \emptyset \in \overline{\mathcal{A}}.$$

COMPLEMENTS. Let $B = E \cup F \in \overline{\mathcal{A}}$, where $E \in \mathcal{A}$ and $F \subseteq N \in \mathcal{N}$. Then

$$B^c = E^c \setminus F = (E^c \setminus N) \cup (E^c \cap (N \setminus F)).$$

Since $E^c \setminus N \in \mathcal{A}$ and $E^c \cap (N \setminus F) \subseteq N \in \mathcal{N}$, the definition of $\overline{\mathcal{A}}$ gives $B^c \in \overline{\mathcal{A}}$.

COUNTABLE UNIONS. Let $B_j = E_j \cup F_j \in \overline{\mathcal{A}}$, where $E_j \in \mathcal{A}$ and $F_j \subseteq N_j \in \mathcal{N}$. By countable subadditivity,

$$\mu\left(\bigcup_{j \in \mathbb{N}} N_j\right) \leq \sum_{j \in \mathbb{N}} \mu(N_j) = 0.$$

Thus $\bigcup_{j \in \mathbb{N}} N_j \in \mathcal{N}$, and

$$\bigcup_{j \in \mathbb{N}} B_j = \left(\bigcup_{j \in \mathbb{N}} E_j\right) \cup \left(\bigcup_{j \in \mathbb{N}} F_j\right) \in \overline{\mathcal{A}},$$

because $\bigcup_j E_j \in \mathcal{A}$ and $\bigcup_j F_j \subseteq \bigcup_j N_j$. Hence $\overline{\mathcal{A}}$ is a σ -algebra.

WELL-DEFINEDNESS. Suppose

$$E \cup F = E' \cup F', \quad F \subseteq N, \quad F' \subseteq N', \quad N, N' \in \mathcal{N}.$$

Then

$$E \setminus E' \subseteq F' \subseteq N', \quad E' \setminus E \subseteq F \subseteq N.$$

Hence $\mu(E \setminus E') = \mu(E' \setminus E) = 0$. By finite additivity,

$$\mu(E) = \mu(E \cap E') + \mu(E \setminus E') = \mu(E \cap E') = \mu(E').$$

Therefore

$$\overline{\mu}(E \cup F) := \mu(E)$$

is well defined.

COUNTABLE ADDITIVITY. Let $(B_j)_{j \in \mathbb{N}} \subseteq \overline{\mathcal{A}}$ be pairwise disjoint, with $B_j = E_j \cup F_j$, where $E_j \in \mathcal{A}$ and $F_j \subseteq N_j \in \mathcal{N}$. Since $E_j \subseteq B_j$, the sets E_j are pairwise disjoint. Moreover,

$$\bigcup_{j \in \mathbb{N}} B_j = \left(\bigcup_{j \in \mathbb{N}} E_j\right) \cup \left(\bigcup_{j \in \mathbb{N}} F_j\right), \quad \bigcup_{j \in \mathbb{N}} F_j \subseteq \bigcup_{j \in \mathbb{N}} N_j \in \mathcal{N}.$$

Thus, by the definition of $\bar{\mu}$ and countable additivity of μ ,

$$\bar{\mu}\left(\bigcup_{j \in \mathbb{N}} B_j\right) = \mu\left(\bigcup_{j \in \mathbb{N}} E_j\right) = \sum_{j \in \mathbb{N}} \mu(E_j) = \sum_{j \in \mathbb{N}} \bar{\mu}(B_j).$$

Therefore $\bar{\mu}$ is a measure. Since $E = E \cup \emptyset$ for every $E \in \mathcal{A}$, $\bar{\mu}|_{\mathcal{A}} = \mu$.

COMPLETENESS. Let $B = E \cup F \in \bar{\mathcal{A}}$ satisfy $\bar{\mu}(B) = 0$. By definition, $\mu(E) = 0$, so $E \in \mathcal{N}$. If $F \subseteq N \in \mathcal{N}$, then

$$B \subseteq E \cup N, \quad \mu(E \cup N) = 0.$$

Hence every $S \subseteq B$ satisfies $S \subseteq E \cup N$, and therefore

$$S = \emptyset \cup S \in \bar{\mathcal{A}}.$$

Thus $\bar{\mu}$ is complete.

UNIQUENESS. Let ν be a measure on $\bar{\mathcal{A}}$ extending μ . For $E \in \mathcal{A}$ and $F \subseteq N \in \mathcal{N}$, monotonicity gives

$$0 \leq \nu(F) \leq \nu(N) = \mu(N) = 0.$$

Thus $\nu(F) = 0$, and finite additivity gives

$$\nu(E \cup F) = \nu(E \sqcup (F \setminus E)) = \nu(E) = \mu(E) = \bar{\mu}(E \cup F).$$

Hence $\nu = \bar{\mu}$. □

3.5 Nonatomic Measures

3.5.1. Definition: Nonatomic measure space. A measure space $(X, \mathcal{M}, \mu)$ is *nonatomic* if every $A \in \mathcal{M}$ with $\mu(A) > 0$ contains some $B \in \mathcal{M}$ satisfying

$$B \subseteq A, \quad 0 < \mu(B) < \mu(A).$$

3.5.2. PROBLEM. Let $(X, \mathcal{M}, \mu)$ be nonatomic and finite. Show that for every $\varepsilon > 0$ and every $A \in \mathcal{M}$ with $\mu(A) > 0$, there exists $B \in \mathcal{M}$ such that $B \subseteq A$ and $0 < \mu(B) < \varepsilon$.

Solution. REPEATED HALVING. We construct decreasing positive-measure subsets whose measures decrease at least by a factor of 2.

Put $A_0 = A$. Suppose that $A_n \in \mathcal{M}$ satisfies $\mu(A_n) > 0$. Since μ is nonatomic, A_n is not an atom. Hence there is $C_n \in \mathcal{M}$ such that

$$C_n \subseteq A_n, \quad 0 < \mu(C_n) < \mu(A_n).$$

Since $\mu(A_n) < \infty$, finite additivity gives

$$\mu(A_n \setminus C_n) = \mu(A_n) - \mu(C_n) > 0.$$

Define

$$A_{n+1} = \begin{cases} C_n, & \mu(C_n) \leq \mu(A_n \setminus C_n), \\ A_n \setminus C_n, & \mu(A_n \setminus C_n) < \mu(C_n). \end{cases}$$

Then $A_{n+1} \subseteq A_n$, $\mu(A_{n+1}) > 0$, and

$$\mu(A_{n+1}) \leq \frac{\mu(A_n)}{2}.$$

By induction,

$$0 < \mu(A_n) \leq \frac{\mu(A)}{2^n} \quad (n \in \mathbb{N}).$$

By the Archimedean Property, choose n such that $2^n > \mu(A)/\varepsilon$. Then

$$0 < \mu(A_n) \leq \frac{\mu(A)}{2^n} < \varepsilon.$$

Thus $B = A_n$ satisfies

$$B \subseteq A, \quad 0 < \mu(B) < \varepsilon.$$

□

3.5.3. PROBLEM (Intermediate values for nonatomic measures). Let $(X, \mathcal{M}, \mu)$ be nonatomic and finite. Show that for every $A \in \mathcal{M}$ and every $\theta \in [0, \mu(A)]$, there exists $B \in \mathcal{M}$ such that $B \subseteq A$ and $\mu(B) = \theta$. *Hint.* Set $H_0 = \emptyset$ and $\mathcal{H}_0 = \{\emptyset\}$. Inductively, for $n \in \mathbb{N}$, define

$$\mathcal{H}_n = \left\{ H \in \mathcal{M} : H \subseteq A \setminus \bigcup_{k=0}^{n-1} H_k, \mu \left(\bigcup_{k=0}^{n-1} H_k \right) + \mu(H) \leq \theta \right\},$$

$$h_n = \sup\{\mu(H) : H \in \mathcal{H}_n\},$$

and choose $H_n \in \mathcal{H}_n$ satisfying

$$\mu(H_n) \geq h_n - \min \left\{ h_n, \frac{1}{n} \right\}.$$

Finally, consider $\bigcup_{n \in \mathbb{N}} H_n$.

Solution. GREEDY APPROXIMATION BY DISJOINT PIECES. For $\theta = 0$, take $B = \emptyset$. Assume $\theta > 0$. Put $H_0 = \emptyset$. Inductively, for $n \geq 1$, define

$$\mathcal{H}_n = \left\{ H \in \mathcal{M} : H \subseteq A \setminus \bigcup_{k=0}^{n-1} H_k, \mu \left(\bigcup_{k=0}^{n-1} H_k \right) + \mu(H) \leq \theta \right\},$$

and let

$$h_n = \sup\{\mu(H) : H \in \mathcal{H}_n\}.$$

Since $\emptyset \in \mathcal{H}_n$ and every $H \in \mathcal{H}_n$ satisfies $\mu(H) \leq \mu(A) < \infty$, the number h_n is defined and finite. Choose $H_n \in \mathcal{H}_n$ such that

$$\mu(H_n) \geq h_n - \min \left\{ h_n, \frac{1}{n} \right\}.$$

By construction, the sets H_n are pairwise disjoint and

$$\mu \left(\bigcup_{k=0}^n H_k \right) \leq \theta \quad (n \in \mathbb{N}).$$

Set

$$B = \bigcup_{n=0}^{\infty} H_n.$$

Then $B \subseteq A$, and continuity from below gives

$$\mu(B) = \lim_{n \rightarrow \infty} \mu \left(\bigcup_{k=0}^n H_k \right) \leq \theta.$$

NO POSITIVE GAP REMAINS. Suppose not. Then $\mu(B) < \theta$. Put

$$r = \theta - \mu(B) > 0.$$

Since $B \subseteq A$, finite additivity gives

$$\mu(A \setminus B) = \mu(A) - \mu(B) \geq \theta - \mu(B) = r > 0.$$

By the preceding small-set result for nonatomic finite measures, there is $C \in \mathcal{M}$ such that

$$C \subseteq A \setminus B, \quad 0 < \mu(C) < r.$$

For every n , since $C \subseteq A \setminus \bigcup_{k=0}^{n-1} H_k$,

$$\mu\left(\bigcup_{k=0}^{n-1} H_k\right) + \mu(C) \leq \mu(B) + \mu(C) < \theta.$$

Thus $C \in \mathcal{H}_n$, so $h_n \geq \mu(C)$. Hence, for every $n > 2/\mu(C)$,

$$\mu(H_n) \geq h_n - \frac{1}{n} \geq \mu(C) - \frac{1}{n} > \frac{\mu(C)}{2}.$$

This is impossible: the H_n 's are pairwise disjoint subsets of A , while $\mu(A) < \infty$. Indeed, countable additivity would give

$$\mu(A) \geq \sum_{n > 2/\mu(C)} \mu(H_n) \geq \sum_{n > 2/\mu(C)} \frac{\mu(C)}{2} = \infty.$$

Therefore $\mu(B) = \theta$. □

3.6 Finite Borel Measures

3.6.1. Definition: Borel measure. Let X be a topological space. A measure on $\text{Bor}(X)$ is a *Borel measure*. The pair $(X, \text{Bor}(X))$ is a *Borel space*, and $(X, \text{Bor}(X), \mu)$ is a *Borel measure space*.

3.6.2. PROBLEM (Regularity of finite Borel measures). Let X be a metric space and $(X, \text{Bor}(X), \mu)$ a finite Borel measure space. Show that $\text{Bor}(X) = \mathcal{M}$, where

$$\mathcal{M} = \left\{ B \in \text{Bor}(X) : \mu(B) = \sup_{\substack{F \subseteq B \\ F \text{ closed}}} \mu(F) = \inf_{\substack{U \supseteq B \\ U \text{ open}}} \mu(U) \right\}.$$

Solution. CLOSED APPROXIMATION OF OPEN SETS. We first show that every open set belongs to $\mathcal{M}$. Let $U \subseteq X$ be open. If $U = X$, then the conclusion is immediate. Otherwise, define

$$F_n = \left\{ x \in X : d(x, U^c) \geq \frac{1}{n} \right\}.$$

Since the distance function $x \mapsto d(x, U^c)$ is continuous, each F_n is closed. By the definition of an open set,

$$F_n \subseteq U, \quad \bigcup_{n \in \mathbb{N}} F_n = U.$$

Thus, by continuity from below,

$$\mu(U) = \lim_{n \rightarrow \infty} \mu(F_n) = \sup_{\substack{F \subseteq U \\ F \text{ closed}}} \mu(F).$$

Also U itself is an open superset of U , so

$$\mu(U) = \inf_{\substack{V \supseteq U \\ V \text{ open}}} \mu(V).$$

Hence $U \in \mathcal{M}$.

TAKING COMPLEMENTS. Let $B \in \mathcal{M}$. Since $\mu(X) < \infty$, for every closed $F \subseteq B$, finite additivity gives

$$\mu(F) = \mu(X) - \mu(F^c).$$

Since F^c is open and contains B^c ,

$$\mu(B) = \sup_{\substack{F \subseteq B \\ F \text{ closed}}} \mu(F) = \mu(X) - \inf_{\substack{V \supseteq B^c \\ V \text{ open}}} \mu(V).$$

Therefore

$$\mu(B^c) = \inf_{\substack{V \supseteq B^c \\ V \text{ open}}} \mu(V).$$

Applying the same argument to the outer approximation of B gives

$$\mu(B^c) = \sup_{\substack{F \subseteq B^c \\ F \text{ closed}}} \mu(F).$$

Thus $B^c \in \mathcal{M}$.

COUNTABLE UNIONS: OUTER APPROXIMATION. Let $(B_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$, and put $B = \bigcup_{n \in \mathbb{N}} B_n$. Let $\varepsilon > 0$. By the outer approximation of B_n , choose open $U_n \supseteq B_n$ such that

$$\mu(U_n) - \mu(B_n) < \frac{\varepsilon}{2^n}.$$

Put $U = \bigcup_{n \in \mathbb{N}} U_n$. Then U is open, $B \subseteq U$, and

$$U \setminus B \subseteq \bigcup_{n \in \mathbb{N}} (U_n \setminus B_n).$$

Thus, by countable subadditivity,

$$\begin{aligned} \mu(U) &= \mu(B) + \mu(U \setminus B) \\ &\leq \mu(B) + \sum_{n \in \mathbb{N}} \mu(U_n \setminus B_n) \\ &= \mu(B) + \sum_{n \in \mathbb{N}} (\mu(U_n) - \mu(B_n)) \\ &< \mu(B) + \varepsilon. \end{aligned}$$

Since every open $V \supseteq B$ satisfies $\mu(B) \leq \mu(V)$,

$$\mu(B) = \inf_{\substack{V \supseteq B \\ V \text{ open}}} \mu(V).$$

COUNTABLE UNIONS: INNER APPROXIMATION. By continuity from below, choose N such that

$$\mu(B) - \mu\left(\bigcup_{n=1}^N B_n\right) < \frac{\varepsilon}{2}.$$

For $1 \leq n \leq N$, by the inner approximation of B_n , choose closed $F_n \subseteq B_n$ such that

$$\mu(B_n) - \mu(F_n) < \frac{\varepsilon}{2N}.$$

Then $F = \bigcup_{n=1}^N F_n$ is closed and $F \subseteq B$. Moreover,

$$\left(\bigcup_{n=1}^N B_n\right) \setminus F \subseteq \bigcup_{n=1}^N (B_n \setminus F_n).$$

Therefore

$$\begin{aligned} \mu(B) - \mu(F) &= \mu(B) - \mu\left(\bigcup_{n=1}^N B_n\right) + \mu\left(\left(\bigcup_{n=1}^N B_n\right) \setminus F\right) \\ &\leq \mu(B) - \mu\left(\bigcup_{n=1}^N B_n\right) + \sum_{n=1}^N (\mu(B_n) - \mu(F_n)) \\ &< \frac{\varepsilon}{2} + N \frac{\varepsilon}{2N} = \varepsilon. \end{aligned}$$

Since every closed $F \subseteq B$ satisfies $\mu(F) \leq \mu(B)$,

$$\mu(B) = \sup_{\substack{F \subseteq B \\ F \text{ closed}}} \mu(F).$$

Hence $B \in \mathcal{M}$.

We have proved that $\mathcal{M}$ is a σ -algebra containing every open subset of X . By the definition of the Borel σ -algebra,

$$\text{Bor}(X) \subseteq \mathcal{M}.$$

The reverse inclusion holds by the definition of $\mathcal{M}$. Therefore

$$\text{Bor}(X) = \mathcal{M}.$$

□

Outer Measures and Carathéodory Measurability

4.1 Outer Measures

4.1.1. Definition: Outer measure. Let $X \neq \emptyset$. An *outer measure* on X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ such that

- (a) $\mu^*(\emptyset) = 0$;
- (b) $A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$;
- (c) $\mu^*\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} \mu^*(A_n)$ for every $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}(X)$.

4.1.2. Remark. Thus an outer measure is a monotone, countably subadditive set function on $\mathcal{P}(X)$ that vanishes at $\emptyset$.

The most common construction starts from a family $\mathcal{E}$ of *elementary sets* on which a notion of size is known—for example, rectangles in the plane—and approximates arbitrary sets from the outside by countable unions of members of $\mathcal{E}$.

4.1.3. Proposition (Outer measure induced by a set function). — Let $\mathcal{E} \subseteq \mathcal{P}(X)$, with $\emptyset, X \in \mathcal{E}$, and let $\rho : \mathcal{E} \rightarrow [0, \infty]$ satisfy $\rho(\emptyset) = 0$. For $A \subseteq X$, define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : (E_j)_{j \in \mathbb{N}} \subseteq \mathcal{E}, A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}.$$

Then μ^* is an outer measure on X .

Proof. THE EMPTY SET. Since $\emptyset \in \mathcal{E}$, the sequence

$$E_1 = \emptyset, \quad E_j = \emptyset \quad (j \geq 2)$$

covers $\emptyset$. Hence, by the definition of μ^* ,

$$\mu^*(\emptyset) \leq \sum_{j=1}^{\infty} \rho(\emptyset) = 0.$$

Since $\mu^* \geq 0$ by definition,

$$\mu^*(\emptyset) = 0.$$

MONOTONICITY BY ENLARGING THE COVERED SET. Let $A \subseteq B \subseteq X$. Every sequence $(E_j)_{j \in \mathbb{N}} \subseteq \mathcal{E}$ that covers B also covers A . Thus the family of admissible covers of B is contained in the family of admissible covers of A . By the definition of the infimum,

$$\mu^*(A) \leq \mu^*(B).$$

COMBINING NEARLY OPTIMAL COVERS. Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}(X)$. If $\sum_{n \in \mathbb{N}} \mu^*(A_n) = \infty$, then

$$\mu^* \left(\bigcup_{n \in \mathbb{N}} A_n \right) \leq \infty = \sum_{n \in \mathbb{N}} \mu^*(A_n),$$

so the required inequality holds.

Assume now that $\sum_{n \in \mathbb{N}} \mu^*(A_n) < \infty$, and let $\varepsilon > 0$. By the definition of $\mu^*(A_n)$, for every $n \in \mathbb{N}$, there is a sequence $(E_{n,j})_{j \in \mathbb{N}} \subseteq \mathcal{E}$ such that

$$A_n \subseteq \bigcup_{j \in \mathbb{N}} E_{n,j}, \quad \sum_{j \in \mathbb{N}} \rho(E_{n,j}) < \mu^*(A_n) + \frac{\varepsilon}{2^n}.$$

Therefore

$$\bigcup_{n \in \mathbb{N}} A_n \subseteq \bigcup_{n \in \mathbb{N}} \bigcup_{j \in \mathbb{N}} E_{n,j}.$$

By the countability of $\mathbb{N} \times \mathbb{N}$, the doubly indexed family $(E_{n,j})_{(n,j) \in \mathbb{N} \times \mathbb{N}}$ can be enumerated as a sequence in $\mathcal{E}$. Hence it is an admissible cover of $\bigcup_n A_n$. By the definition of μ^* ,

$$\begin{aligned} \mu^* \left(\bigcup_{n \in \mathbb{N}} A_n \right) &\leq \sum_{n \in \mathbb{N}} \sum_{j \in \mathbb{N}} \rho(E_{n,j}) \\ &< \sum_{n \in \mathbb{N}} \left(\mu^*(A_n) + \frac{\varepsilon}{2^n} \right) \\ &= \sum_{n \in \mathbb{N}} \mu^*(A_n) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu^* \left(\bigcup_{n \in \mathbb{N}} A_n \right) \leq \sum_{n \in \mathbb{N}} \mu^*(A_n).$$

Thus μ^* satisfies the three defining properties of an outer measure. $\square$

4.2 Lebesgue Outer Measure on $\mathbb{R}^d$

4.2.1. PROBLEM (Outer measures induced by volume). For $E \subseteq \mathbb{R}^d$, define

$$\lambda_{d,o}^*(E) = \inf \left\{ \sum_{n \in \mathbb{N}} v_d(I_n) : \begin{array}{l} (I_n)_{n \in \mathbb{N}} \subseteq \mathcal{I}_o^d, \\ E \subseteq \bigcup_{n \in \mathbb{N}} I_n \end{array} \right\},$$

Using respectively $\mathcal{I}_{oc}^d$, $\mathcal{I}_{co}^d$, and $\mathcal{I}_c^d$, define $\lambda_{d,oc}^*$, $\lambda_{d,co}^*$, and $\lambda_{d,c}^*$ analogously. Prove that all four are outer measures and that

$$\lambda_{d,o}^*(E) = \lambda_{d,oc}^*(E) = \lambda_{d,co}^*(E) = \lambda_{d,c}^*(E) \quad (E \subseteq \mathbb{R}^d).$$

Solution. Write λ_C^* for the outer set function obtained from v_d using one of the four families

$$\mathcal{C} \in \{\mathcal{I}_o^d, \mathcal{I}_{oc}^d, \mathcal{I}_{co}^d, \mathcal{I}_c^d\}.$$

OUTER-MEASURE CONSTRUCTION. For each of the four families, $\mathbb{R}^d$ has a countable cover by members of that family; for example,

$$\mathbb{R}^d = \bigcup_{n \in \mathbb{N}} (-n, n)^d$$

for $\mathcal{J}_o^d$, and analogous expanding rectangles work for the other three families. Thus the admissible covers in the definition of $\lambda_{\mathcal{C}}^*$ are nonempty.

The proof of the preceding proposition now applies verbatim, with $\rho = v_d$. Hence each of

$$\lambda_{d,o}^*, \quad \lambda_{d,oc}^*, \quad \lambda_{d,co}^*, \quad \lambda_{d,c}^*$$

is an outer measure.

CHANGING ENDPOINT CONVENTIONS BY ENLARGEMENT. Let $\mathcal{C}, \mathcal{D}$ be any two of the four families. We claim that, for every $I \in \mathcal{C}$ and every $\eta > 0$, there is $J \in \mathcal{D}$ such that

$$I \subseteq J, \quad v_d(J) < v_d(I) + \eta.$$

Indeed, write I using endpoints $a_i < b_i$. For $\delta > 0$, let J_δ be the rectangle of the endpoint type prescribed by $\mathcal{D}$, with endpoints

$$a_i - \delta, \quad b_i + \delta \quad (1 \leq i \leq d).$$

Then $I \subseteq J_\delta$, and

$$v_d(J_\delta) = \prod_{i=1}^d ((b_i - a_i) + 2\delta) \longrightarrow \prod_{i=1}^d (b_i - a_i) = v_d(I) \quad (\delta \downarrow 0).$$

Hence some $\delta > 0$ gives the claim.

Now let $(I_n)_{n \in \mathbb{N}} \subseteq \mathcal{C}$ cover $E \subseteq \mathbb{R}^d$, and let $\varepsilon > 0$. By the claim, choose $J_n \in \mathcal{D}$ such that

$$I_n \subseteq J_n, \quad v_d(J_n) < v_d(I_n) + \frac{\varepsilon}{2^n}.$$

Then $(J_n)_{n \in \mathbb{N}}$ covers E . Therefore

$$\begin{aligned} \lambda_{\mathcal{D}}^*(E) &\leq \sum_{n \in \mathbb{N}} v_d(J_n) \\ &< \sum_{n \in \mathbb{N}} v_d(I_n) + \varepsilon. \end{aligned}$$

Taking the infimum over all $\mathcal{C}$ -covers of E , then using that $\varepsilon > 0$ is arbitrary, gives

$$\lambda_{\mathcal{D}}^*(E) \leq \lambda_{\mathcal{C}}^*(E).$$

Since $\mathcal{C}$ and $\mathcal{D}$ were arbitrary, interchanging them gives the reverse inequality. Thus

$$\lambda_{d,o}^*(E) = \lambda_{d,oc}^*(E) = \lambda_{d,co}^*(E) = \lambda_{d,c}^*(E) \quad (E \subseteq \mathbb{R}^d).$$

□

4.2.2. Definition: Lebesgue outer measure. The d -dimensional Lebesgue outer measure is

$$\lambda_d^*(E) = \inf \left\{ \sum_{n \in \mathbb{N}} v_d(E_n) : (E_n)_{n \in \mathbb{N}} \subseteq \mathcal{J}_{co}^d, E \subseteq \bigcup_{n \in \mathbb{N}} E_n \right\}.$$

Thus $\lambda_d^* = \lambda_{d,co}^*$. For $d = 1$, write λ^* for λ_1^* .

4.2.3. PROBLEM. For every rectangle $E \in \mathcal{J}^d$, prove

$$v_d(E) = \lambda_{d,o}^*(E) = \lambda_{d,oc}^*(E) = \lambda_{d,co}^*(E) = \lambda_{d,c}^*(E).$$

Solution. By the preceding problem, the four outer measures are equal. It therefore suffices to prove

$$\lambda_{d,o}^*(E) = v_d(E) \quad (E \in \mathcal{J}^d).$$

OPEN ENLARGEMENT. For every $\varepsilon > 0$, the endpoint-enlargement technique gives an open rectangle $U \in \mathcal{J}_o^d$ such that

$$E \subseteq U, \quad v_d(U) < v_d(E) + \varepsilon,$$

provided $v_d(E) < \infty$. Indeed, enlarge each finite endpoint of E by a sufficiently small amount. Hence, by the definition of $\lambda_{d,o}^*$,

$$\lambda_{d,o}^*(E) \leq v_d(U) < v_d(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\lambda_{d,o}^*(E) \leq v_d(E).$$

If $v_d(E) = \infty$, this inequality is automatic.

FINITE-COVERING INEQUALITY. We use the following consequence of the finite subadditivity of volume: if $K \in \mathcal{J}_c^d$ and

$$K \subseteq \bigcup_{j=1}^N I_j, \quad I_j \in \mathcal{J}_o^d,$$

then

$$v_d(K) \leq \sum_{j=1}^N v_d(I_j).$$

Indeed, by the endpoint-grid disjointification technique, decompose K into finitely many pairwise almost disjoint rectangles, each contained in one of the I_j 's. Finite additivity of volume on almost disjoint rectangles then gives the displayed inequality.

COMPACT EXHAUSTION FROM INSIDE. Choose closed rectangles $K_m \in \mathcal{J}_c^d$ such that

$$K_m \subseteq E, \quad v_d(K_m) \uparrow v_d(E).$$

Such a sequence is obtained by moving finite endpoints inward by $1/m$ and truncating any infinite endpoints at $\pm m$.

Let $(I_n)_{n \in \mathbb{N}} \subseteq \mathcal{J}_o^d$ cover E . Since $K_m \subseteq E$, it is also covered by the I_n 's. Since K_m is compact, the Heine–Borel Theorem gives a finite subcover

$$K_m \subseteq \bigcup_{n=1}^{N_m} I_n.$$

By the finite-covering inequality,

$$v_d(K_m) \leq \sum_{n=1}^{N_m} v_d(I_n) \leq \sum_{n \in \mathbb{N}} v_d(I_n).$$

Taking the supremum over $\mathbf{m}$ gives

$$v_d(E) = \sup_{\mathbf{m} \in \mathbb{N}} v_d(K_{\mathbf{m}}) \leq \sum_{n \in \mathbb{N}} v_d(I_n).$$

Since this holds for every open-rectangle cover of E , the definition of the infimum gives

$$v_d(E) \leq \lambda_{d,o}^*(E).$$

Therefore

$$v_d(E) = \lambda_{d,o}^*(E).$$

By the equality of the four induced outer measures,

$$v_d(E) = \lambda_{d,o}^*(E) = \lambda_{d,oc}^*(E) = \lambda_{d,co}^*(E) = \lambda_{d,c}^*(E).$$

□

4.3 Carathéodory Measurability

4.3.1. Definition: Carathéodory condition. Let μ^* be an outer measure on X . A set $A \subseteq X$ is μ^* -measurable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \text{for every } E \subseteq X.$$

4.3.2. Remark. Subadditivity always gives $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$. Hence, to prove μ^* -measurability, it suffices to prove the reverse inequality; it is enough to do this for sets E with $\mu^*(E) < \infty$.

4.3.3. Remark: The splitting principle. Suppose $X = (a, b) \times (c, d) \subseteq \mathbb{R}^2$, $\mathcal{C}$ is a family of open rectangles in X , and ρ is the area function. The induced outer measure μ^* approximates the area of a bounded set A from the outside by coverings with rectangles. If $\mu^*(X) < \infty$, define the corresponding inner quantity by

$$\mu_*(A) := \mu^*(X) - \mu^*(A^c).$$

It approximates the area of A from the inside. Thus it is natural to regard their common value as the area of A when $\mu^*(A) = \mu_*(A)$. Equivalently,

$$\mu^*(X) = \mu^*(X \cap A) + \mu^*(X \cap A^c).$$

Carathéodory's condition requires this exact splitting for every test set $E \subseteq X$.

4.3.4. Theorem (Carathéodory's theorem). — Let μ^* be an outer measure on X , and put

$$\mathcal{M}(\mu^*) = \{A \subseteq X : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subseteq X\}.$$

Then $\mathcal{M}(\mu^*)$ is a σ -algebra, and $(X, \mathcal{M}(\mu^*), \mu^*)$ is a complete measure space.

Proof. THE WHOLE SPACE AND COMPLEMENTS. By the definition of an outer measure, $\mu^*(\emptyset) = 0$. Thus, for every $E \subseteq X$,

$$\mu^*(E) = \mu^*(E \cap X) + \mu^*(E \cap \emptyset),$$

so $X \in \mathcal{M}(\mu^*)$.

If $A \in \mathcal{M}(\mu^*)$, then

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E \cap A^c) + \mu^*(E \cap (A^c)^c)$$

for every $E \subseteq X$. Hence $A^c \in \mathcal{M}(\mu^*)$.

FINITE UNIONS. Let $A, B \in \mathcal{M}(\mu^*)$. By the defining equality for A , then for B applied to $E \cap A^c$,

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c).\end{aligned}$$

By countable subadditivity,

$$\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B).$$

Therefore

$$\mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c) \leq \mu^*(E).$$

The reverse inequality follows from the subadditivity of μ^* , since

$$E = (E \cap (A \cup B)) \cup (E \cap (A \cup B)^c).$$

Hence $A \cup B \in \mathcal{M}(\mu^*)$. By complements and De Morgan's law, $\mathcal{M}(\mu^*)$ is also closed under finite intersections and differences.

COUNTABLE DISJOINT UNIONS. Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}(\mu^*)$ be pairwise disjoint, and set

$$A = \bigcup_{n \in \mathbb{N}} A_n, \quad A^{(N)} = \bigcup_{n=1}^N A_n.$$

By finite-union closure, $A^{(N)} \in \mathcal{M}(\mu^*)$. Repeatedly applying the defining equality to $A_1, \dots, A_N$ gives

$$\mu^*(E) = \sum_{n=1}^N \mu^*(E \cap A_n) + \mu^*(E \cap (A^{(N)})^c).$$

Since $A^c \subseteq (A^{(N)})^c$, monotonicity of μ^* gives

$$\sum_{n=1}^N \mu^*(E \cap A_n) + \mu^*(E \cap A^c) \leq \mu^*(E).$$

Taking the supremum over N , then using countable subadditivity,

$$\begin{aligned}\mu^*(E \cap A) + \mu^*(E \cap A^c) &\leq \sum_{n \in \mathbb{N}} \mu^*(E \cap A_n) + \mu^*(E \cap A^c) \\ &\leq \mu^*(E).\end{aligned}$$

The reverse inequality follows from subadditivity applied to $E = (E \cap A) \cup (E \cap A^c)$. Thus $A \in \mathcal{M}(\mu^*)$.

DISJOINTIFICATION. Let $(B_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}(\mu^*)$. Define

$$D_1 = B_1, \quad D_n = B_n \setminus \bigcup_{k=1}^{n-1} B_k \quad (n \geq 2).$$

By the preceding finite closure properties, $D_n \in \mathcal{M}(\mu^*)$. The sets D_n are pairwise disjoint and

$$\bigcup_{n \in \mathbb{N}} D_n = \bigcup_{n \in \mathbb{N}} B_n.$$

By countable disjoint-union closure, $\bigcup_{n \in \mathbb{N}} B_n \in \mathcal{M}(\mu^*)$. Hence $\mathcal{M}(\mu^*)$ is a σ -algebra.

COUNTABLE ADDITIVITY ON MEASURABLE SETS. Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}(\mu^*)$ be pairwise disjoint, and set $A = \bigcup_n A_n$. Applying the preceding finite decompositions with $E = A$ gives

$$\mu^*(A) = \sum_{n=1}^N \mu^*(A_n) + \mu^*(A \setminus A^{(N)}) \geq \sum_{n=1}^N \mu^*(A_n).$$

Therefore

$$\mu^*(A) \geq \sum_{n \in \mathbb{N}} \mu^*(A_n).$$

The reverse inequality is the countable subadditivity of μ^* . Thus

$$\mu^*\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu^*(A_n).$$

Hence $\mu^*|_{\mathcal{M}(\mu^*)}$ is a measure.

COMPLETENESS. Let $N \in \mathcal{M}(\mu^*)$ satisfy $\mu^*(N) = 0$, and let $S \subseteq N$. For every $E \subseteq X$, monotonicity gives

$$\mu^*(E \cap S) \leq \mu^*(N) = 0, \quad \mu^*(E \cap S^c) \leq \mu^*(E).$$

Hence

$$\mu^*(E \cap S) + \mu^*(E \cap S^c) \leq \mu^*(E).$$

The reverse inequality follows from subadditivity. Thus $S \in \mathcal{M}(\mu^*)$, and $\mu^*(S) = 0$. Therefore $(X, \mathcal{M}(\mu^*), \mu^*)$ is complete. $\square$

4.3.5. PROBLEM. If $A, B \in \mathcal{M}(\mu^*)$, prove that the verification of the Carathéodory condition for $A \cup B$ does not require subadditivity: the relevant inequality is in fact an equality.

Solution. SUCCESSIVE MEASURABLE SPLITTING. Let $E \subseteq X$. Since $A \in \mathcal{M}(\mu^*)$, apply the Carathéodory condition to the set $E \cap (A \cup B)$:

$$\begin{aligned} \mu^*(E \cap (A \cup B)) &= \mu^*(E \cap (A \cup B) \cap A) + \mu^*(E \cap (A \cup B) \cap A^c) \\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B). \end{aligned}$$

Thus the inequality previously obtained from subadditivity is actually the equality

$$\mu^*(E \cap (A \cup B)) = \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B).$$

Now apply the Carathéodory condition first to A , then to B with $E \cap A^c$:

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) \\ &= \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c). \end{aligned}$$

Hence $A \cup B \in \mathcal{M}(\mu^*)$, without using subadditivity. $\square$

4.3.6. PROBLEM (Outer measure from a measure). Let $(X, \mathcal{M}, \mu)$ be a measure space. For $E \subseteq X$, set

$$\mu^*(E) = \inf\{\mu(A) : A \supseteq E, A \in \mathcal{M}\}.$$

- (a) Prove that μ^* is an outer measure on X .
- (b) Prove $\mathcal{M} \subseteq \mathcal{M}(\mu^*)$.

(c) Prove $\mu^*(E) = \mu(E)$ for every $E \in \mathcal{M}$.

Solution. (a) MEASURABLE ENLARGEMENT. Since $\emptyset \in \mathcal{M}$,

$$\mu^*(\emptyset) \leq \mu(\emptyset) = 0.$$

Since $\mu^* \geq 0$, this gives $\mu^*(\emptyset) = 0$.

If $E \subseteq F \subseteq X$, every measurable superset of F is also a measurable superset of E . By the definition of the infimum,

$$\mu^*(E) \leq \mu^*(F).$$

Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}(X)$. If $\sum_n \mu^*(E_n) = \infty$, countable subadditivity is immediate. Assume that this sum is finite, and let $\varepsilon > 0$. By the definition of $\mu^*(E_n)$, choose $A_n \in \mathcal{M}$ such that

$$E_n \subseteq A_n, \quad \mu(A_n) < \mu^*(E_n) + \frac{\varepsilon}{2^n}.$$

Then $\bigcup_n A_n \in \mathcal{M}$ and $\bigcup_n E_n \subseteq \bigcup_n A_n$. Thus, by countable subadditivity of μ ,

$$\begin{aligned} \mu^*\left(\bigcup_{n \in \mathbb{N}} E_n\right) &\leq \mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) \\ &\leq \sum_{n \in \mathbb{N}} \mu(A_n) \\ &< \sum_{n \in \mathbb{N}} \mu^*(E_n) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu^*\left(\bigcup_{n \in \mathbb{N}} E_n\right) \leq \sum_{n \in \mathbb{N}} \mu^*(E_n).$$

Therefore μ^* is an outer measure.

(b) SPLITTING EVERY MEASURABLE ENLARGEMENT. Let $B \in \mathcal{M}$ and $E \subseteq X$. By subadditivity of the outer measure μ^* ,

$$\mu^*(E) \leq \mu^*(E \cap B) + \mu^*(E \cap B^c).$$

Let $A \in \mathcal{M}$ satisfy $E \subseteq A$. Since $A = (A \cap B) \sqcup (A \cap B^c)$, countable additivity gives

$$\mu(A) = \mu(A \cap B) + \mu(A \cap B^c).$$

Since $A \cap B \supseteq E \cap B$ and $A \cap B^c \supseteq E \cap B^c$, the definition of μ^* gives

$$\mu(A) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c).$$

Taking the infimum over all measurable $A \supseteq E$, we obtain

$$\mu^*(E) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c).$$

Hence equality holds, so $B \in \mathcal{M}(\mu^*)$. Therefore

$$\mathcal{M} \subseteq \mathcal{M}(\mu^*).$$

- (c) **THE SET ITSELF IS THE OPTIMAL ENLARGEMENT.** Let $E \in \mathcal{M}$. Since $E \supseteq E$, the definition of μ^* gives

$$\mu^*(E) \leq \mu(E).$$

Conversely, if $A \in \mathcal{M}$ satisfies $A \supseteq E$, then monotonicity of μ gives $\mu(E) \leq \mu(A)$. Taking the infimum over such A ,

$$\mu(E) \leq \mu^*(E).$$

Therefore

$$\mu^*(E) = \mu(E).$$

□

4.3.7. Remark: Purpose of the construction. The preceding construction yields a complete extension

$$(X, \mathcal{M}(\mu^*), \mu^*)$$

of the original measure space $(X, \mathcal{M}, \mu)$.

4.4 Lebesgue Measure

4.4.1. Definition: Lebesgue σ -algebra and measure. The Lebesgue σ -algebra on $\mathbb{R}^d$ is

$$\mathcal{L}(\mathbb{R}^d) = \mathcal{M}(\lambda_d^*).$$

Its members are the *Lebesgue measurable sets*, and the restriction

$$\lambda_d = \lambda_d^*|_{\mathcal{L}(\mathbb{R}^d)}$$

is the *d -dimensional Lebesgue measure*. For $d = 1$, write λ for λ_1 .

4.4.2. Remark. The measure space $(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^d), \lambda_d)$ is complete. Moreover, $\lambda_d(E) = \nu_d(E)$ for every $E \in \mathcal{J}^d$, and λ_d is σ -finite.

4.4.3. Theorem (Borel sets are Lebesgue measurable). —

$$\text{Bor}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d).$$

Proof. Put

$$\lambda_d^* = \lambda_{d,o}^*, \quad \mathcal{L}(\mathbb{R}^d) = \mathcal{M}(\lambda_d^*).$$

By Carathéodory's theorem, it suffices to show that every open rectangle belongs to $\mathcal{M}(\lambda_d^*)$.

CUTTING EVERY COVERING RECTANGLE ALONG THE BOUNDARY. Let $I \in \mathcal{J}_o^d$, let $E \subseteq \mathbb{R}^d$, and assume first that $\lambda_d^*(E) < \infty$. Let $\varepsilon > 0$. By the definition of λ_d^* , choose $(R_n)_{n \in \mathbb{N}} \subseteq \mathcal{J}_o^d$ such that

$$E \subseteq \bigcup_{n \in \mathbb{N}} R_n, \quad \sum_{n \in \mathbb{N}} \nu_d(R_n) < \lambda_d^*(E) + \varepsilon.$$

For each n , use the endpoint-grid disjointification technique to decompose R_n along the endpoints of I :

$$R_n = (R_n \cap I) \cup \bigcup_{k=1}^{m_n} Q_{n,k},$$

where the displayed rectangles are almost disjoint,

$$R_n \cap I \in \mathcal{J}_o^d, \quad Q_{n,k} \in \mathcal{J}^d,$$

and finite additivity of volume gives

$$v_d(R_n) = v_d(R_n \cap I) + \sum_{k=1}^{m_n} v_d(Q_{n,k}).$$

Since

$$E \cap I \subseteq \bigcup_{n \in \mathbb{N}} (R_n \cap I), \quad E \cap I^c \subseteq \bigcup_{n \in \mathbb{N}} \bigcup_{k=1}^{m_n} Q_{n,k},$$

and all four rectangle-induced outer measures agree, the definition of λ_d^* gives

$$\begin{aligned} \lambda_d^*(E \cap I) + \lambda_d^*(E \cap I^c) &\leq \sum_{n \in \mathbb{N}} v_d(R_n \cap I) + \sum_{n \in \mathbb{N}} \sum_{k=1}^{m_n} v_d(Q_{n,k}) \\ &= \sum_{n \in \mathbb{N}} v_d(R_n) \\ &< \lambda_d^*(E) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\lambda_d^*(E \cap I) + \lambda_d^*(E \cap I^c) \leq \lambda_d^*(E).$$

The reverse inequality is the subadditivity of λ_d^* , since

$$E = (E \cap I) \cup (E \cap I^c).$$

Thus $I \in \mathcal{M}(\lambda_d^*)$. If $\lambda_d^*(E) = \infty$, the same equality follows immediately from subadditivity.

Therefore

$$\mathcal{J}_o^d \subseteq \mathcal{L}(\mathbb{R}^d).$$

Since $\mathcal{L}(\mathbb{R}^d)$ is a σ -algebra,

$$\sigma(\mathcal{J}_o^d) \subseteq \mathcal{L}(\mathbb{R}^d).$$

Finally, every open subset of $\mathbb{R}^d$ is a countable union of open rectangles with rational endpoints. Hence

$$\text{Bor}(\mathbb{R}^d) = \sigma(\mathcal{J}_o^d),$$

and consequently

$$\text{Bor}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d).$$

□

4.4.4. Remark: A generating family for $\text{Bor}(\mathbb{R}^d)$. For $i \in \{1, \dots, d\}$ and $\xi \in \mathbb{R}$, let

$$H_{i,\xi} = \mathbb{R}^{i-1} \times (-\infty, \xi) \times \mathbb{R}^{d-i}.$$

Then

$$\text{Bor}(\mathbb{R}^d) = \sigma(\{H_{i,\xi} : i \in \{1, \dots, d\}, \xi \in \mathbb{Q}\}).$$

For $\mathbf{a} = (a_1, \dots, a_d), \mathbf{b} = (b_1, \dots, b_d) \in [-\infty, \infty]^d$, write

$$[\mathbf{a}, \mathbf{b}] = \prod_{i=1}^d [a_i, b_i] \in \mathcal{J}_{co}^d.$$

4.4.5. Proposition (Measurability of half-spaces). — For $i \in \{1, \dots, d\}$ and $\xi \in \mathbb{R}$, the half-space

$$H_{i,\xi} = \mathbb{R}^{i-1} \times (-\infty, \xi) \times \mathbb{R}^{d-i}$$

is Lebesgue measurable.

Proof. CONTINUOUS-COORDINATE PREIMAGE. Let

$$\pi_i : \mathbb{R}^d \longrightarrow \mathbb{R}, \quad \pi_i(x_1, \dots, x_d) = x_i.$$

The coordinate projection π_i is continuous, and $(-\infty, \xi) \subseteq \mathbb{R}$ is open. Therefore

$$H_{i,\xi} = \pi_i^{-1}((-\infty, \xi))$$

is open in $\mathbb{R}^d$. Hence

$$H_{i,\xi} \in \text{Bor}(\mathbb{R}^d).$$

By the preceding theorem,

$$\text{Bor}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d),$$

so $H_{i,\xi} \in \mathcal{L}(\mathbb{R}^d)$. □

4.4.6. Definition: Metric outer measure. Let (X, ρ) be a metric space. An outer measure μ^* is a metric outer measure if

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B) \quad \text{whenever } \rho(A, B) > 0,$$

where $\rho(A, B) = \inf\{\rho(a, b) : a \in A, b \in B\}$.

4.4.7. PROBLEM. Let μ^* be an outer measure on a metric space (X, ρ) . Prove that μ^* is a metric outer measure if and only if $\text{Bor}(X) \subseteq \mathcal{M}(\mu^*)$.

Solution. Recall that μ^* is a metric outer measure when

$$\text{dist}(A, B) > 0 \implies \mu^*(A \cup B) = \mu^*(A) + \mu^*(B).$$

METRIC OUTER MEASURE $\implies$ BOREL MEASURABLE. We use the annular-separation lemma: if μ^* is a metric outer measure and $F \subseteq X$ is closed, then

$$\mu^*(E) = \mu^*(E \cap F) + \mu^*(E \cap F^c) \quad (E \subseteq X).$$

Indeed, when $\mu^*(E) < \infty$, one separates $E \cap F$ from the parts of $E \cap F^c$ lying a positive distance from F , and then uses the annuli

$$\left\{ x \in E : \frac{1}{n+1} < \rho(x, F) \leq \frac{1}{n} \right\}.$$

The even annuli are pairwise positively separated. By metric additivity, their finite unions have measure equal to the sum of their measures; since they are contained in E , these sums are bounded by $\mu^*(E)$. Thus the annular error can be made arbitrarily small, which yields the displayed equality. If $\mu^*(E) = \infty$, the equality follows immediately from subadditivity.

Hence every closed set belongs to $\mathcal{M}(\mu^*)$. Since $\mathcal{M}(\mu^*)$ is a σ -algebra by Carathéodory's Theorem and every open set is the complement of a closed set,

$$\text{Bor}(X) \subseteq \mathcal{M}(\mu^*).$$

BOREL MEASURABLE $\implies$ METRIC OUTER MEASURE. Assume now that

$$\text{Bor}(X) \subseteq \mathcal{M}(\mu^*).$$

Let $A, B \subseteq X$ satisfy

$$\delta := \text{dist}(A, B) > 0.$$

Define

$$U = \left\{ x \in X : \rho(x, A) < \frac{\delta}{2} \right\}.$$

By the continuity of $x \mapsto \rho(x, A)$, the set U is open; hence $U \in \text{Bor}(X) \subseteq \mathcal{M}(\mu^*)$. Moreover,

$$A \subseteq U, \quad B \subseteq U^c.$$

Apply the Carathéodory condition for U to $A \cup B$. By monotonicity,

$$\begin{aligned} \mu^*(A \cup B) &= \mu^*((A \cup B) \cap U) + \mu^*((A \cup B) \cap U^c) \\ &\geq \mu^*(A) + \mu^*(B). \end{aligned}$$

The reverse inequality is the subadditivity of μ^* :

$$\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B).$$

Therefore

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B).$$

Thus μ^* is a metric outer measure. □

4.4.8. PROBLEM. Prove that λ_d^* is a metric outer measure.

Solution. SMALL-DIAMETER REFINEMENT. Let $A, B \subseteq \mathbb{R}^d$ satisfy

$$\delta := \text{dist}(A, B) > 0.$$

The subadditivity of λ_d^* gives

$$\lambda_d^*(A \cup B) \leq \lambda_d^*(A) + \lambda_d^*(B).$$

We prove the reverse inequality.

If $\lambda_d^*(A \cup B) = \infty$, the preceding inequality already implies

$$\lambda_d^*(A) + \lambda_d^*(B) = \infty,$$

so there is nothing to prove. Assume $\lambda_d^*(A \cup B) < \infty$, and let $\varepsilon > 0$.

Choose an open-rectangle cover $(R_n)_{n \in \mathbb{N}}$ of $A \cup B$ such that

$$\sum_{n \in \mathbb{N}} v_d(R_n) < \lambda_d^*(A \cup B) + \varepsilon.$$

Every R_n has finite volume. By subdividing each coordinate interval into sufficiently many pieces, then enlarging the resulting pieces by an arbitrarily small amount, we obtain finite families

$$\{Q_{n,1}, \dots, Q_{n,m_n}\} \subseteq \mathcal{J}_o^d$$

such that

$$R_n \subseteq \bigcup_{k=1}^{m_n} Q_{n,k}, \quad \text{diam}(Q_{n,k}) < \delta, \quad \sum_{k=1}^{m_n} v_d(Q_{n,k}) < v_d(R_n) + \frac{\varepsilon}{2^n}.$$

Thus the rectangles $Q_{n,k}$ cover $A \cup B$, and

$$\sum_{n \in \mathbb{N}} \sum_{k=1}^{m_n} v_d(Q_{n,k}) < \lambda_d^*(A \cup B) + 2\varepsilon.$$

SEPARATE THE REFINED COVER. No $Q_{n,k}$ can meet both A and B . Indeed, if $x \in Q_{n,k} \cap A$ and $y \in Q_{n,k} \cap B$, then

$$\delta \leq |x - y| < \delta,$$

a contradiction.

Let $\mathcal{Q}_A$ be the rectangles $Q_{n,k}$ that meet A , and define $\mathcal{Q}_B$ analogously. Then $\mathcal{Q}_A$ covers A , $\mathcal{Q}_B$ covers B , and the two families are disjoint. By the definition of λ_d^* ,

$$\begin{aligned} \lambda_d^*(A) + \lambda_d^*(B) &\leq \sum_{Q \in \mathcal{Q}_A} v_d(Q) + \sum_{Q \in \mathcal{Q}_B} v_d(Q) \\ &\leq \sum_{n \in \mathbb{N}} \sum_{k=1}^{m_n} v_d(Q_{n,k}) \\ &< \lambda_d^*(A \cup B) + 2\varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\lambda_d^*(A) + \lambda_d^*(B) \leq \lambda_d^*(A \cup B).$$

Together with subadditivity,

$$\lambda_d^*(A \cup B) = \lambda_d^*(A) + \lambda_d^*(B).$$

Therefore λ_d^* is a metric outer measure. $\square$

4.5 Lebesgue–Stieltjes Measures

4.5.1. Remark: Distribution functions. If μ is a finite Borel measure on $\mathbb{R}$, then $F(x) = \mu((-\infty, x])$ is increasing and right-continuous. Moreover,

$$\mu((a, b]) = F(b) - F(a) \quad (-\infty < a < b \leq \infty),$$

where $(a, \infty] = (a, \infty)$, $F(-\infty) = 0$, and $F(\infty) = \lim_{x \rightarrow \infty} F(x) = \mu(\mathbb{R})$. In particular, $F(\infty) = 1$ only when μ is a probability measure.

Conversely, an increasing right-continuous function F produces a Lebesgue–Stieltjes measure μ_F . The choice $F(x) = x$ produces Lebesgue measure on $\mathbb{R}$.

4.5.2. PROBLEM (Lebesgue–Stieltjes outer measure). Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right-continuous. Define

$$\mu_F^*(E) = \inf \left\{ \sum_{n \in \mathbb{N}} (F(b_n) - F(a_n)) : E \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n] \right\}.$$

Prove that μ_F^* is an outer measure.

Solution. THE EMPTY SET: RIGHT-CONTINUITY PRODUCES ARBITRARILY CHEAP COVERS. Since F is increasing,

$$F(b) - F(a) \geq 0 \quad (a < b).$$

Hence $\mu_F^*(E) \geq 0$ for every $E \subseteq \mathbb{R}$.

Let $\varepsilon > 0$. By the right-continuity of F at 0, for every $n \in \mathbb{N}$, there is $b_n > 0$ such that

$$0 \leq F(b_n) - F(0) < \frac{\varepsilon}{2^n}.$$

The sequence $((0, b_n])_{n \in \mathbb{N}}$ covers $\emptyset$. Thus, by the definition of μ_F^* ,

$$\mu_F^*(\emptyset) \leq \sum_{n \in \mathbb{N}} (F(b_n) - F(0)) < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu_F^*(\emptyset) = 0.$$

MONOTONICITY BY ENLARGING THE COVERED SET. Let $E \subseteq G \subseteq \mathbb{R}$. Every interval cover of G is also an interval cover of E . By the definition of the infimum,

$$\mu_F^*(E) \leq \mu_F^*(G).$$

COMBINING NEARLY OPTIMAL INTERVAL COVERS. Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}(\mathbb{R})$. If

$$\sum_{n \in \mathbb{N}} \mu_F^*(E_n) = \infty,$$

then countable subadditivity is immediate. Assume that this sum is finite, and let $\varepsilon > 0$. By the definition of $\mu_F^*(E_n)$, choose intervals

$$(a_{n,j}, b_{n,j}] \quad (j \in \mathbb{N})$$

such that

$$E_n \subseteq \bigcup_{j \in \mathbb{N}} (a_{n,j}, b_{n,j}], \quad \sum_{j \in \mathbb{N}} (F(b_{n,j}) - F(a_{n,j})) < \mu_F^*(E_n) + \frac{\varepsilon}{2^n}.$$

Then

$$\bigcup_{n \in \mathbb{N}} E_n \subseteq \bigcup_{n \in \mathbb{N}} \bigcup_{j \in \mathbb{N}} (a_{n,j}, b_{n,j}].$$

By the countability of $\mathbb{N} \times \mathbb{N}$, this doubly indexed family can be enumerated as a sequence of intervals. Hence it is an admissible cover of $\bigcup_n E_n$. Therefore

$$\begin{aligned} \mu_F^*\left(\bigcup_{n \in \mathbb{N}} E_n\right) &\leq \sum_{n \in \mathbb{N}} \sum_{j \in \mathbb{N}} (F(b_{n,j}) - F(a_{n,j})) \\ &< \sum_{n \in \mathbb{N}} \left(\mu_F^*(E_n) + \frac{\varepsilon}{2^n}\right) \\ &= \sum_{n \in \mathbb{N}} \mu_F^*(E_n) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu_F^*\left(\bigcup_{n \in \mathbb{N}} E_n\right) \leq \sum_{n \in \mathbb{N}} \mu_F^*(E_n).$$

Thus μ_F^* is an outer measure on $\mathbb{R}$. □

4.5.3. Definition: Lebesgue–Stieltjes measure. The Lebesgue–Stieltjes σ -algebra of F is $\mathcal{L}_{\mu_F} = \mathcal{M}(\mu_F^*)$. The restriction $\mu_F = \mu_F^*|_{\mathcal{L}_{\mu_F}}$ is the Lebesgue–Stieltjes measure of F . The space $(\mathbb{R}, \mathcal{L}_{\mu_F}, \mu_F)$ is complete.

4.5.4. Remark: Endpoints for a general increasing function. For an increasing $F : \mathbb{R} \rightarrow \mathbb{R}$, write

$$F(-\infty) := \lim_{x \rightarrow -\infty} F(x), \quad F(\infty) := \lim_{x \rightarrow \infty} F(x),$$

with values allowed in $[-\infty, \infty]$. These are the endpoint values used in expressions $F(b) - F(a)$ for extended intervals.

4.5.5. PROBLEM. Prove that μ_F^* is a metric outer measure and deduce $\text{Bor}(\mathbb{R}) \subseteq \mathcal{L}_{\mu_F}$.

Solution. SUBDIVISION PRESERVES THE F -COST. Let $A, B \subseteq \mathbb{R}$ satisfy

$$\delta := \text{dist}(A, B) > 0.$$

The subadditivity of μ_F^* gives

$$\mu_F^*(A \cup B) \leq \mu_F^*(A) + \mu_F^*(B).$$

We prove the reverse inequality.

If $\mu_F^*(A \cup B) = \infty$, the displayed inequality already implies $\mu_F^*(A) + \mu_F^*(B) = \infty$. Assume instead that $\mu_F^*(A \cup B) < \infty$, and let $\varepsilon > 0$. Choose a cover

$$A \cup B \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n]$$

such that

$$\sum_{n \in \mathbb{N}} (F(b_n) - F(a_n)) < \mu_F^*(A \cup B) + \varepsilon.$$

For every n , choose

$$a_n = t_{n,0} < t_{n,1} < \cdots < t_{n,r_n} = b_n$$

such that

$$t_{n,k} - t_{n,k-1} < \delta \quad (1 \leq k \leq r_n).$$

Then

$$(a_n, b_n] = \bigcup_{k=1}^{r_n} (t_{n,k-1}, t_{n,k}],$$

and the F -cost telescopes:

$$\sum_{k=1}^{r_n} (F(t_{n,k}) - F(t_{n,k-1})) = F(b_n) - F(a_n).$$

SEPARATE THE REFINED INTERVAL COVER. No interval $(t_{n,k-1}, t_{n,k}]$ can meet both A and B . Indeed, if it contained $x \in A$ and $y \in B$, then

$$\delta \leq |x - y| < t_{n,k} - t_{n,k-1} < \delta,$$

a contradiction.

Let $\mathcal{I}_A$ be the refined intervals that meet A , and define $\mathcal{I}_B$ analogously. Then $\mathcal{I}_A$ covers A , $\mathcal{I}_B$ covers B , and the two families are disjoint. Hence, by the definition of μ_F^* ,

$$\begin{aligned}\mu_F^*(A) + \mu_F^*(B) &\leq \sum_{I \in \mathcal{I}_A} \text{cost}_F(I) + \sum_{I \in \mathcal{I}_B} \text{cost}_F(I) \\ &\leq \sum_{n \in \mathbb{N}} (F(b_n) - F(a_n)) \\ &< \mu_F^*(A \cup B) + \varepsilon,\end{aligned}$$

where

$$\text{cost}_F((a, b]) = F(b) - F(a).$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu_F^*(A) + \mu_F^*(B) \leq \mu_F^*(A \cup B).$$

Therefore

$$\mu_F^*(A \cup B) = \mu_F^*(A) + \mu_F^*(B),$$

so μ_F^* is a metric outer measure.

By the preceding characterization of metric outer measures,

$$\text{Bor}(\mathbb{R}) \subseteq \mathcal{M}(\mu_F^*) = \mathcal{L}_{\mu_F}.$$

□

4.5.6. PROBLEM. Prove that $\mu_F((a, b]) = F(b) - F(a)$ for $-\infty \leq a \leq b \leq \infty$.

Solution. Extend F to $[-\infty, \infty]$ by

$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x), \quad F(\infty) = \lim_{x \rightarrow \infty} F(x).$$

For a distribution function, these are $F(-\infty) = 0$ and $F(\infty) = 1$.

Since $(a, b]$ is Borel, it belongs to $\mathcal{L}_{\mu_F}$. Hence

$$\mu_F((a, b]) = \mu_F^*((a, b]).$$

We prove the formula first for $a, b \in \mathbb{R}$.

THE EASY INEQUALITY. The single interval $(a, b]$ covers itself. Thus, by the definition of μ_F^* ,

$$\mu_F^*((a, b]) \leq F(b) - F(a).$$

FINITE INTERVAL-CHAIN ESTIMATE. We use the following elementary fact. If finitely many open intervals (α_j, β_j) cover $[c, b]$, then

$$F(b) - F(c) \leq \sum_j (F(\beta_j) - F(\alpha_j)).$$

Indeed, by the interval-chain construction, choose a subfamily

$$(\alpha_{j_1}, \beta_{j_1}), \dots, (\alpha_{j_r}, \beta_{j_r})$$

such that

$$\alpha_{j_1} < c < \beta_{j_1}, \quad \alpha_{j_{\ell+1}} < \beta_{j_\ell} < \beta_{j_{\ell+1}}, \quad b < \beta_{j_r}.$$

Since F is increasing,

$$\begin{aligned} F(b) - F(c) &\leq \sum_{\ell=1}^r (F(\beta_{j_\ell}) - F(\beta_{j_{\ell-1}})) \\ &\leq \sum_{\ell=1}^r (F(\beta_{j_\ell}) - F(\alpha_{j_\ell})), \end{aligned}$$

where $\beta_{j_0} = c$.

THE REVERSE INEQUALITY. Let

$$(a, b] \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n],$$

and let $\varepsilon > 0$. Fix $c \in (a, b)$. By the right-continuity of F at b_n , choose $d_n > b_n$ such that

$$F(d_n) - F(b_n) < \frac{\varepsilon}{2^n}.$$

The open intervals (a_n, d_n) cover the compact interval $[c, b]$. By compactness, finitely many of them cover $[c, b]$. By the finite interval-chain estimate,

$$\begin{aligned} F(b) - F(c) &\leq \sum_{n \in J} (F(d_n) - F(a_n)) \\ &\leq \sum_{n \in J} (F(b_n) - F(a_n)) + \varepsilon \\ &\leq \sum_{n \in \mathbb{N}} (F(b_n) - F(a_n)) + \varepsilon, \end{aligned}$$

where $J \subseteq \mathbb{N}$ is finite. Letting $\varepsilon \downarrow 0$, then $c \downarrow a$, right-continuity at a gives

$$F(b) - F(a) \leq \sum_{n \in \mathbb{N}} (F(b_n) - F(a_n)).$$

Taking the infimum over all covers yields

$$F(b) - F(a) \leq \mu_F^*((a, b]).$$

Therefore

$$\mu_F((a, b]) = F(b) - F(a) \quad (a, b \in \mathbb{R}).$$

INFINITE ENDPOINTS. Let $r_n \downarrow -\infty$. The intervals

$$(r_1, b], \quad (r_{n+1}, r_n] \quad (n \in \mathbb{N})$$

cover $(-\infty, b]$, and their costs telescope to

$$F(b) - F(r_1) + \sum_{n \in \mathbb{N}} (F(r_n) - F(r_{n+1})) = F(b) - F(-\infty).$$

The finite-endpoint formula and monotonicity give the reverse inequality. Hence

$$\mu_F((-\infty, b]) = F(b) - F(-\infty).$$

Similarly, if $s_n \uparrow \infty$, the intervals

$$(a, s_1], \quad (s_n, s_{n+1}] \quad (n \in \mathbb{N})$$

cover $(a, \infty]$, and their costs telescope to

$$F(\infty) - F(a).$$

Thus

$$\mu_F((a, \infty]) = F(\infty) - F(a).$$

Combining the two endpoint cases gives

$$\boxed{\mu_F((a, b]) = F(b) - F(a)} \quad (-\infty \leq a \leq b \leq \infty).$$

□

4.5.7. PROBLEM (Uniqueness). Prove that

$$\mu_F = \mu_G \iff F - G \text{ is constant.}$$

Conversely, let μ be a Borel measure on $\mathbb{R}$, finite on bounded Borel sets. Define

$$F(x) = \begin{cases} \mu((0, x]), & x > 0, \\ 0, & x = 0, \\ -\mu((x, 0]), & x < 0. \end{cases}$$

Prove that F is increasing and right-continuous, and that

$$\mu = \mu_F.$$

Solution. INTERVALS DETERMINE THE INCREMENTS. By the preceding result,

$$\mu_F((a, b]) = F(b) - F(a), \quad \mu_G((a, b]) = G(b) - G(a).$$

We prove both implications.

($\implies$) Suppose $\mu_F = \mu_G$. Then, for every $a < b$,

$$F(b) - F(a) = \mu_F((a, b]) = \mu_G((a, b]) = G(b) - G(a).$$

Hence

$$F(b) - G(b) = F(a) - G(a).$$

Thus $F - G$ is constant on $\mathbb{R}$.

($\impliedby$) Suppose $F - G = c$ for some constant c . Then, for every $a < b$,

$$\mu_F((a, b]) = F(b) - F(a) = G(b) - G(a) = \mu_G((a, b]).$$

Since both measures are finite on bounded intervals, the uniqueness theorem on each bounded interval gives $\mu_F = \mu_G$.

CONSTRUCTING THE NORMALIZED DISTRIBUTION FUNCTION. Let μ be a Borel measure finite on bounded Borel sets. Define

$$F(x) = \begin{cases} \mu((0, x]), & x > 0, \\ 0, & x = 0, \\ -\mu((x, 0]), & x < 0. \end{cases}$$

For every $a < b$, finite additivity gives

$$F(b) - F(a) = \mu((a, b]).$$

Indeed, this is immediate when $0 \leq a < b$ or $a < b \leq 0$; if $a < 0 < b$, then

$$F(b) - F(a) = \mu((0, b]) + \mu((a, 0]) = \mu((a, b]).$$

Since $\mu((a, b]) \geq 0$, this identity gives

$$a < b \implies F(a) \leq F(b).$$

Thus F is increasing.

RIGHT-CONTINUITY FROM ABOVE. Let $x \in \mathbb{R}$ and let $x_n \downarrow x$. By the increment formula,

$$F(x_n) - F(x) = \mu((x, x_n]).$$

The sets $(x, x_n]$ decrease to $\emptyset$, and $\mu((x, x_1]) < \infty$ because this interval is bounded. By continuity from above,

$$\mu((x, x_n]) \longrightarrow 0.$$

Hence $F(x_n) \rightarrow F(x)$, so F is right-continuous.

By the construction of the Lebesgue–Stieltjes measure,

$$\mu_F((a, b]) = F(b) - F(a) = \mu((a, b]) \quad (a < b).$$

For $n \in \mathbb{N}$, restrict both measures to $(-n, n]$. They are finite there and agree on the algebra generated by half-open intervals. Therefore, by uniqueness from an algebra,

$$\mu_F|_{(-n, n]} = \mu|_{(-n, n]}.$$

Since $(-n, n] \uparrow \mathbb{R}$, continuity from below gives

$$\mu_F(E) = \lim_{n \rightarrow \infty} \mu_F(E \cap (-n, n]) = \lim_{n \rightarrow \infty} \mu(E \cap (-n, n]) = \mu(E)$$

for every $E \in \text{Bor}(\mathbb{R})$. Hence

$$\mu = \mu_F.$$

□

4.6 Hausdorff Measure and Dimension

4.6.1. Definition: Hausdorff outer measure. Let (X, ρ) be a metric space, $p \geq 0$, and $\delta > 0$. For $A \subseteq X$, set

$$\mathcal{H}_\delta^p(A) = \inf \left\{ \sum_{j \in \mathbb{N}} (\text{diam } B_j)^p : A \subseteq \bigcup_{j \in \mathbb{N}} B_j, \text{diam } B_j \leq \delta \right\},$$

with $\inf \emptyset = \infty$. The p -dimensional Hausdorff outer measure is

$$\mathcal{H}^p(A) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^p(A).$$

Here $\text{diam } B = \sup\{\rho(x, y) : x, y \in B\}$, $\text{diam } \emptyset = 0$, and $(\text{diam } B)^0 = 1$ for $B \neq \emptyset$.

4.6.2. Remark. If $0 < \delta' \leq \delta$, then $\mathcal{H}_\delta^p(A) \leq \mathcal{H}_{\delta'}^p(A)$: decreasing δ restricts the covers allowed in the infimum. Hence $\mathcal{H}^p(A)$ exists and equals $\sup_{\delta > 0} \mathcal{H}_\delta^p(A)$.

4.6.3. PROBLEM. Let (X, ρ) be a metric space and $p \geq 0$. Prove that

- (a) $\mathcal{H}_\delta^p$ is an outer measure for every $\delta > 0$;
 (b) $0 < \delta' \leq \delta \Rightarrow \mathcal{H}_\delta^p(A) \leq \mathcal{H}_{\delta'}^p(A)$, hence

$$\mathcal{H}^p(A) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^p(A) = \sup_{\delta > 0} \mathcal{H}_\delta^p(A);$$

- (c) $\mathcal{H}^p$ is an outer measure on X .

Solution. For $A \subseteq X$, recall that

$$\mathcal{H}_\delta^p(A) = \inf \left\{ \sum_{n \in \mathbb{N}} \text{diam}(E_n)^p : A \subseteq \bigcup_{n \in \mathbb{N}} E_n, \text{diam}(E_n) \leq \delta \right\}.$$

- (a) **COMBINING ADMISSIBLE COVERS.** Since $\emptyset$ is covered by the sequence of empty sets,

$$\mathcal{H}_\delta^p(\emptyset) = 0.$$

If $A \subseteq B$, every admissible δ -cover of B is also an admissible δ -cover of A . Hence

$$\mathcal{H}_\delta^p(A) \leq \mathcal{H}_\delta^p(B).$$

Let $(A_j)_{j \in \mathbb{N}} \subseteq \mathcal{P}(X)$. If $\sum_j \mathcal{H}_\delta^p(A_j) = \infty$, countable subadditivity is immediate. Otherwise, let $\varepsilon > 0$. By the definition of $\mathcal{H}_\delta^p(A_j)$, choose an admissible δ -cover $(E_{j,n})_{n \in \mathbb{N}}$ of A_j such that

$$\sum_{n \in \mathbb{N}} \text{diam}(E_{j,n})^p < \mathcal{H}_\delta^p(A_j) + \frac{\varepsilon}{2^j}.$$

The doubly indexed family $(E_{j,n})_{j,n \in \mathbb{N}}$ is an admissible δ -cover of $\bigcup_j A_j$. Thus

$$\begin{aligned} \mathcal{H}_\delta^p \left(\bigcup_{j \in \mathbb{N}} A_j \right) &\leq \sum_{j \in \mathbb{N}} \sum_{n \in \mathbb{N}} \text{diam}(E_{j,n})^p \\ &< \sum_{j \in \mathbb{N}} \mathcal{H}_\delta^p(A_j) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\mathcal{H}_\delta^p \left(\bigcup_{j \in \mathbb{N}} A_j \right) \leq \sum_{j \in \mathbb{N}} \mathcal{H}_\delta^p(A_j).$$

Therefore $\mathcal{H}_\delta^p$ is an outer measure.

- (b) **SMALLER SCALES ALLOW FEWER COVERS.** Suppose $0 < \delta' \leq \delta$. Every admissible δ' -cover is also an admissible δ -cover. Hence, by the definition of the infimum,

$$\mathcal{H}_\delta^p(A) \leq \mathcal{H}_{\delta'}^p(A).$$

Thus $\mathcal{H}_\delta^p(A)$ is increasing as $\delta \downarrow 0$. Therefore

$$\mathcal{H}^p(A) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^p(A) = \sup_{\delta > 0} \mathcal{H}_\delta^p(A).$$

(c) SUPREMUM OVER SCALES. Since every $\mathcal{H}_\delta^p$ vanishes at $\emptyset$,

$$\mathcal{H}^p(\emptyset) = \sup_{\delta > 0} \mathcal{H}_\delta^p(\emptyset) = 0.$$

Since every $\mathcal{H}_\delta^p$ is monotone, so is $\mathcal{H}^p$.

Finally, by the countable subadditivity of each $\mathcal{H}_\delta^p$,

$$\begin{aligned} \mathcal{H}^p\left(\bigcup_{j \in \mathbb{N}} A_j\right) &= \sup_{\delta > 0} \mathcal{H}_\delta^p\left(\bigcup_{j \in \mathbb{N}} A_j\right) \\ &\leq \sup_{\delta > 0} \sum_{j \in \mathbb{N}} \mathcal{H}_\delta^p(A_j) \\ &\leq \sum_{j \in \mathbb{N}} \sup_{\delta > 0} \mathcal{H}_\delta^p(A_j) \\ &= \sum_{j \in \mathbb{N}} \mathcal{H}^p(A_j). \end{aligned}$$

Hence $\mathcal{H}^p$ is an outer measure on X .

□

4.6.4. PROBLEM. Prove that $\mathcal{H}_\delta^p(A)$ is unchanged if the covering sets are required to be closed, and also if they are required to be open. If $X = \mathbb{R}$, prove that the covering sets may be restricted to closed intervals or to open intervals.

Solution. We use the defining condition

$$\text{diam}(E_n) < \delta.$$

CLOSED ENLARGEMENTS DO NOT CHANGE DIAMETER. Let $(E_n)_{n \in \mathbb{N}}$ be an admissible δ -cover of $A \subseteq X$. Since

$$E_n \subseteq \overline{E_n},$$

the closed sets $\overline{E_n}$ still cover A . Moreover,

$$\text{diam}(\overline{E_n}) = \text{diam}(E_n).$$

Indeed, the inequality $\geq$ follows from $E_n \subseteq \overline{E_n}$; the reverse inequality follows by approximating two points of $\overline{E_n}$ by points of E_n .

Thus every admissible cover can be replaced by a closed admissible cover with exactly the same cost. Since closed covers are a special case of arbitrary covers,

$$\mathcal{H}_\delta^p(A) = \inf \left\{ \sum_{n \in \mathbb{N}} \text{diam}(F_n)^p : A \subseteq \bigcup_{n \in \mathbb{N}} F_n, F_n \text{ closed, } \text{diam}(F_n) < \delta \right\}.$$

SMALL OPEN NEIGHBOURHOODS. Let $(E_n)_{n \in \mathbb{N}}$ be an admissible δ -cover of A , and let $\varepsilon > 0$. Put

$$d_n = \text{diam}(E_n) < \delta.$$

For $p > 0$, choose $r_n > 0$ such that

$$d_n + 2r_n < \delta, \quad (d_n + 2r_n)^p < d_n^p + \frac{\varepsilon}{2^n}.$$

Define the open r_n -neighbourhood

$$U_n = \{x \in X : \rho(x, E_n) < r_n\}.$$

Then $E_n \subseteq U_n$, so the U_n 's cover A . By the triangle inequality,

$$\text{diam}(U_n) \leq d_n + 2r_n < \delta.$$

Therefore

$$\sum_{n \in \mathbb{N}} \text{diam}(U_n)^p < \sum_{n \in \mathbb{N}} \text{diam}(E_n)^p + \varepsilon.$$

For $p = 0$, choose only $r_n > 0$ with $d_n + 2r_n < \delta$; every nonempty set and its nonempty open neighbourhood have the same 0-dimensional cost.

Thus every admissible cover can be replaced by an open admissible cover with arbitrarily close cost. Since open covers are a special case of arbitrary covers,

$$\mathcal{H}_\delta^p(A) = \inf \left\{ \sum_{n \in \mathbb{N}} \text{diam}(U_n)^p : A \subseteq \bigcup_{n \in \mathbb{N}} U_n, U_n \text{ open, } \text{diam}(U_n) < \delta \right\}.$$

INTERVALS WHEN $X = \mathbb{R}$. Let $(F_n)_{n \in \mathbb{N}}$ be a closed admissible cover of $A \subseteq \mathbb{R}$. For every nonempty F_n , put

$$I_n = [\inf F_n, \sup F_n].$$

Since F_n is closed and bounded, I_n is a closed interval, $F_n \subseteq I_n$, and

$$\text{diam}(I_n) = \sup F_n - \inf F_n = \text{diam}(F_n).$$

Hence closed covering sets may be replaced by closed intervals without changing the cost.

Likewise, let $(U_n)_{n \in \mathbb{N}}$ be an open admissible cover. For every nonempty U_n , put

$$J_n = (\inf U_n, \sup U_n).$$

Since $\text{diam}(U_n) < \delta < \infty$, the endpoints are finite. Moreover,

$$U_n \subseteq J_n, \quad \text{diam}(J_n) = \text{diam}(U_n).$$

Thus open covering sets may be replaced by open intervals without changing the cost. $\square$

4.6.5. Remark: Comments on the definition. For closed covering sets,

$$\text{diam } B_j = \text{diam } \overline{B_j}.$$

For open covering sets, replace B_j by

$$U_j = \{x \in X : \rho(x, B_j) < \varepsilon 2^{-j-1}\},$$

whose diameter is at most $\text{diam } B_j + \varepsilon 2^{-j}$.

When p is an integer, a p -dimensional set should have size comparable to r^p inside a region of diameter r . For instance, this applies to a set relatively open in a p -dimensional linear subspace of $\mathbb{R}^d$.

Requiring arbitrarily small covering diameters is essential for $\mathcal{H}^p$ to capture local geometry. Without this restriction, one could cover A by A itself and obtain only the crude bound

$$\mathcal{H}^p(A) \leq (\text{diam } A)^p,$$

which generally does not reflect its true p -dimensional size.

4.6.6. PROBLEM. Prove that $\mathcal{H}^p$ is a metric outer measure on (X, ρ) , and deduce $\text{Bor}(X) \subseteq \mathcal{M}(\mathcal{H}^p)$.

Solution. AT SCALES BELOW THE SEPARATION DISTANCE, NO COVERING SET CAN MEET BOTH SETS. Let $A, B \subseteq X$ satisfy

$$\eta := \text{dist}(A, B) > 0.$$

Fix $0 < \delta < \eta$. Let $(E_n)_{n \in \mathbb{N}}$ be an admissible δ -cover of $A \cup B$. Since

$$\text{diam}(E_n) < \delta < \eta,$$

no E_n can meet both A and B . Indeed, if $x \in E_n \cap A$ and $y \in E_n \cap B$, then

$$\eta \leq \rho(x, y) \leq \text{diam}(E_n) < \eta,$$

a contradiction.

Let $\mathcal{E}_A$ be the members of the cover that meet A , and define $\mathcal{E}_B$ analogously. Then $\mathcal{E}_A$ covers A , $\mathcal{E}_B$ covers B , and the two families are disjoint. Therefore

$$\sum_{n \in \mathbb{N}} \text{diam}(E_n)^p \geq \mathcal{H}_\delta^p(A) + \mathcal{H}_\delta^p(B).$$

Taking the infimum over all admissible δ -covers of $A \cup B$ gives

$$\mathcal{H}_\delta^p(A \cup B) \geq \mathcal{H}_\delta^p(A) + \mathcal{H}_\delta^p(B).$$

The reverse inequality is the subadditivity of the outer measure $\mathcal{H}_\delta^p$. Hence

$$\mathcal{H}_\delta^p(A \cup B) = \mathcal{H}_\delta^p(A) + \mathcal{H}_\delta^p(B) \quad (0 < \delta < \eta).$$

PASS TO ARBITRARILY SMALL SCALES. By the definition of $\mathcal{H}^p$,

$$\begin{aligned} \mathcal{H}^p(A \cup B) &= \sup_{0 < \delta < \eta} \mathcal{H}_\delta^p(A \cup B) \\ &= \sup_{0 < \delta < \eta} (\mathcal{H}_\delta^p(A) + \mathcal{H}_\delta^p(B)) \\ &= \sup_{\delta > 0} \mathcal{H}_\delta^p(A) + \sup_{\delta > 0} \mathcal{H}_\delta^p(B) \\ &= \mathcal{H}^p(A) + \mathcal{H}^p(B). \end{aligned}$$

For the third equality, use the monotonicity

$$0 < \delta' \leq \delta \implies \mathcal{H}_{\delta'}^p \leq \mathcal{H}_\delta^p,$$

given scales for A and B , their minimum works simultaneously for both sets.

Thus $\mathcal{H}^p$ is a metric outer measure. By the preceding characterization of metric outer measures,

$$\text{Bor}(X) \subseteq \mathcal{M}(\mathcal{H}^p).$$

□

4.6.7. Definition: Hausdorff measure. The restriction of $\mathcal{H}^p$ to $\text{Bor}(X)$ is a measure, called the *Hausdorff measure of dimension p*. It is also denoted by $\mathcal{H}^p$.

4.6.8. PROBLEM. Prove that $\mathcal{H}^p$ is invariant under isometries of (X, ρ) . If $f, g : Y \rightarrow X$ satisfy

$$\rho(f(y), f(z)) \leq C \rho(g(y), g(z)) \quad (y, z \in Y),$$

prove $\mathcal{H}^p(f(A)) \leq C^p \mathcal{H}^p(g(A))$ for every $A \subseteq Y$.

Solution. ISOMETRIES PRESERVE EVERY COVERING COST. Let $T : X \rightarrow X$ be an isometry. Thus T is bijective and

$$\rho(Tx, Ty) = \rho(x, y) \quad (x, y \in X).$$

If $(E_n)_{n \in \mathbb{N}}$ is an admissible δ -cover of $A \subseteq X$, then $(T(E_n))_{n \in \mathbb{N}}$ is an admissible δ -cover of $T(A)$, since

$$\text{diam}(T(E_n)) = \text{diam}(E_n).$$

Hence

$$\mathcal{H}_\delta^p(T(A)) \leq \mathcal{H}_\delta^p(A).$$

Applying the same argument to the isometry T^{-1} gives the reverse inequality. Therefore

$$\mathcal{H}_\delta^p(T(A)) = \mathcal{H}_\delta^p(A).$$

Taking the supremum over $\delta > 0$,

$$\mathcal{H}^p(T(A)) = \mathcal{H}^p(A).$$

PUSH A COVER THROUGH THE DISTANCE COMPARISON. Assume $C > 0$ and

$$\rho(f(y), f(z)) \leq C\rho(g(y), g(z)) \quad (y, z \in Y).$$

Let $(E_n)_{n \in \mathbb{N}}$ be an admissible δ -cover of $g(A)$. Define

$$D_n = f(A \cap g^{-1}(E_n)).$$

Since the E_n 's cover $g(A)$, the sets D_n cover $f(A)$. Moreover, if $u = f(y)$ and $v = f(z)$ belong to D_n , then $g(y), g(z) \in E_n$. By hypothesis,

$$\rho(u, v) = \rho(f(y), f(z)) \leq C\rho(g(y), g(z)) \leq C \text{diam}(E_n).$$

Thus

$$\text{diam}(D_n) \leq C \text{diam}(E_n).$$

Hence $(D_n)_{n \in \mathbb{N}}$ is an admissible $C\delta$ -cover of $f(A)$, and

$$\begin{aligned} \mathcal{H}_{C\delta}^p(f(A)) &\leq \sum_{n \in \mathbb{N}} \text{diam}(D_n)^p \\ &\leq C^p \sum_{n \in \mathbb{N}} \text{diam}(E_n)^p. \end{aligned}$$

Taking the infimum over all admissible δ -covers of $g(A)$,

$$\mathcal{H}_{C\delta}^p(f(A)) \leq C^p \mathcal{H}_\delta^p(g(A)).$$

Finally, as $\delta \rightarrow 0^+$, also $C\delta \rightarrow 0^+$. Therefore

$$\begin{aligned} \mathcal{H}^p(f(A)) &= \sup_{\delta > 0} \mathcal{H}_{C\delta}^p(f(A)) \\ &\leq C^p \sup_{\delta > 0} \mathcal{H}_\delta^p(g(A)) \\ &= C^p \mathcal{H}^p(g(A)). \end{aligned}$$

□

4.6.9. PROBLEM. Let $A \subseteq X$. Prove that

- (a) $\mathcal{H}^p(A) < \infty \Rightarrow \mathcal{H}^q(A) = 0$ for every $q > p$;
- (b) $\mathcal{H}^p(A) > 0 \Rightarrow \mathcal{H}^q(A) = \infty$ for every $q < p$.

Solution. (a) **HIGHER POWERS MAKE SMALL SETS CHEAPER.** Let $q > p$. If $\text{diam}(E) < \delta$, then

$$\text{diam}(E)^q = \text{diam}(E)^{q-p} \text{diam}(E)^p \leq \delta^{q-p} \text{diam}(E)^p.$$

Applying this inequality to every admissible δ -cover of A , then taking infima, gives

$$\mathcal{H}_\delta^q(A) \leq \delta^{q-p} \mathcal{H}_\delta^p(A) \leq \delta^{q-p} \mathcal{H}^p(A).$$

If $\mathcal{H}^p(A) < \infty$, then

$$\lim_{\delta \rightarrow 0^+} \delta^{q-p} \mathcal{H}_\delta^p(A) = 0.$$

Hence

$$\mathcal{H}^q(A) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^q(A) = 0.$$

- (b) **LOWER POWERS MAKE SMALL SETS MORE EXPENSIVE.** Let $q < p$. If $\text{diam}(E) < \delta$, then

$$\text{diam}(E)^p = \text{diam}(E)^{p-q} \text{diam}(E)^q \leq \delta^{p-q} \text{diam}(E)^q.$$

Thus, by taking infima over admissible δ -covers,

$$\mathcal{H}_\delta^p(A) \leq \delta^{p-q} \mathcal{H}_\delta^q(A),$$

or equivalently,

$$\mathcal{H}_\delta^q(A) \geq \delta^{q-p} \mathcal{H}_\delta^p(A).$$

Since $\mathcal{H}^p(A) > 0$, choose $c > 0$ such that $c < \mathcal{H}^p(A)$. By

$$\mathcal{H}^p(A) = \sup_{\delta > 0} \mathcal{H}_\delta^p(A),$$

there is $\delta_0 > 0$ such that

$$\mathcal{H}_{\delta_0}^p(A) > c.$$

If $0 < \delta \leq \delta_0$, the monotonicity in δ gives

$$\mathcal{H}_\delta^p(A) \geq \mathcal{H}_{\delta_0}^p(A) > c.$$

Therefore

$$\mathcal{H}_\delta^q(A) > c \delta^{q-p}.$$

Since $q - p < 0$,

$$c \delta^{q-p} \longrightarrow \infty \quad (\delta \rightarrow 0^+).$$

Hence

$$\mathcal{H}^q(A) = \infty.$$

□

4.6.10. Definition: Hausdorff dimension. The Hausdorff dimension of $A \subseteq X$ is the common value

$$\dim_H(A) = \inf\{p \geq 0 : \mathcal{H}^p(A) = 0\} = \sup\{p \geq 0 : \mathcal{H}^p(A) = \infty\}.$$

4.6.11. PROBLEM. If $X = \mathbb{R}^d$, prove that there is a constant $\gamma_d > 0$ such that $\gamma_d \mathcal{H}^d$ is Lebesgue measure on $\mathbb{R}^d$. Deduce $\dim_H(\mathbb{R}^d) = d$.

Solution. Let

$$Q = [0, 1]^d.$$

THE d -DIMENSIONAL HAUSDORFF MEASURE OF A UNIT CUBE IS FINITE AND POSITIVE. For $n \in \mathbb{N}$, partition Q into n^d half-open cubes of side length $1/n$. Each such cube has diameter $\sqrt{d}/n$. Hence, for every $\delta > 0$, choosing n sufficiently large gives

$$\mathcal{H}_\delta^d(Q) \leq n^d \left(\frac{\sqrt{d}}{n} \right)^d = d^{d/2}.$$

Therefore

$$\mathcal{H}^d(Q) \leq d^{d/2} < \infty.$$

For the reverse inequality, let $(E_n)_{n \in \mathbb{N}}$ cover Q . For each nonempty E_n , choose $x_n \in E_n$. Then

$$E_n \subseteq \bar{B}(x_n, \text{diam}(E_n)).$$

If ω_d denotes the Lebesgue measure of the Euclidean unit ball, then

$$\lambda_d^*(E_n) \leq \omega_d \text{diam}(E_n)^d.$$

Thus, by the subadditivity of Lebesgue outer measure,

$$1 = \lambda_d(Q) \leq \sum_{n \in \mathbb{N}} \lambda_d^*(E_n) \leq \omega_d \sum_{n \in \mathbb{N}} \text{diam}(E_n)^d.$$

Taking the infimum over all δ -covers gives

$$\mathcal{H}_\delta^d(Q) \geq \frac{1}{\omega_d}.$$

Hence

$$0 < c_d := \mathcal{H}^d(Q) < \infty.$$

TRANSLATION AND SCALING. By invariance under isometries, $\mathcal{H}^d$ is translation invariant. By the preceding Lipschitz comparison, for $r > 0$,

$$\mathcal{H}^d(rA) \leq r^d \mathcal{H}^d(A).$$

Applying the same inequality to the dilation $x \mapsto x/r$ gives the reverse inequality. Therefore

$$\mathcal{H}^d(rA) = r^d \mathcal{H}^d(A).$$

Put

$$\gamma_d = \frac{1}{c_d}.$$

Then $\gamma_d \mathcal{H}^d$ is a translation-invariant Borel measure, finite on bounded sets, and

$$\gamma_d \mathcal{H}^d(Q) = 1 = \lambda_d(Q).$$

By the uniqueness theorem for locally finite translation-invariant Borel measures on $\mathbb{R}^d$,

$$\gamma_d \mathcal{H}^d = \lambda_d.$$

HAUSDORFF DIMENSION. Since

$$\mathcal{H}^d([-n, n]^d) < \infty,$$

the preceding dimension comparison gives, for every $q > d$,

$$\mathcal{H}^q([-n, n]^d) = 0.$$

Since

$$\mathbb{R}^d = \bigcup_{n \in \mathbb{N}} [-n, n]^d,$$

countable subadditivity gives

$$\mathcal{H}^q(\mathbb{R}^d) = 0 \quad (q > d).$$

On the other hand,

$$\mathcal{H}^d(\mathbb{R}^d) = \gamma_d^{-1} \lambda_d(\mathbb{R}^d) = \infty.$$

Thus, for $q < d$, the preceding comparison result gives

$$\mathcal{H}^q(\mathbb{R}^d) = \infty.$$

Therefore

$$\dim_H(\mathbb{R}^d) = d.$$

□

Regularity and Invariance of Lebesgue Measure

5.1 Regularity of Lebesgue Measure

5.1.1. Theorem (Regularity of Lebesgue measure). — For $E \subseteq \mathbb{R}^d$, the following are equivalent:

- (a) $E \in \mathcal{L}(\mathbb{R}^d)$;
- (b) for every $\varepsilon > 0$, there is an open $O \supseteq E$ with $\lambda_d^*(O \setminus E) \leq \varepsilon$;
- (c) there is a G_δ set $G \supseteq E$ with $\lambda_d^*(G \setminus E) = 0$;
- (d) for every $\varepsilon > 0$, there is a closed $C \subseteq E$ with $\lambda_d^*(E \setminus C) \leq \varepsilon$;
- (e) there is an F_σ set $F \subseteq E$ with $\lambda_d^*(E \setminus F) = 0$.

Moreover, for $E \in \mathcal{L}(\mathbb{R}^d)$,

$$\lambda_d(E) = \inf_{\substack{U \supseteq E \\ U \text{ open}}} \lambda_d(U) = \sup_{\substack{F \subseteq E \\ F \text{ closed}}} \lambda_d(F) = \sup_{\substack{K \subseteq E \\ K \text{ compact}}} \lambda_d(K).$$

Proof. OUTER APPROXIMATION OF MEASURABLE SETS. Assume first that $E \in \mathcal{L}(\mathbb{R}^d)$, and let $\varepsilon > 0$. Put

$$E_m = E \cap [-m, m]^d.$$

Since $\lambda_d(E_m) < \infty$, choose open rectangles $(R_{m,j})_{j \in \mathbb{N}}$ such that

$$E_m \subseteq \bigcup_{j \in \mathbb{N}} R_{m,j}, \quad \sum_{j \in \mathbb{N}} v_d(R_{m,j}) < \lambda_d(E_m) + \frac{\varepsilon}{2^m}.$$

Let

$$O_m = \bigcup_{j \in \mathbb{N}} R_{m,j}.$$

Then O_m is open and $E_m \subseteq O_m$. Since E_m is Lebesgue measurable,

$$\begin{aligned} \lambda_d(O_m \setminus E_m) &= \lambda_d(O_m) - \lambda_d(E_m) \\ &\leq \sum_{j \in \mathbb{N}} v_d(R_{m,j}) - \lambda_d(E_m) \\ &< \frac{\varepsilon}{2^m}. \end{aligned}$$

Now put

$$O = \bigcup_{m \in \mathbb{N}} O_m.$$

Then O is open and $E \subseteq O$. Moreover,

$$O \setminus E \subseteq \bigcup_{m \in \mathbb{N}} (O_m \setminus E_m).$$

Hence, by countable subadditivity,

$$\lambda_d^*(O \setminus E) \leq \sum_{m \in \mathbb{N}} \lambda_d(O_m \setminus E_m) \leq \varepsilon.$$

Thus (1) $\Rightarrow$ (2).

INTERSECTING OPEN APPROXIMATIONS. Assume (2). For every $n \in \mathbb{N}$, choose an open set $O_n \supseteq E$ such that

$$\lambda_d^*(O_n \setminus E) \leq \frac{1}{n}.$$

Define

$$G = \bigcap_{n \in \mathbb{N}} O_n.$$

Then G is a G_δ set, $E \subseteq G$, and

$$G \setminus E \subseteq O_n \setminus E \quad (n \in \mathbb{N}).$$

Therefore

$$\lambda_d^*(G \setminus E) \leq \frac{1}{n} \quad (n \in \mathbb{N}),$$

so

$$\lambda_d^*(G \setminus E) = 0.$$

Hence (2) $\Rightarrow$ (3).

COMPLETENESS REMOVES A NULL DIFFERENCE. Assume (3). Since G is Borel,

$$G \in \mathcal{L}(\mathbb{R}^d).$$

Since $\lambda_d^*(G \setminus E) = 0$, the set $G \setminus E$ is Lebesgue measurable by completeness. Thus

$$E = G \setminus (G \setminus E) \in \mathcal{L}(\mathbb{R}^d).$$

Hence (3) $\Rightarrow$ (1).

INNER APPROXIMATION BY TAKING COMPLEMENTS. Assume (1), and let $\varepsilon > 0$. By the already proved implication (1) $\Rightarrow$ (2), applied to E^c , choose an open set $O \supseteq E^c$ such that

$$\lambda_d^*(O \setminus E^c) \leq \varepsilon.$$

Put

$$C = O^c.$$

Then C is closed and $C \subseteq E$. Moreover,

$$E \setminus C = E \cap O = O \setminus E^c.$$

Thus

$$\lambda_d^*(E \setminus C) \leq \varepsilon.$$

Hence (1) $\Rightarrow$ (4).

UNION OF CLOSED APPROXIMATIONS. Assume (4). For every $n \in \mathbb{N}$, choose a closed set $C_n \subseteq E$ such that

$$\lambda_d^*(E \setminus C_n) \leq \frac{1}{n}.$$

Put

$$F = \bigcup_{n \in \mathbb{N}} C_n.$$

Then F is an F_σ set and $F \subseteq E$. Since

$$E \setminus F \subseteq E \setminus C_n \quad (n \in \mathbb{N}),$$

we have

$$\lambda_d^*(E \setminus F) \leq \frac{1}{n} \quad (n \in \mathbb{N}).$$

Therefore

$$\lambda_d^*(E \setminus F) = 0.$$

Hence (4) $\Rightarrow$ (5).

A BOREL SET PLUS A NULL SET. Assume (5). Since F is Borel,

$$F \in \mathcal{L}(\mathbb{R}^d).$$

Since $\lambda_d^*(E \setminus F) = 0$, completeness gives $E \setminus F \in \mathcal{L}(\mathbb{R}^d)$. Therefore

$$E = F \cup (E \setminus F) \in \mathcal{L}(\mathbb{R}^d).$$

Hence (5) $\Rightarrow$ (1), and all five conditions are equivalent.

THE OPEN AND CLOSED FORMULAS. Assume $E \in \mathcal{L}(\mathbb{R}^d)$. If $U \supseteq E$ is open, then monotonicity gives

$$\lambda_d(E) \leq \lambda_d(U).$$

By (2), for every $\varepsilon > 0$, there is an open $O \supseteq E$ such that

$$\lambda_d(O) = \lambda_d(E) + \lambda_d(O \setminus E) \leq \lambda_d(E) + \varepsilon.$$

Therefore

$$\lambda_d(E) = \inf_{\substack{U \supseteq E \\ U \text{ open}}} \lambda_d(U).$$

Likewise, if $C \subseteq E$ is closed, then $\lambda_d(C) \leq \lambda_d(E)$. By (4), for every $\varepsilon > 0$, there is a closed $C \subseteq E$ such that

$$\lambda_d(E) = \lambda_d(C) + \lambda_d(E \setminus C) \leq \lambda_d(C) + \varepsilon.$$

Hence

$$\lambda_d(E) = \sup_{\substack{C \subseteq E \\ C \text{ closed}}} \lambda_d(C).$$

Finally, let $C \subseteq E$ be closed. The compact sets

$$K_n = C \cap [-n, n]^d$$

increase to C . By continuity from below,

$$\lambda_d(C) = \lim_{n \rightarrow \infty} \lambda_d(K_n).$$

Thus every closed-set value is the supremum of values of compact subsets. Combining this with the closed approximation formula gives

$$\lambda_d(E) = \sup_{\substack{K \subseteq E \\ K \text{ compact}}} \lambda_d(K).$$

□

5.1.2. Theorem (Approximation by finite unions of rectangles). — Let $E \in \mathcal{L}(\mathbb{R}^d)$ with $\lambda_d(E) < \infty$. For every $\varepsilon > 0$, there is a finite union A of open rectangles such that

$$\lambda_d(E \Delta A) < \varepsilon, \quad E \Delta A = (E \setminus A) \cup (A \setminus E).$$

Thus every finite-measure Lebesgue measurable set is, up to a set of arbitrarily small measure, a finite union of open rectangles.

Proof. FIRST ENLARGE E BY A SMALL OPEN ERROR. Let $\varepsilon > 0$. By the outer regularity of Lebesgue measure, there is an open set $O \supseteq E$ such that

$$\lambda_d(O \setminus E) < \frac{\varepsilon}{2}.$$

Since $\lambda_d(E) < \infty$,

$$\lambda_d(O) = \lambda_d(E) + \lambda_d(O \setminus E) < \infty.$$

APPROXIMATE THE OPEN SET FROM INSIDE BY FINITELY MANY RECTANGLES. Every open subset of $\mathbb{R}^d$ is a countable union of open rectangles with rational endpoints. Choose such an enumeration

$$O = \bigcup_{n \in \mathbb{N}} R_n, \quad R_n \in \mathcal{J}_o^d,$$

and define

$$A_N = \bigcup_{n=1}^N R_n.$$

Then $A_N \subseteq O$, each A_N is a finite union of open rectangles, and $A_N \uparrow O$. By continuity from below,

$$\lambda_d(O) = \lim_{N \rightarrow \infty} \lambda_d(A_N).$$

Since $\lambda_d(O) < \infty$, choose N such that

$$\lambda_d(O \setminus A_N) = \lambda_d(O) - \lambda_d(A_N) < \frac{\varepsilon}{2}.$$

Put $A = A_N$.

CONTROL THE TWO PARTS OF THE SYMMETRIC DIFFERENCE. Since $E \subseteq O$ and $A \subseteq O$,

$$E \setminus A \subseteq O \setminus A, \quad A \setminus E \subseteq O \setminus E.$$

The two sets in the symmetric difference are disjoint. Therefore

$$\begin{aligned} \lambda_d(E \triangle A) &= \lambda_d(E \setminus A) + \lambda_d(A \setminus E) \\ &\leq \lambda_d(O \setminus A) + \lambda_d(O \setminus E) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

□

5.2 Translation and Dilation Invariance

5.2.1. Theorem (Translation invariance). — For every $E \subseteq \mathbb{R}^d$ and $x \in \mathbb{R}^d$,

$$\lambda_d^*(x + E) = \lambda_d^*(E), \quad x + E = \{x + y : y \in E\}.$$

If $E \in \mathcal{L}(\mathbb{R}^d)$, then $x + E \in \mathcal{L}(\mathbb{R}^d)$ and

$$\lambda_d(x + E) = \lambda_d(E).$$

Proof. TRANSLATE EVERY RECTANGLE IN A COVER. Let $(R_n)_{n \in \mathbb{N}} \subseteq \mathcal{J}_o^d$ cover E . Then

$$x + E \subseteq \bigcup_{n \in \mathbb{N}} (x + R_n).$$

Since translation preserves open rectangles and their volumes,

$$x + R_n \in \mathcal{J}_o^d, \quad v_d(x + R_n) = v_d(R_n).$$

Thus, by the definition of λ_d^* ,

$$\lambda_d^*(x + E) \leq \sum_{n \in \mathbb{N}} v_d(x + R_n) = \sum_{n \in \mathbb{N}} v_d(R_n).$$

Taking the infimum over all covers of E gives

$$\lambda_d^*(x + E) \leq \lambda_d^*(E).$$

Applying this inequality to $x + E$ and the translation by $-x$,

$$\lambda_d^*(E) = \lambda_d^*((-x) + (x + E)) \leq \lambda_d^*(x + E).$$

Hence

$$\lambda_d^*(x + E) = \lambda_d^*(E).$$

TRANSPORT THE CARATHÉODORY CONDITION. Assume $E \in \mathcal{L}(\mathbb{R}^d)$, and let $A \subseteq \mathbb{R}^d$. By translation invariance of λ_d^* ,

$$\lambda_d^*(A) = \lambda_d^*(A - x).$$

Since E is Lebesgue measurable, the Carathéodory condition gives

$$\lambda_d^*(A - x) = \lambda_d^*((A - x) \cap E) + \lambda_d^*((A - x) \cap E^c).$$

By the elementary identities

$$x + ((A - x) \cap E) = A \cap (x + E),$$

and

$$x + ((A - x) \cap E^c) = A \cap (x + E)^c,$$

translation invariance gives

$$\lambda_d^*(A) = \lambda_d^*(A \cap (x + E)) + \lambda_d^*(A \cap (x + E)^c).$$

Thus $x + E \in \mathcal{L}(\mathbb{R}^d)$.

Finally, since E and $x + E$ are measurable,

$$\lambda_d(x + E) = \lambda_d^*(x + E) = \lambda_d^*(E) = \lambda_d(E).$$

□

5.2.2. Remark: Translation of rectangles. For every $I \in \mathcal{J}_{co}^d$ and $x \in \mathbb{R}^d$, one has $x + I \in \mathcal{J}_{co}^d$ and

$$v_d(x + I) = v_d(I).$$

A translation shifts every vertex of a rectangle by the same vector.

5.2.3. Theorem (Homogeneity). — For every $E \subseteq \mathbb{R}^d$ and $\alpha \in \mathbb{R}$,

$$\lambda_d^*(\alpha E) = |\alpha|^d \lambda_d^*(E), \quad \alpha E = \{\alpha y : y \in E\}.$$

If $E \in \mathcal{L}(\mathbb{R}^d)$, then $\alpha E \in \mathcal{L}(\mathbb{R}^d)$ and

$$\lambda_d(\alpha E) = |\alpha|^d \lambda_d(E).$$

Proof. THE CASE $\alpha = 0$. If $E = \emptyset$, then $0E = \emptyset$. If $E \neq \emptyset$, then $0E = \{0\}$. Since

$$\{0\} \subseteq \left(-\frac{1}{n}, \frac{1}{n}\right)^d,$$

we have

$$0 \leq \lambda_d^*(\{0\}) \leq \left(\frac{2}{n}\right)^d \quad (n \in \mathbb{N}).$$

Hence

$$\lambda_d^*(0E) = 0 = 0^d \lambda_d^*(E),$$

where $0 \cdot \infty = 0$.

DILATING A RECTANGLE COVER. Assume $\alpha \neq 0$. Let

$$E \subseteq \bigcup_{n \in \mathbb{N}} R_n, \quad R_n \in \mathcal{J}_o^d.$$

Then

$$\alpha E \subseteq \bigcup_{n \in \mathbb{N}} \alpha R_n.$$

Since $\alpha R_n \in \mathcal{J}_o^d$ and

$$v_d(\alpha R_n) = |\alpha|^d v_d(R_n),$$

the definition of λ_d^* gives

$$\begin{aligned} \lambda_d^*(\alpha E) &\leq \sum_{n \in \mathbb{N}} v_d(\alpha R_n) \\ &= |\alpha|^d \sum_{n \in \mathbb{N}} v_d(R_n). \end{aligned}$$

Taking the infimum over all open-rectangle covers of E ,

$$\lambda_d^*(\alpha E) \leq |\alpha|^d \lambda_d^*(E).$$

Apply this inequality to αE and the dilation by α^{-1} :

$$\lambda_d^*(E) \leq |\alpha|^{-d} \lambda_d^*(\alpha E).$$

Thus

$$\lambda_d^*(\alpha E) = |\alpha|^d \lambda_d^*(E).$$

TRANSPORTING THE CARATHÉODORY CONDITION. Assume $E \in \mathcal{L}(\mathbb{R}^d)$ and $\alpha \neq 0$. Let $A \subseteq \mathbb{R}^d$. By the homogeneity just proved,

$$\lambda_d^*(A) = |\alpha|^d \lambda_d^*(\alpha^{-1}A).$$

Since E is Lebesgue measurable,

$$\lambda_d^*(\alpha^{-1}A) = \lambda_d^*(\alpha^{-1}A \cap E) + \lambda_d^*(\alpha^{-1}A \cap E^c).$$

Multiplying by $|\alpha|^d$, then applying homogeneity to both terms,

$$\begin{aligned} \lambda_d^*(A) &= \lambda_d^*(\alpha(\alpha^{-1}A \cap E)) + \lambda_d^*(\alpha(\alpha^{-1}A \cap E^c)) \\ &= \lambda_d^*(A \cap \alpha E) + \lambda_d^*(A \cap (\alpha E)^c). \end{aligned}$$

Thus

$$\alpha E \in \mathcal{L}(\mathbb{R}^d).$$

If $\alpha = 0$, then αE is either $\emptyset$ or $\{0\}$, hence is Lebesgue measurable. Therefore, for every $E \in \mathcal{L}(\mathbb{R}^d)$,

$$\lambda_d(\alpha E) = \lambda_d^*(\alpha E) = |\alpha|^d \lambda_d^*(E) = |\alpha|^d \lambda_d(E).$$

□

5.2.4. Remark: Why open rectangles are convenient for dilation. Since $\lambda_d^* = \lambda_{d,o}^*$, it may be computed from coverings by open rectangles. For $\alpha \neq 0$ and $I \in \mathcal{J}_o^d$,

$$\alpha I \in \mathcal{J}_o^d, \quad \nu_d(\alpha I) = |\alpha|^d \nu_d(I).$$

In contrast, $\mathcal{J}_{co}^d$ is not preserved by negative dilations: $\alpha[a, b] = (\alpha b, \alpha a]$ when $\alpha < 0$.

5.2.5. Theorem (Completion of the Borel measure). — The Lebesgue measure space $(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^d), \lambda_d)$ is the completion of the Borel measure space $(\mathbb{R}^d, \text{Bor}(\mathbb{R}^d), \lambda_d)$.

Proof. Let

$$\overline{\text{Bor}}(\mathbb{R}^d) = \{B \cup N : B \in \text{Bor}(\mathbb{R}^d), N \subseteq Z \text{ for some } Z \in \text{Bor}(\mathbb{R}^d) \text{ with } \lambda_d(Z) = 0\}.$$

By definition, this is the σ -algebra of the completion of the Borel measure space.

THE COMPLETION IS CONTAINED IN THE LEBESGUE σ -ALGEBRA. Let

$$E = B \cup N \in \overline{\text{Bor}}(\mathbb{R}^d),$$

where B is Borel and $N \subseteq Z$ for a Borel null set Z . By the inclusion

$$\text{Bor}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d),$$

we have $B, Z \in \mathcal{L}(\mathbb{R}^d)$. Since Lebesgue measure is complete, $N \in \mathcal{L}(\mathbb{R}^d)$. Hence

$$E = B \cup N \in \mathcal{L}(\mathbb{R}^d).$$

Therefore

$$\overline{\text{Bor}}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d).$$

EVERY LEBESGUE MEASURABLE SET DIFFERS FROM A BOREL SET BY A BOREL NULL SET. Let $E \in \mathcal{L}(\mathbb{R}^d)$. By the regularity theorem, there is a G_δ set $G \supseteq E$ such that

$$\lambda_d^*(G \setminus E) = 0.$$

Since G is Borel, it remains to place $G \setminus E$ inside a Borel null set.

For every $n \in \mathbb{N}$, by the definition of Lebesgue outer measure, choose an open set $U_n \supseteq G \setminus E$ such that

$$\lambda_d(U_n) < \frac{1}{n}.$$

Put

$$Z = \bigcap_{n \in \mathbb{N}} U_n.$$

Then Z is Borel,

$$G \setminus E \subseteq Z,$$

and, by monotonicity,

$$\lambda_d(Z) \leq \lambda_d(U_n) < \frac{1}{n} \quad (n \in \mathbb{N}).$$

Hence

$$\lambda_d(Z) = 0.$$

Now define

$$B = G \setminus Z.$$

Since G and Z are Borel, B is Borel. Since $G \setminus E \subseteq Z$,

$$B = G \setminus Z \subseteq E.$$

Moreover,

$$E = B \cup (E \cap Z),$$

and

$$E \cap Z \subseteq Z, \quad \lambda_d(Z) = 0.$$

Thus

$$E \in \overline{\text{Bor}}(\mathbb{R}^d).$$

Therefore

$$\mathcal{L}(\mathbb{R}^d) \subseteq \overline{\text{Bor}}(\mathbb{R}^d).$$

We have proved

$$\mathcal{L}(\mathbb{R}^d) = \overline{\text{Bor}}(\mathbb{R}^d).$$

Finally, if $E = B \cup N$, with B Borel and $N \subseteq Z$ for a Borel null set Z , then

$$\lambda_d(N) = 0,$$

so

$$\lambda_d(E) = \lambda_d(B).$$

Thus the measure λ_d on $\mathcal{L}(\mathbb{R}^d)$ is exactly the completed Borel measure. Hence

$$(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^d), \lambda_d)$$

is the completion of

$$(\mathbb{R}^d, \text{Bor}(\mathbb{R}^d), \lambda_d).$$

□

5.3 Lebesgue–Stieltjes Regularity

5.3.1. Remark. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right-continuous. Its Lebesgue–Stieltjes measure μ_F is σ -finite, $\text{Bor}(\mathbb{R}) \subseteq \mathcal{L}_{\mu_F}$, and

$$\mu_F((a, b]) = F(b) - F(a) \quad (-\infty \leq a \leq b \leq \infty).$$

5.3.2. PROBLEM (Regularity of Lebesgue–Stieltjes measure). For $E \subseteq \mathbb{R}$, prove

$$\mu_F^*(E) = \inf \left\{ \sum_{n \in \mathbb{N}} \mu_F((a_n, b_n)) : E \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n) \right\}.$$

Use this to prove that the following are equivalent:

- (a) $E \in \mathcal{L}_{\mu_F}$;
- (b) for every $\varepsilon > 0$, there is an open $O \supseteq E$ with $\mu_F^*(O \setminus E) \leq \varepsilon$;
- (c) there is a G_δ set $G \supseteq E$ with $\mu_F^*(G \setminus E) = 0$;
- (d) for every $\varepsilon > 0$, there is a closed $C \subseteq E$ with $\mu_F^*(E \setminus C) \leq \varepsilon$;
- (e) there is an F_σ set $F \subseteq E$ with $\mu_F^*(E \setminus F) = 0$.

Show also that, for $E \in \mathcal{L}_{\mu_F}$,

$$\mu_F(E) = \inf_{\substack{U \supseteq E \\ U \text{ open}}} \mu_F(U) = \sup_{\substack{C \subseteq E \\ C \text{ closed}}} \mu_F(C) = \sup_{\substack{K \subseteq E \\ K \text{ compact}}} \mu_F(K).$$

Solution. REPLACING HALF-OPEN INTERVALS BY OPEN INTERVALS. Put

$$\nu(E) = \inf \left\{ \sum_{n \in \mathbb{N}} \mu_F((a_n, b_n)) : E \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n) \right\}.$$

We prove both inequalities.

($\leq$) Let

$$E \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n).$$

Since every open interval is Borel,

$$\mu_F^*((a_n, b_n)) = \mu_F((a_n, b_n)).$$

Thus, by the subadditivity of μ_F^* ,

$$\mu_F^*(E) \leq \sum_{n \in \mathbb{N}} \mu_F^*((a_n, b_n)) = \sum_{n \in \mathbb{N}} \mu_F((a_n, b_n)).$$

Taking the infimum over all open-interval covers gives

$$\mu_F^*(E) \leq \nu(E).$$

($\geq$) Let

$$E \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n],$$

and let $\varepsilon > 0$. Since

$$(a_n, b_n + \delta) \downarrow (a_n, b_n] \quad (\delta \downarrow 0),$$

and μ_F is finite on bounded intervals, continuity from above gives

$$\mu_F((a_n, b_n + \delta)) \longrightarrow \mu_F((a_n, b_n]).$$

Choose $\delta_n > 0$ such that

$$\mu_F((a_n, b_n + \delta_n)) < \mu_F((a_n, b_n]) + \frac{\varepsilon}{2^n}.$$

Then the open intervals $(a_n, b_n + \delta_n)$ cover E . Hence

$$\begin{aligned} \nu(E) &\leq \sum_{n \in \mathbb{N}} \mu_F((a_n, b_n + \delta_n)) \\ &< \sum_{n \in \mathbb{N}} \mu_F((a_n, b_n]) + \varepsilon \\ &= \sum_{n \in \mathbb{N}} (F(b_n) - F(a_n)) + \varepsilon. \end{aligned}$$

Taking the infimum over half-open interval covers, then letting $\varepsilon \downarrow 0$, gives

$$\nu(E) \leq \mu_F^*(E).$$

Therefore

$$\mu_F^*(E) = \inf \left\{ \sum_{n \in \mathbb{N}} \mu_F((a_n, b_n)) : E \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n) \right\}.$$

OUTER APPROXIMATION OF MEASURABLE SETS. Assume $E \in \mathcal{L}_{\mu_F}$, and let $\varepsilon > 0$. Put

$$E_m = E \cap [-m, m].$$

Since μ_F is finite on bounded intervals,

$$\mu_F(E_m) < \infty.$$

By the open-interval formula just proved, choose open intervals $(a_{m,j}, b_{m,j})$ such that

$$E_m \subseteq \bigcup_{j \in \mathbb{N}} (a_{m,j}, b_{m,j}), \quad \sum_{j \in \mathbb{N}} \mu_F((a_{m,j}, b_{m,j})) < \mu_F(E_m) + \frac{\varepsilon}{2^m}.$$

Let

$$O_m = \bigcup_{j \in \mathbb{N}} (a_{m,j}, b_{m,j}).$$

Then O_m is open, $E_m \subseteq O_m$, and

$$\begin{aligned} \mu_F(O_m \setminus E_m) &= \mu_F(O_m) - \mu_F(E_m) \\ &\leq \sum_{j \in \mathbb{N}} \mu_F((a_{m,j}, b_{m,j})) - \mu_F(E_m) \\ &< \frac{\varepsilon}{2^m}. \end{aligned}$$

Put

$$O = \bigcup_{m \in \mathbb{N}} O_m.$$

Then O is open, $E \subseteq O$, and

$$O \setminus E \subseteq \bigcup_{m \in \mathbb{N}} (O_m \setminus E_m).$$

Hence

$$\mu_F^*(O \setminus E) \leq \sum_{m \in \mathbb{N}} \mu_F(O_m \setminus E_m) \leq \varepsilon.$$

Thus (1) $\Rightarrow$ (2).

INTERSECTING OPEN APPROXIMATIONS. Assume (2). For every $n \in \mathbb{N}$, choose an open set $O_n \supseteq E$ such that

$$\mu_F^*(O_n \setminus E) \leq \frac{1}{n}.$$

Define

$$G = \bigcap_{n \in \mathbb{N}} O_n.$$

Then G is a G_δ set, $E \subseteq G$, and

$$G \setminus E \subseteq O_n \setminus E \quad (n \in \mathbb{N}).$$

Therefore

$$\mu_F^*(G \setminus E) = 0.$$

Hence (2) $\Rightarrow$ (3).

COMPLETENESS REMOVES A NULL DIFFERENCE. Assume (3). Since G is Borel,

$$G \in \mathcal{L}_{\mu_F}.$$

Since $\mu_F^*(G \setminus E) = 0$, completeness gives

$$G \setminus E \in \mathcal{L}_{\mu_F}.$$

Thus

$$E = G \setminus (G \setminus E) \in \mathcal{L}_{\mu_F}.$$

Hence (3) $\Rightarrow$ (1).

INNER APPROXIMATION BY COMPLEMENTS. Assume (1). Apply (1) $\Rightarrow$ (2) to E^c . For every $\varepsilon > 0$, there is an open set $O \supseteq E^c$ such that

$$\mu_F^*(O \setminus E^c) \leq \varepsilon.$$

Set

$$C = O^c.$$

Then C is closed, $C \subseteq E$, and

$$E \setminus C = E \cap O = O \setminus E^c.$$

Therefore

$$\mu_F^*(E \setminus C) \leq \varepsilon.$$

Hence (1) $\Rightarrow$ (4).

UNION OF CLOSED APPROXIMATIONS. Assume (4). Choose closed sets $C_n \subseteq E$ such that

$$\mu_F^*(E \setminus C_n) \leq \frac{1}{n}.$$

Let

$$F = \bigcup_{n \in \mathbb{N}} C_n.$$

Then F is an F_σ set, $F \subseteq E$, and

$$E \setminus F \subseteq E \setminus C_n \quad (n \in \mathbb{N}).$$

Thus

$$\mu_F^*(E \setminus F) = 0.$$

Hence (4) $\Rightarrow$ (5).

A BOREL SET PLUS A NULL SET. Assume (5). Since F is Borel,

$$F \in \mathcal{L}_{\mu_F}.$$

Since $\mu_F^*(E \setminus F) = 0$, completeness gives

$$E \setminus F \in \mathcal{L}_{\mu_F}.$$

Therefore

$$E = F \cup (E \setminus F) \in \mathcal{L}_{\mu_F}.$$

Hence (5) $\Rightarrow$ (1).

THE REGULARITY FORMULAS. Let $E \in \mathcal{L}_{\mu_F}$. By monotonicity,

$$\mu_F(E) \leq \mu_F(U) \quad (U \supseteq E, U \text{ open}).$$

By (2), for every $\varepsilon > 0$, there is an open $O \supseteq E$ such that

$$\mu_F(O) = \mu_F(E) + \mu_F(O \setminus E) \leq \mu_F(E) + \varepsilon.$$

Thus

$$\mu_F(E) = \inf_{\substack{U \supseteq E \\ U \text{ open}}} \mu_F(U).$$

Likewise, by monotonicity,

$$\mu_F(C) \leq \mu_F(E) \quad (C \subseteq E, C \text{ closed}).$$

By (4), for every $\varepsilon > 0$, there is a closed $C \subseteq E$ such that

$$\mu_F(E) = \mu_F(C) + \mu_F(E \setminus C) \leq \mu_F(C) + \varepsilon.$$

Therefore

$$\mu_F(E) = \sup_{\substack{C \subseteq E \\ C \text{ closed}}} \mu_F(C).$$

Finally, for every closed $C \subseteq E$, define

$$K_n = C \cap [-n, n].$$

Then K_n is compact, $K_n \uparrow C$, and continuity from below gives

$$\mu_F(C) = \lim_{n \rightarrow \infty} \mu_F(K_n).$$

Hence

$$\mu_F(E) = \sup_{\substack{K \subseteq E \\ K \text{ compact}}} \mu_F(K).$$

□

5.4 General Induced Outer Measures

5.4.1. Remark: *Carathéodory measure induced by a set function.* Let $X \neq \emptyset$, $\emptyset, X \in \mathcal{E} \subseteq \mathcal{P}(X)$, and $\rho : \mathcal{E} \rightarrow [0, \infty]$ satisfy $\rho(\emptyset) = 0$. The induced outer measure is

$$\mu^*(A) = \inf \left\{ \sum_{j \in \mathbb{N}} \rho(E_j) : (E_j)_{j \in \mathbb{N}} \subseteq \mathcal{E}, A \subseteq \bigcup_{j \in \mathbb{N}} E_j \right\} \quad (A \subseteq X).$$

Its Carathéodory σ -algebra is

$$\mathcal{M}(\mu^*) = \{A \subseteq X : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subseteq X\}.$$

The restriction $\mu^*|_{\mathcal{M}(\mu^*)}$ is a complete measure, called the *Carathéodory measure induced by ρ* .

5.4.2. Definition: *Operations on a family of sets.* For $\mathcal{E} \subseteq \mathcal{P}(X)$, put

$$\mathcal{E}_\sigma = \left\{ \bigcup_{n \in \mathbb{N}} E_n : (E_n)_{n \in \mathbb{N}} \subseteq \mathcal{E} \right\}, \quad \mathcal{E}_\delta = \left\{ \bigcap_{n \in \mathbb{N}} E_n : (E_n)_{n \in \mathbb{N}} \subseteq \mathcal{E} \right\},$$

$$\mathcal{E}_{\sigma\delta} = (\mathcal{E}_\sigma)_\delta, \quad \mathcal{E}_{\delta\sigma} = (\mathcal{E}_\delta)_\sigma, \quad \mathcal{E}^c = \{E^c : E \in \mathcal{E}\}.$$

Thus $\mathcal{E}_\sigma$ and $\mathcal{E}_\delta$ are the families of countable unions and countable intersections of members of $\mathcal{E}$; $\mathcal{E}_{\sigma\delta}$ and $\mathcal{E}_{\delta\sigma}$ apply these operations in succession. Then $(\mathcal{E}_\sigma)^c = (\mathcal{E}^c)_\delta$, $(\mathcal{E}_\delta)^c = (\mathcal{E}^c)_\sigma$, and $(\mathcal{E}_{\sigma\delta})^c = (\mathcal{E}^c)_{\delta\sigma}$.

5.4.3. PROBLEM (Induced outer measure). Under the hypotheses in the preceding remark, prove that μ^* is an outer measure on X . Where are the assumptions $\emptyset, X \in \mathcal{E}$ used?

Solution. ADMISSIBLE COVERS EXIST. Since $X \in \mathcal{E}$, every $A \subseteq X$ has an admissible $\mathcal{E}$ -cover, namely

$$A \subseteq \bigcup_{j \in \mathbb{N}} X.$$

Thus the infimum defining $\mu^*(A)$ is taken over a nonempty subset of $[0, \infty]$, so $\mu^*(A)$ is well defined.

THE EMPTY SET. Since $\emptyset \in \mathcal{E}$ and $\rho(\emptyset) = 0$, the sequence $E_j = \emptyset$ covers $\emptyset$. Therefore

$$\mu^*(\emptyset) \leq \sum_{j \in \mathbb{N}} \rho(\emptyset) = 0.$$

Since $\mu^*(\emptyset) \geq 0$,

$$\mu^*(\emptyset) = 0.$$

MONOTONICITY. Let $A \subseteq B \subseteq X$. Every admissible cover of B is also an admissible cover of A . By the definition of the infimum,

$$\mu^*(A) \leq \mu^*(B).$$

COMBINING NEARLY OPTIMAL COVERS. Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}(X)$. If

$$\sum_{n \in \mathbb{N}} \mu^*(A_n) = \infty,$$

then

$$\mu^* \left(\bigcup_{n \in \mathbb{N}} A_n \right) \leq \infty = \sum_{n \in \mathbb{N}} \mu^*(A_n).$$

Assume now that

$$\sum_{n \in \mathbb{N}} \mu^*(A_n) < \infty,$$

and let $\varepsilon > 0$. Then $\mu^*(A_n) < \infty$ for every n . By the defining property of the infimum, choose $(E_{n,j})_{j \in \mathbb{N}} \subseteq \mathcal{E}$ such that

$$A_n \subseteq \bigcup_{j \in \mathbb{N}} E_{n,j}, \quad \sum_{j \in \mathbb{N}} \rho(E_{n,j}) < \mu^*(A_n) + \frac{\varepsilon}{2^n}.$$

Then

$$\bigcup_{n \in \mathbb{N}} A_n \subseteq \bigcup_{(n,j) \in \mathbb{N} \times \mathbb{N}} E_{n,j}.$$

Since $\mathbb{N} \times \mathbb{N}$ is countable, this is an admissible sequence in $\mathcal{E}$. Hence

$$\begin{aligned} \mu^* \left(\bigcup_{n \in \mathbb{N}} A_n \right) &\leq \sum_{n \in \mathbb{N}} \sum_{j \in \mathbb{N}} \rho(E_{n,j}) \\ &< \sum_{n \in \mathbb{N}} \left(\mu^*(A_n) + \frac{\varepsilon}{2^n} \right) \\ &= \sum_{n \in \mathbb{N}} \mu^*(A_n) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu^* \left(\bigcup_{n \in \mathbb{N}} A_n \right) \leq \sum_{n \in \mathbb{N}} \mu^*(A_n).$$

Therefore μ^* is an outer measure.

USE OF THE ASSUMPTIONS. The assumption $X \in \mathcal{E}$ guarantees that every subset of X has an admissible cover, so the defining infimum is meaningful.

The assumptions $\emptyset \in \mathcal{E}$ and $\rho(\emptyset) = 0$ give the zero-cost cover of $\emptyset$, which proves

$$\mu^*(\emptyset) = 0.$$

□

5.4.4. PROBLEM. Let $\mathcal{E} \subseteq \mathcal{P}(X)$. Prove

$$(\mathcal{E}_\sigma)^c = (\mathcal{E}^c)_\delta, \quad (\mathcal{E}_\delta)^c = (\mathcal{E}^c)_\sigma, \quad (\mathcal{E}_{\sigma\delta})^c = (\mathcal{E}^c)_{\delta\sigma}.$$

Solution. COMPLEMENTS OF COUNTABLE UNIONS. We prove both inclusions.

$$(\mathcal{E}_\sigma)^c = (\mathcal{E}^c)_\delta.$$

($\subseteq$) Let $A \in (\mathcal{E}_\sigma)^c$. By the definition of $(\mathcal{E}_\sigma)^c$, there are $E_n \in \mathcal{E}$ such that

$$A = \left(\bigcup_{n \in \mathbb{N}} E_n \right)^c.$$

By De Morgan's law,

$$A = \bigcap_{n \in \mathbb{N}} E_n^c.$$

Since $E_n^c \in \mathcal{E}^c$ for every n ,

$$A \in (\mathcal{E}^c)_\delta.$$

($\supseteq$) Let $A \in (\mathcal{E}^c)_\delta$. By definition, there are $F_n \in \mathcal{E}^c$ such that

$$A = \bigcap_{n \in \mathbb{N}} F_n.$$

By the definition of $\mathcal{E}^c$, write

$$F_n = E_n^c \quad \text{for some } E_n \in \mathcal{E}.$$

Then De Morgan's law gives

$$A = \bigcap_{n \in \mathbb{N}} E_n^c = \left(\bigcup_{n \in \mathbb{N}} E_n \right)^c.$$

Since $\bigcup_n E_n \in \mathcal{E}_\sigma$,

$$A \in (\mathcal{E}_\sigma)^c.$$

Hence

$$(\mathcal{E}_\sigma)^c = (\mathcal{E}^c)_\delta.$$

COMPLEMENTS OF COUNTABLE INTERSECTIONS. We prove both inclusions.

$$(\mathcal{E}_\delta)^c = (\mathcal{E}^c)_\sigma.$$

($\subseteq$) Let $A \in (\mathcal{E}_\delta)^c$. Then

$$A = \left(\bigcap_{n \in \mathbb{N}} E_n \right)^c = \bigcup_{n \in \mathbb{N}} E_n^c$$

for some $E_n \in \mathcal{E}$. Since $E_n^c \in \mathcal{E}^c$,

$$A \in (\mathcal{E}^c)_\sigma.$$

($\supseteq$) Let $A \in (\mathcal{E}^c)_\sigma$. Then

$$A = \bigcup_{n \in \mathbb{N}} F_n$$

for some $F_n \in \mathcal{E}^c$. Write $F_n = E_n^c$ with $E_n \in \mathcal{E}$. By De Morgan's law,

$$A = \bigcup_{n \in \mathbb{N}} E_n^c = \left(\bigcap_{n \in \mathbb{N}} E_n \right)^c.$$

Thus

$$A \in (\mathcal{E}_\delta)^c.$$

Hence

$$(\mathcal{E}_\delta)^c = (\mathcal{E}^c)_\sigma.$$

COMPLEMENTS AFTER SUCCESSIVE OPERATIONS. By the second identity, applied to the family $\mathcal{E}_\sigma$,

$$\begin{aligned} (\mathcal{E}_{\sigma\delta})^c &= ((\mathcal{E}_\sigma)_\delta)^c \\ &= ((\mathcal{E}_\sigma)^c)_\sigma. \end{aligned}$$

By the first identity,

$$(\mathcal{E}_\sigma)^c = (\mathcal{E}^c)_\delta.$$

Therefore

$$(\mathcal{E}_{\sigma\delta})^c = ((\mathcal{E}^c)_\delta)_\sigma = (\mathcal{E}^c)_{\delta\sigma}.$$

□

5.4.5. Remark: *The Lebesgue outer measure in this framework.* The Lebesgue outer measure λ_d^* is induced by the volume function v_d on $\mathcal{E} = \mathcal{J}_o^d$. In this case $\mathcal{E}_{\sigma\delta}$ plays the role of the G_δ sets, while $(\mathcal{E}^c)_{\delta\sigma}$ plays the role of the F_σ sets.

5.4.6. Note: *Standing assumptions.* Let $X \neq \emptyset$, $\emptyset, X \in \mathcal{E} \subseteq \mathcal{P}(X)$, and $\rho : \mathcal{E} \rightarrow [0, \infty]$ satisfy $\rho(\emptyset) = 0$. Let μ^* be the induced outer measure

$$\mu^*(A) = \inf \left\{ \sum_{j \in \mathbb{N}} \rho(E_j) : (E_j)_{j \in \mathbb{N}} \subseteq \mathcal{E}, A \subseteq \bigcup_{j \in \mathbb{N}} E_j \right\}.$$

In the following problems, the hypotheses used are indicated among:

- (a) $\mu^* = \rho$ on $\mathcal{E}$;
- (b) $\mathcal{E} \subseteq \mathcal{M}(\mu^*)$;
- (c) μ^* is σ -finite on $(X, \mathcal{M}(\mu^*), \mu^*)$.

5.4.7. PROBLEM (Outer approximation). Assume $\mu^* = \rho$ on $\mathcal{E}$. Prove:

- (a) for every $A \subseteq X$ and $\varepsilon > 0$, there is $U \supseteq A$ with $U \in \mathcal{E}_\sigma$ and

$$\mu^*(A) \leq \mu^*(U) \leq \mu^*(A) + \varepsilon;$$

- (b) for every $A \subseteq X$, there is $B \supseteq A$ with $B \in \mathcal{E}_{\sigma\delta}$ and $\mu^*(B) = \mu^*(A)$.

Solution. (a) A NEARLY OPTIMAL COVER BECOMES A COUNTABLE UNION. Let $A \subseteq X$ and $\varepsilon > 0$.

First suppose that $\mu^*(A) < \infty$. By the definition of the infimum, choose $(E_j)_{j \in \mathbb{N}} \subseteq \mathcal{E}$ such that

$$A \subseteq \bigcup_{j \in \mathbb{N}} E_j, \quad \sum_{j \in \mathbb{N}} \rho(E_j) < \mu^*(A) + \varepsilon.$$

Define

$$U = \bigcup_{j \in \mathbb{N}} E_j.$$

By the definition of $\mathcal{E}_\sigma$,

$$U \in \mathcal{E}_\sigma, \quad A \subseteq U.$$

By monotonicity,

$$\mu^*(A) \leq \mu^*(U).$$

By countable subadditivity and the hypothesis $\mu^* = \rho$ on $\mathcal{E}$,

$$\begin{aligned}\mu^*(U) &\leq \sum_{j \in \mathbb{N}} \mu^*(E_j) \\ &= \sum_{j \in \mathbb{N}} \rho(E_j) \\ &< \mu^*(A) + \varepsilon.\end{aligned}$$

Hence

$$\mu^*(A) \leq \mu^*(U) \leq \mu^*(A) + \varepsilon.$$

If $\mu^*(A) = \infty$, take $U = X$. Since $X \in \mathcal{E}$, repeating X gives $X \in \mathcal{E}_\sigma$. By monotonicity,

$$\infty = \mu^*(A) \leq \mu^*(X),$$

so $\mu^*(X) = \infty$. Therefore

$$\mu^*(A) \leq \mu^*(X) \leq \mu^*(A) + \varepsilon.$$

(b) INTERSECT COUNTABLY MANY OUTER APPROXIMATIONS. First suppose that $\mu^*(A) < \infty$. By part (a), for every $n \in \mathbb{N}$, choose $U_n \in \mathcal{E}_\sigma$ such that

$$A \subseteq U_n, \quad \mu^*(A) \leq \mu^*(U_n) \leq \mu^*(A) + \frac{1}{n}.$$

Define

$$B = \bigcap_{n \in \mathbb{N}} U_n.$$

By the definition of $\mathcal{E}_{\sigma\delta}$,

$$B \in \mathcal{E}_{\sigma\delta}.$$

Since $A \subseteq U_n$ for every n ,

$$A \subseteq B.$$

By monotonicity,

$$\mu^*(A) \leq \mu^*(B).$$

Since $B \subseteq U_n$ for every n , monotonicity also gives

$$\mu^*(B) \leq \mu^*(U_n) \leq \mu^*(A) + \frac{1}{n} \quad (n \in \mathbb{N}).$$

Hence

$$\mu^*(B) = \mu^*(A).$$

If $\mu^*(A) = \infty$, take $B = X$. Since $X \in \mathcal{E}$, we have $X \in \mathcal{E}_\sigma$, and therefore

$$X = \bigcap_{n \in \mathbb{N}} X \in \mathcal{E}_{\sigma\delta}.$$

By monotonicity,

$$\mu^*(B) = \mu^*(X) = \infty = \mu^*(A).$$

□

5.4.8. PROBLEM (Outer regularity). Assume $\mu^* = \rho$ on $\mathcal{E}$. Prove that μ^* is outer regular with respect to $\mathcal{E}_\sigma$:

$$\mu^*(E) = \inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\} \quad (E \subseteq X).$$

Solution. EVERY ADMISSIBLE ENLARGEMENT HAS AT LEAST THE OUTER MEASURE OF E .
Let

$$U \supseteq E, \quad U \in \mathcal{E}_\sigma.$$

By monotonicity of μ^* ,

$$\mu^*(E) \leq \mu^*(U).$$

Taking the infimum over all such U ,

$$\mu^*(E) \leq \inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\}.$$

OUTER APPROXIMATION GIVES THE REVERSE INEQUALITY. Let $\varepsilon > 0$. By the preceding outer-approximation result, there is $U_\varepsilon \in \mathcal{E}_\sigma$ such that

$$E \subseteq U_\varepsilon, \quad \mu^*(U_\varepsilon) \leq \mu^*(E) + \varepsilon.$$

Therefore

$$\inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\} \leq \mu^*(U_\varepsilon) \leq \mu^*(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\} \leq \mu^*(E).$$

Hence

$$\mu^*(E) = \inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\}.$$

□

5.4.9. PROBLEM (Criterion). Assume all three standing hypotheses. Prove that, for $E \subseteq X$, the following are equivalent:

- (a) $E \in \mathcal{M}(\mu^*)$;
- (b) for every $\varepsilon > 0$, there is $U \supseteq E$ with $U \in \mathcal{E}_\sigma$ and $\mu^*(U \setminus E) \leq \varepsilon$;
- (c) there is $G \supseteq E$ with $G \in \mathcal{E}_{\sigma\delta}$ and $\mu^*(G \setminus E) = 0$;
- (d) for every $\varepsilon > 0$, there is $C \subseteq E$ with $C \in (\mathcal{E}^c)_\delta$ and $\mu^*(E \setminus C) \leq \varepsilon$;
- (e) there is $F \subseteq E$ with $F \in (\mathcal{E}^c)_{\delta\sigma}$ and $\mu^*(E \setminus F) = 0$.

Solution. We prove the implications

$$(1) \implies (2) \implies (3) \implies (1), \quad (1) \implies (4) \implies (5) \implies (1).$$

MEASURABLE SETS HAVE ARBITRARILY SMALL OUTER ENLARGEMENTS. Assume (1), so $E \in \mathcal{M}(\mu^*)$. By σ -finiteness, choose $(X_m)_{m \in \mathbb{N}} \subseteq \mathcal{M}(\mu^*)$ such that

$$X = \bigcup_{m \in \mathbb{N}} X_m, \quad \mu^*(X_m) < \infty.$$

Put

$$E_m = E \cap X_m.$$

Since $E, X_m \in \mathcal{M}(\mu^*)$,

$$E_m \in \mathcal{M}(\mu^*), \quad \mu^*(E_m) \leq \mu^*(X_m) < \infty.$$

Let $\varepsilon > 0$. By outer approximation, for every m , there is $U_m \in \mathcal{E}_\sigma$ such that

$$E_m \subseteq U_m, \quad \mu^*(U_m) \leq \mu^*(E_m) + \frac{\varepsilon}{2^m}.$$

Since $\mathcal{E} \subseteq \mathcal{M}(\mu^*)$, the set $U_m \in \mathcal{E}_\sigma$ is measurable. Hence, by finite additivity,

$$\mu^*(U_m \setminus E_m) = \mu^*(U_m) - \mu^*(E_m) \leq \frac{\varepsilon}{2^m}.$$

Define

$$U = \bigcup_{m \in \mathbb{N}} U_m.$$

Since every U_m is a countable union of members of $\mathcal{E}$, the countable-flattening technique gives

$$U \in \mathcal{E}_\sigma.$$

Since $X = \bigcup_m X_m$,

$$E = \bigcup_{m \in \mathbb{N}} E_m \subseteq U.$$

Moreover,

$$U \setminus E \subseteq \bigcup_{m \in \mathbb{N}} (U_m \setminus E_m).$$

Thus, by countable subadditivity,

$$\mu^*(U \setminus E) \leq \sum_{m \in \mathbb{N}} \mu^*(U_m \setminus E_m) \leq \sum_{m \in \mathbb{N}} \frac{\varepsilon}{2^m} = \varepsilon.$$

Hence (1) $\Rightarrow$ (2).

INTERSECT OUTER APPROXIMATIONS. Assume (2). For every $n \in \mathbb{N}$, choose $U_n \in \mathcal{E}_\sigma$ such that

$$E \subseteq U_n, \quad \mu^*(U_n \setminus E) \leq \frac{1}{n}.$$

Define

$$G = \bigcap_{n \in \mathbb{N}} U_n.$$

Then

$$G \in \mathcal{E}_{\sigma\delta}, \quad E \subseteq G.$$

Since

$$G \setminus E \subseteq U_n \setminus E \quad (n \in \mathbb{N}),$$

monotonicity gives

$$\mu^*(G \setminus E) \leq \frac{1}{n} \quad (n \in \mathbb{N}).$$

Therefore

$$\mu^*(G \setminus E) = 0.$$

Hence (2) $\Rightarrow$ (3).

REMOVE A NULL DIFFERENCE. Assume (3). Since $\mathcal{E} \subseteq \mathcal{M}(\mu^*)$ and $\mathcal{M}(\mu^*)$ is a σ -algebra,

$$G \in \mathcal{E}_{\sigma\delta} \implies G \in \mathcal{M}(\mu^*).$$

Since $\mu^*(G \setminus E) = 0$, completeness of the Carathéodory measure gives

$$G \setminus E \in \mathcal{M}(\mu^*).$$

Thus

$$E = G \setminus (G \setminus E) \in \mathcal{M}(\mu^*).$$

Hence (3) $\Rightarrow$ (1).

INNER APPROXIMATION BY COMPLEMENTS. Assume (1). Then $E^c \in \mathcal{M}(\mu^*)$. By (1) $\Rightarrow$ (2), for every $\varepsilon > 0$, there is $U \in \mathcal{E}_\sigma$ such that

$$E^c \subseteq U, \quad \mu^*(U \setminus E^c) \leq \varepsilon.$$

Put

$$C = U^c.$$

By the complement identity proved previously,

$$C \in (\mathcal{E}_\sigma)^c = (\mathcal{E}^c)_\delta.$$

Since $E^c \subseteq U$,

$$C \subseteq E.$$

Moreover,

$$E \setminus C = E \cap U = U \setminus E^c.$$

Therefore

$$\mu^*(E \setminus C) \leq \varepsilon.$$

Hence (1) $\Rightarrow$ (4).

UNION OF INNER APPROXIMATIONS. Assume (4). For every $n \in \mathbb{N}$, choose $C_n \in (\mathcal{E}^c)_\delta$ such that

$$C_n \subseteq E, \quad \mu^*(E \setminus C_n) \leq \frac{1}{n}.$$

Define

$$F = \bigcup_{n \in \mathbb{N}} C_n.$$

Then

$$F \in (\mathcal{E}^c)_{\delta\sigma}, \quad F \subseteq E.$$

Since

$$E \setminus F \subseteq E \setminus C_n \quad (n \in \mathbb{N}),$$

we obtain

$$\mu^*(E \setminus F) \leq \frac{1}{n} \quad (n \in \mathbb{N}).$$

Thus

$$\mu^*(E \setminus F) = 0.$$

Hence (4) $\Rightarrow$ (5).

ADD BACK A NULL DIFFERENCE. Assume (5). Since $\mathcal{E} \subseteq \mathcal{M}(\mu^*)$ and $\mathcal{M}(\mu^*)$ is a σ -algebra,

$$F \in (\mathcal{E}^c)_{\delta\sigma} \implies F \in \mathcal{M}(\mu^*).$$

Since $\mu^*(E \setminus F) = 0$, completeness gives

$$E \setminus F \in \mathcal{M}(\mu^*).$$

Therefore

$$E = F \cup (E \setminus F) \in \mathcal{M}(\mu^*).$$

Hence (5) $\Rightarrow$ (1). □

5.4.10. PROBLEM (Outer and inner regularity). Assume all three standing hypotheses. Prove that for $E \in \mathcal{M}(\mu^*)$,

$$\mu^*(E) = \inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\}$$

and

$$\mu^*(E) = \sup\{\mu^*(C) : C \subseteq E, C \in (\mathcal{E}^c)_\delta\}.$$

Compare this with the regularity theorem for λ_d .

Solution. OUTER REGULARITY. Let

$$U \supseteq E, \quad U \in \mathcal{E}_\sigma.$$

By monotonicity,

$$\mu^*(E) \leq \mu^*(U).$$

Therefore

$$\mu^*(E) \leq \inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\}.$$

For the reverse inequality, let $\varepsilon > 0$. Since $E \in \mathcal{M}(\mu^*)$, the Criterion gives $U_\varepsilon \in \mathcal{E}_\sigma$ such that

$$E \subseteq U_\varepsilon, \quad \mu^*(U_\varepsilon \setminus E) \leq \varepsilon.$$

Since $\mathcal{E} \subseteq \mathcal{M}(\mu^*)$,

$$U_\varepsilon \in \mathcal{M}(\mu^*).$$

Thus finite additivity gives

$$\mu^*(U_\varepsilon) = \mu^*(E) + \mu^*(U_\varepsilon \setminus E) \leq \mu^*(E) + \varepsilon.$$

Hence

$$\inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\} \leq \mu^*(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu^*(E) = \inf\{\mu^*(U) : U \supseteq E, U \in \mathcal{E}_\sigma\}.$$

INNER REGULARITY BY COMPLEMENTARY OUTER APPROXIMATION. Let

$$C \subseteq E, \quad C \in (\mathcal{E}^c)_\delta.$$

Since $\mathcal{E} \subseteq \mathcal{M}(\mu^*)$ and $\mathcal{M}(\mu^*)$ is a σ -algebra,

$$C \in \mathcal{M}(\mu^*).$$

By monotonicity,

$$\mu^*(C) \leq \mu^*(E).$$

Therefore

$$\sup\{\mu^*(C) : C \subseteq E, C \in (\mathcal{E}^c)_\delta\} \leq \mu^*(E).$$

For the reverse inequality, let $\varepsilon > 0$. By the Criterion, there is $C_\varepsilon \in (\mathcal{E}^c)_\delta$ such that

$$C_\varepsilon \subseteq E, \quad \mu^*(E \setminus C_\varepsilon) \leq \varepsilon.$$

Since C_ε and $E \setminus C_\varepsilon$ are measurable and disjoint,

$$\mu^*(E) = \mu^*(C_\varepsilon) + \mu^*(E \setminus C_\varepsilon) \leq \mu^*(C_\varepsilon) + \varepsilon.$$

Hence

$$\mu^*(E) \leq \sup\{\mu^*(C) : C \subseteq E, C \in (\mathcal{E}^c)_\delta\} + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu^*(E) = \sup\{\mu^*(C) : C \subseteq E, C \in (\mathcal{E}^c)_\delta\}.$$

COMPARISON WITH LEBESGUE MEASURE. For Lebesgue measure, take

$$X = \mathbb{R}^d, \quad \mathcal{E} = \mathcal{J}_o^d, \quad \rho = \nu_d.$$

Then $\mathcal{E}_\sigma$ is the family of open subsets of $\mathbb{R}^d$, while

$$(\mathcal{E}^c)_\delta = (\mathcal{E}_\sigma)^c$$

is the family of closed subsets of $\mathbb{R}^d$. Thus the two formulas become

$$\lambda_d(E) = \inf_{\substack{U \supseteq E \\ U \text{ open}}} \lambda_d(U), \quad \lambda_d(E) = \sup_{\substack{C \subseteq E \\ C \text{ closed}}} \lambda_d(C),$$

which are exactly the outer and inner regularity statements for Lebesgue measure. $\square$

5.4.11. PROBLEM (Approximation). Assume the first two standing hypotheses and that

$$\mu^*(X) < \infty.$$

Show that, for every $E \in \mathcal{M}(\mu^*)$ and every $\varepsilon > 0$, there is $A \in \mathcal{E}_\sigma$ such that

$$\mu^*(E \Delta A) < \varepsilon.$$

Show moreover that A can be chosen to be a finite union of members of $\mathcal{E}$.

For $X = \mathbb{R}^d$, $\mathcal{E} = \mathcal{J}_o^d$, and $\rho = \nu_d$, the corresponding statement is: if

$$E \in \mathcal{L}(\mathbb{R}^d), \quad \lambda_d(E) < \infty,$$

then, for every $\varepsilon > 0$, there is a finite union A of open rectangles such that

$$\lambda_d(E \Delta A) < \varepsilon.$$

Solution. FIRST ENLARGE E BY A SMALL ERROR. Let $E \in \mathcal{M}(\mu^*)$ and $\varepsilon > 0$. By outer approximation, there is $U \in \mathcal{E}_\sigma$ such that

$$E \subseteq U, \quad \mu^*(U) \leq \mu^*(E) + \frac{\varepsilon}{2}.$$

Since $\mathcal{E} \subseteq \mathcal{M}(\mu^*)$,

$$U \in \mathcal{M}(\mu^*).$$

Since the measure space is finite, $\mu^*(E) < \infty$. Thus finite additivity gives

$$\mu^*(U \setminus E) = \mu^*(U) - \mu^*(E) \leq \frac{\varepsilon}{2}.$$

TRUNCATE THE COUNTABLE UNION. By the definition of $\mathcal{E}_\sigma$, write

$$U = \bigcup_{n \in \mathbb{N}} E_n, \quad E_n \in \mathcal{E}.$$

For $N \in \mathbb{N}$, define

$$A_N = \bigcup_{n=1}^N E_n.$$

Then $A_N \subseteq U$, each A_N is a finite union of members of $\mathcal{E}$, and

$$A_N \uparrow U.$$

Since $\mu^*(U) \leq \mu^*(X) < \infty$, continuity from below gives

$$\mu^*(U) = \lim_{N \rightarrow \infty} \mu^*(A_N).$$

Therefore, for some N ,

$$\mu^*(U \setminus A_N) = \mu^*(U) - \mu^*(A_N) < \frac{\varepsilon}{2}.$$

Put

$$A = A_N.$$

Since $\emptyset \in \mathcal{E}$, every finite union of members of $\mathcal{E}$ belongs to $\mathcal{E}_\sigma$. Hence

$$A \in \mathcal{E}_\sigma.$$

CONTROL BOTH SIDES OF THE SYMMETRIC DIFFERENCE. Since $E \subseteq U$ and $A \subseteq U$,

$$E \setminus A \subseteq U \setminus A, \quad A \setminus E \subseteq U \setminus E.$$

The two parts of $E \Delta A$ are disjoint. Therefore

$$\begin{aligned} \mu^*(E \Delta A) &= \mu^*(E \setminus A) + \mu^*(A \setminus E) \\ &\leq \mu^*(U \setminus A) + \mu^*(U \setminus E) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Thus $A \in \mathcal{E}_\sigma$ is a finite union of members of $\mathcal{E}$ and satisfies

$$\mu^*(E \Delta A) < \varepsilon.$$

For $X = \mathbb{R}^d$, $\mathcal{E} = \mathcal{J}_o^d$, and $\rho = \nu_d$, the same proof applies to every Lebesgue measurable E with $\lambda_d(E) < \infty$. It gives approximation by finite unions of open rectangles. $\square$

5.4.12. Theorem (Approximation for Lebesgue–Stieltjes measures). — *If $E \in \mathcal{L}_{\mu_F}$ and $\mu_F(E) < \infty$, then for every $\varepsilon > 0$ there is a finite union A of open intervals such that*

$$\mu_F(E \Delta A) < \varepsilon.$$

Proof. FIRST ENLARGE E BY A SMALL OPEN ERROR. Let $\varepsilon > 0$. By the outer regularity of μ_F , choose an open set $O \supseteq E$ such that

$$\mu_F(O \setminus E) < \frac{\varepsilon}{2}.$$

Since $\mu_F(E) < \infty$,

$$\mu_F(O) = \mu_F(E) + \mu_F(O \setminus E) < \infty.$$

TRUNCATE A COUNTABLE INTERVAL DECOMPOSITION OF O . Every open subset of $\mathbb{R}$ is a countable union of open intervals. Write

$$O = \bigcup_{n \in \mathbb{N}} I_n,$$

where every I_n is an open interval, and define

$$A_N = \bigcup_{n=1}^N I_n.$$

Then A_N is a finite union of open intervals and

$$A_N \uparrow O.$$

By continuity from below,

$$\mu_F(O) = \lim_{N \rightarrow \infty} \mu_F(A_N).$$

Since $\mu_F(O) < \infty$, choose N such that

$$\mu_F(O \setminus A_N) = \mu_F(O) - \mu_F(A_N) < \frac{\varepsilon}{2}.$$

Put

$$A = A_N.$$

CONTROL THE SYMMETRIC DIFFERENCE. Since $E \subseteq O$ and $A \subseteq O$,

$$E \setminus A \subseteq O \setminus A, \quad A \setminus E \subseteq O \setminus E.$$

Therefore

$$\begin{aligned} \mu_F(E \Delta A) &= \mu_F(E \setminus A) + \mu_F(A \setminus E) \\ &\leq \mu_F(O \setminus A) + \mu_F(O \setminus E) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

□

5.5 Equivalent Forms of Regularity

5.5.1. PROBLEM (Outer regularity). Let $(X, \mathcal{M}, \mu)$ be a measure space and $\emptyset, X \in \mathcal{E} \subseteq \mathcal{M}$. Consider:

- (a) for every $E \in \mathcal{M}$ and $\varepsilon > 0$, there is $G \in \mathcal{E}$ with $G \supseteq E$ and $\mu(G \setminus E) \leq \varepsilon$;
- (b) $\mu(E) = \inf\{\mu(U) : U \supseteq E, U \in \mathcal{E}\}$ for every $E \in \mathcal{M}$.

Prove that (a) implies (b); that (b) implies (a) if $\mu(X) < \infty$; and that (b) implies (a) if μ is σ -finite and $\mathcal{E}$ is closed under countable unions.

Solution. (a) $\implies$ (b): AN OUTER APPROXIMATION GIVES THE INFIMUM. Assume (a), and let $E \in \mathcal{M}$. For every $U \in \mathcal{E}$ with $E \subseteq U$, monotonicity gives

$$\mu(E) \leq \mu(U).$$

Therefore

$$\mu(E) \leq \inf\{\mu(U) : U \supseteq E, U \in \mathcal{E}\}.$$

Let $\varepsilon > 0$. By (a), choose $G \in \mathcal{E}$ such that

$$E \subseteq G, \quad \mu(G \setminus E) \leq \varepsilon.$$

Since

$$G = E \bigsqcup (G \setminus E),$$

finite additivity gives

$$\mu(G) = \mu(E) + \mu(G \setminus E) \leq \mu(E) + \varepsilon.$$

Hence

$$\inf\{\mu(U) : U \supseteq E, U \in \mathcal{E}\} \leq \mu(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu(E) = \inf\{\mu(U) : U \supseteq E, U \in \mathcal{E}\}.$$

(b) $\implies$ (a) WHEN $\mu(X) < \infty$. Assume (b) and $\mu(X) < \infty$. Let $E \in \mathcal{M}$ and $\varepsilon > 0$. Since $E \subseteq X$, monotonicity gives

$$\mu(E) \leq \mu(X) < \infty.$$

By (b), choose $G \in \mathcal{E}$ such that

$$E \subseteq G, \quad \mu(G) < \mu(E) + \varepsilon.$$

Since

$$G = E \bigsqcup (G \setminus E),$$

finite additivity gives

$$\mu(G \setminus E) = \mu(G) - \mu(E) < \varepsilon.$$

Thus (a) holds.

(b) $\implies$ (a) BY σ -FINITE LOCALIZATION. Assume (b), assume that μ is σ -finite, and assume that $\mathcal{E}$ is closed under countable unions. Choose $(X_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$ such that

$$X = \bigcup_{n \in \mathbb{N}} X_n, \quad \mu(X_n) < \infty.$$

Let $E \in \mathcal{M}$ and $\varepsilon > 0$. For every n ,

$$E \cap X_n \in \mathcal{M}, \quad \mu(E \cap X_n) \leq \mu(X_n) < \infty.$$

By (b), choose $G_n \in \mathcal{E}$ such that

$$E \cap X_n \subseteq G_n, \quad \mu(G_n) < \mu(E \cap X_n) + \frac{\varepsilon}{2^n}.$$

By finite additivity,

$$\mu(G_n \setminus (E \cap X_n)) = \mu(G_n) - \mu(E \cap X_n) < \frac{\varepsilon}{2^n}.$$

Define

$$G = \bigcup_{n \in \mathbb{N}} G_n.$$

By the assumed closure of $\mathcal{E}$ under countable unions,

$$G \in \mathcal{E}.$$

Since $X = \bigcup_n X_n$,

$$E = \bigcup_{n \in \mathbb{N}} (E \cap X_n) \subseteq G.$$

Moreover,

$$G \setminus E \subseteq \bigcup_{n \in \mathbb{N}} (G_n \setminus (E \cap X_n)).$$

Therefore, by countable subadditivity,

$$\begin{aligned} \mu(G \setminus E) &\leq \sum_{n \in \mathbb{N}} \mu(G_n \setminus (E \cap X_n)) \\ &< \sum_{n \in \mathbb{N}} \frac{\varepsilon}{2^n} = \varepsilon. \end{aligned}$$

Thus (a) holds. □

5.5.2. PROBLEM (Inner regularity). Let $(X, \mathcal{M}, \mu)$ be a measure space and $\emptyset, X \in \mathcal{E} \subseteq \mathcal{M}$. Consider:

- (a) for every $E \in \mathcal{M}$ and $\varepsilon > 0$, there is $F \in \mathcal{E}$ with $F \subseteq E$ and $\mu(E \setminus F) \leq \varepsilon$;
- (b) $\mu(E) = \sup\{\mu(C) : C \subseteq E, C \in \mathcal{E}\}$ for every $E \in \mathcal{M}$.

Prove that (a) implies (b); that (b) implies (a) if $\mu(X) < \infty$; and that (b) implies (a) if μ is σ -finite and $\mathcal{E}$ is closed under countable unions.

Solution. (a) $\implies$ (b): AN INNER APPROXIMATION GIVES THE SUPREMUM. Assume (a), and let $E \in \mathcal{M}$. For every $C \in \mathcal{E}$ with $C \subseteq E$, monotonicity gives

$$\mu(C) \leq \mu(E).$$

Therefore

$$\sup\{\mu(C) : C \subseteq E, C \in \mathcal{E}\} \leq \mu(E).$$

Let $\varepsilon > 0$. By (a), choose $F \in \mathcal{E}$ such that

$$F \subseteq E, \quad \mu(E \setminus F) \leq \varepsilon.$$

Since

$$E = F \bigsqcup (E \setminus F),$$

finite additivity gives

$$\mu(E) = \mu(F) + \mu(E \setminus F) \leq \mu(F) + \varepsilon.$$

Hence

$$\mu(E) \leq \sup\{\mu(C) : C \subseteq E, C \in \mathcal{E}\} + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu(E) = \sup\{\mu(C) : C \subseteq E, C \in \mathcal{E}\}.$$

(b) $\implies$ (a) WHEN $\mu(X) < \infty$. Assume (b) and $\mu(X) < \infty$. Let $E \in \mathcal{M}$ and $\varepsilon > 0$. Since $E \subseteq X$,

$$\mu(E) \leq \mu(X) < \infty.$$

By (b), choose $F \in \mathcal{E}$ such that

$$F \subseteq E, \quad \mu(F) > \mu(E) - \varepsilon.$$

Since

$$E = F \sqcup (E \setminus F),$$

finite additivity gives

$$\mu(E \setminus F) = \mu(E) - \mu(F) < \varepsilon.$$

Thus (a) holds.

(b) $\implies$ (a) BY σ -FINITE LOCALIZATION. Assume (b), assume that μ is σ -finite, and assume that $\mathcal{E}$ is closed under countable unions. Choose $(X_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$ such that

$$X = \bigcup_{n \in \mathbb{N}} X_n, \quad \mu(X_n) < \infty.$$

Let $E \in \mathcal{M}$ and $\varepsilon > 0$. For every n ,

$$E \cap X_n \in \mathcal{M}, \quad \mu(E \cap X_n) < \infty.$$

By (b), choose $F_n \in \mathcal{E}$ such that

$$F_n \subseteq E \cap X_n, \quad \mu(F_n) > \mu(E \cap X_n) - \frac{\varepsilon}{2^n}.$$

By finite additivity,

$$\mu((E \cap X_n) \setminus F_n) = \mu(E \cap X_n) - \mu(F_n) < \frac{\varepsilon}{2^n}.$$

Define

$$F = \bigcup_{n \in \mathbb{N}} F_n.$$

By the assumed closure of $\mathcal{E}$ under countable unions,

$$F \in \mathcal{E}.$$

Since $F_n \subseteq E$ for every n ,

$$F \subseteq E.$$

Moreover,

$$E \setminus F \subseteq \bigcup_{n \in \mathbb{N}} ((E \cap X_n) \setminus F_n).$$

Therefore, by countable subadditivity,

$$\begin{aligned} \mu(E \setminus F) &\leq \sum_{n \in \mathbb{N}} \mu((E \cap X_n) \setminus F_n) \\ &< \sum_{n \in \mathbb{N}} \frac{\varepsilon}{2^n} = \varepsilon. \end{aligned}$$

Thus (a) holds. $\square$

5.5.3. PROBLEM (Approximation modulo null sets). Let $(X, \mathcal{M}, \mu)$ be σ -finite and let $\emptyset, X \in \mathcal{E} \subseteq \mathcal{M}$.

- (a) If μ is outer regular with respect to $\mathcal{E}$ and $\mathcal{E}$ is closed under countable unions, prove that for every $E \in \mathcal{M}$ there is $G \in \mathcal{E}_\delta$ with $G \supseteq E$ and $\mu(G \setminus E) = 0$.
- (b) If μ is inner regular with respect to $\mathcal{E}$, prove that for every $E \in \mathcal{M}$ there is $F \in \mathcal{E}_\sigma$ with $F \subseteq E$ and $\mu(E \setminus F) = 0$.

Solution. (a) INTERSECT OUTER APPROXIMATIONS. By the preceding outer-regularity result, the assumptions of σ -finiteness and closure of $\mathcal{E}$ under countable unions imply the following approximation property: for every $E \in \mathcal{M}$ and every $\varepsilon > 0$, there is $U \in \mathcal{E}$ such that

$$E \subseteq U, \quad \mu(U \setminus E) \leq \varepsilon.$$

Let $E \in \mathcal{M}$. For every $n \in \mathbb{N}$, choose $U_n \in \mathcal{E}$ such that

$$E \subseteq U_n, \quad \mu(U_n \setminus E) \leq \frac{1}{n}.$$

Define

$$G = \bigcap_{n \in \mathbb{N}} U_n.$$

By the definition of $\mathcal{E}_\delta$,

$$G \in \mathcal{E}_\delta.$$

Since $E \subseteq U_n$ for every n ,

$$E \subseteq G.$$

Moreover,

$$G \setminus E \subseteq U_n \setminus E \quad (n \in \mathbb{N}).$$

Thus, by monotonicity,

$$\mu(G \setminus E) \leq \frac{1}{n} \quad (n \in \mathbb{N}).$$

Hence

$$\mu(G \setminus E) = 0.$$

(b) LOCAL INNER APPROXIMATIONS AND COUNTABLE UNION. By σ -finiteness, choose $(X_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$ such that

$$X = \bigcup_{n \in \mathbb{N}} X_n, \quad \mu(X_n) < \infty.$$

Let $E \in \mathcal{M}$. Then

$$E \cap X_n \in \mathcal{M}, \quad \mu(E \cap X_n) < \infty.$$

Since μ is inner regular with respect to $\mathcal{E}$, choose $F_n \in \mathcal{E}$ such that

$$F_n \subseteq E \cap X_n, \quad \mu((E \cap X_n) \setminus F_n) \leq \frac{1}{2^n}.$$

Define

$$F = \bigcup_{n \in \mathbb{N}} F_n.$$

By the definition of $\mathcal{E}_\sigma$,

$$F \in \mathcal{E}_\sigma.$$

Since every $F_n \subseteq E$,

$$F \subseteq E.$$

Since $X = \bigcup_{n \in \mathbb{N}} X_n$,

$$E \setminus F \subseteq \bigcup_{n \in \mathbb{N}} ((E \cap X_n) \setminus F_n).$$

Therefore, by countable subadditivity,

$$\begin{aligned}\mu(E \setminus F) &\leq \sum_{n \in \mathbb{N}} \mu((E \cap X_n) \setminus F_n) \\ &\leq \sum_{n \in \mathbb{N}} \frac{1}{2^n}.\end{aligned}$$

Replacing 2^{-n} above by $\varepsilon 2^{-n}$, for arbitrary $\varepsilon > 0$, yields

$$\mu(E \setminus F) = 0.$$

□

5.5.4. Remark: Asymmetry. In the inner approximation statement above, no closure assumption on $\mathcal{E}$ is required; the outer approximation statement needs closure of $\mathcal{E}$ under countable unions.

Cantor Sets, Nonmeasurable Sets, and Kakeya Sets

6.1 Measure and Topological Size

6.1.1. Proposition. — For Lebesgue measure λ on $\mathbb{R}$:

- (a) $\lambda(\{x\}) = 0$ for every $x \in \mathbb{R}$;
- (b) every countable subset of $\mathbb{R}$, in particular $\mathbb{Q}$, has measure zero;
- (c) every nonempty open subset of $\mathbb{R}$ has strictly positive measure;
- (d) if $E \subseteq \mathbb{R}$ is null, then E^c is dense in $\mathbb{R}$.

Proof. (a) COVER A POINT BY ARBITRARILY SHORT INTERVALS. Let $x \in \mathbb{R}$. For every $n \in \mathbb{N}$,

$$\{x\} \subseteq \left(x - \frac{1}{n}, x + \frac{1}{n}\right).$$

By the definition of Lebesgue outer measure,

$$0 \leq \lambda(\{x\}) \leq \frac{2}{n} \quad (n \in \mathbb{N}).$$

Hence

$$\lambda(\{x\}) = 0.$$

(b) COUNTABLE SUBADDITIVITY. Let

$$E = \{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}.$$

By countable subadditivity and part (a),

$$\lambda(E) \leq \sum_{n \in \mathbb{N}} \lambda(\{x_n\}) = 0.$$

Therefore

$$\lambda(E) = 0.$$

Since $\mathbb{Q}$ is countable,

$$\lambda(\mathbb{Q}) = 0.$$

(c) EVERY NONEMPTY OPEN SET CONTAINS AN INTERVAL. Let $U \subseteq \mathbb{R}$ be nonempty and open. Choose $x \in U$. By the definition of an open set, there is $r > 0$ such that

$$(x - r, x + r) \subseteq U.$$

By monotonicity,

$$\lambda(U) \geq \lambda((x - r, x + r)) = 2r > 0.$$

- (d) A NULL SET CANNOT CONTAIN AN OPEN INTERVAL. Let $E \subseteq \mathbb{R}$ satisfy $\lambda(E) = 0$. Suppose not. Then E^c is not dense in $\mathbb{R}$. By the definition of failure of density, there is a nonempty open set $U \subseteq \mathbb{R}$ such that

$$U \cap E^c = \emptyset.$$

Hence

$$U \subseteq E.$$

By monotonicity,

$$0 < \lambda(U) \leq \lambda(E) = 0,$$

where $\lambda(U) > 0$ follows from part (c). This is a contradiction. Therefore E^c is dense in $\mathbb{R}$. $\square$

6.1.2. Remark. For every $\varepsilon > 0$, enumerate $\mathbb{Q} \cap [0, 1] = \{r_j : j \in \mathbb{N}\}$, and let I_j be an interval centered at r_j of length $\varepsilon 2^{-j}$. Then

$$U = (0, 1) \cap \bigcup_{j \in \mathbb{N}} I_j$$

is open and dense in $(0, 1)$, with $\lambda(U) \leq \varepsilon$. Thus $K = [0, 1] \setminus U$ is closed and nowhere dense, while $\lambda(K) \geq 1 - \varepsilon$.

6.2 The Cantor Set

6.2.1. Definition: Accumulation and isolated points. Let (X, ρ) be a metric space and $E \subseteq X$. A point $x \in X$ is an *accumulation point* of E if $(E \setminus \{x\}) \cap U \neq \emptyset$ for every open neighborhood $U \ni x$. A point $x \in E$ that is not an accumulation point of E is *isolated*.

6.2.2. Definition: Perfect set. A subset of a metric space is *perfect* if it is closed and has no isolated points.

6.2.3. Construction: Ternary Cantor set. Set $C_0 = [0, 1]$,

$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right],$$

and

$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right].$$

In general, obtain C_{n+1} from C_n by removing the open middle third from every component interval of C_n . Thus C_n is a union of 2^n closed intervals of length 3^{-n} , and

$$C_n = \left\{ \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j} : \varepsilon_j \in \{0, 1, 2\}, \varepsilon_j \neq 1 \text{ for } 1 \leq j \leq n \right\}.$$

The Cantor set is

$$C = \bigcap_{n=0}^{\infty} C_n.$$

6.2.4. Remark: Deleted intervals. For every $k, m \in \mathbb{N}$,

$$\left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m} \right) \cap \mathcal{C} = \emptyset.$$

Every nontrivial interval in $[0, 1]$ contains an interval of this form; this is the concrete reason that $\mathcal{C}$ has empty interior.

6.2.5. Proposition (Properties of the Cantor set). — *The set $\mathcal{C}$ is nonempty, compact, perfect, nowhere dense, contains no nontrivial interval, and*

$$\lambda(\mathcal{C}) = 0.$$

Moreover,

$$\mathcal{C} = \left\{ \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j} : \varepsilon_j \in \{0, 2\} \right\},$$

and $|\mathcal{C}| = \mathfrak{c}$.

Proof. Let

$$\mathcal{C}_0 = [0, 1],$$

and, for $n \geq 1$, let $\mathcal{C}_n$ be obtained from $\mathcal{C}_{n-1}$ by removing the open middle third of every component interval. By definition,

$$\mathcal{C} = \bigcap_{n \in \mathbb{N}} \mathcal{C}_n.$$

THE FINITE-STAGE GEOMETRY. For every $n \in \mathbb{N}$, the set $\mathcal{C}_n$ is a union of 2^n pairwise disjoint closed intervals, each of length 3^{-n} . Moreover,

$$\mathcal{C}_{n+1} \subseteq \mathcal{C}_n, \quad 0 \in \mathcal{C}_n.$$

Hence

$$0 \in \mathcal{C},$$

so $\mathcal{C} \neq \emptyset$. Since every $\mathcal{C}_n$ is closed and $\mathcal{C} \subseteq [0, 1]$,

$$\mathcal{C} = \bigcap_{n \in \mathbb{N}} \mathcal{C}_n$$

is closed in the compact set $[0, 1]$. Therefore $\mathcal{C}$ is compact.

THE MEASURE TENDS TO ZERO. By finite additivity of Lebesgue measure,

$$\lambda(\mathcal{C}_n) = 2^n \frac{1}{3^n} = \left(\frac{2}{3} \right)^n.$$

Since $\mathcal{C} \subseteq \mathcal{C}_n$, monotonicity gives

$$0 \leq \lambda(\mathcal{C}) \leq \left(\frac{2}{3} \right)^n \quad (n \in \mathbb{N}).$$

Hence

$$\lambda(\mathcal{C}) = 0.$$

NO NONTRIVIAL INTERVAL SURVIVES. Suppose that $I \subseteq \mathcal{C}$ is an interval with positive length. Choose n such that

$$3^{-n} < \text{length}(I).$$

Since $I \subseteq \mathcal{C} \subseteq \mathcal{C}_n$, the connected set I must lie in one component interval of $\mathcal{C}_n$. Every such component has length 3^{-n} , which contradicts

$$\text{length}(I) > 3^{-n}.$$

Thus $\mathcal{C}$ contains no nontrivial interval.

Since $\mathcal{C}$ is closed and contains no nonempty open interval,

$$\text{int}(\mathcal{C}) = \emptyset.$$

Therefore

$$\overline{\mathcal{C}} = \mathcal{C}$$

has empty interior, so $\mathcal{C}$ is nowhere dense.

TERNARY DESCRIPTION. For $\varepsilon_1, \dots, \varepsilon_n \in \{0, 2\}$, the corresponding component of $\mathcal{C}_n$ is

$$\left[\sum_{j=1}^n \frac{\varepsilon_j}{3^j}, \sum_{j=1}^n \frac{\varepsilon_j}{3^j} + \frac{1}{3^n} \right].$$

Thus, if

$$x = \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j}, \quad \varepsilon_j \in \{0, 2\},$$

then

$$0 \leq x - \sum_{j=1}^n \frac{\varepsilon_j}{3^j} \leq \sum_{j=n+1}^{\infty} \frac{2}{3^j} = \frac{1}{3^n}.$$

Hence $x \in \mathcal{C}_n$ for every n , so $x \in \mathcal{C}$.

Conversely, let $x \in \mathcal{C}$. At every stage n , the point x lies in one component interval of $\mathcal{C}_n$. The nested component intervals determine digits $\varepsilon_j \in \{0, 2\}$ such that

$$x \in \left[\sum_{j=1}^n \frac{\varepsilon_j}{3^j}, \sum_{j=1}^n \frac{\varepsilon_j}{3^j} + \frac{1}{3^n} \right] \quad (n \in \mathbb{N}).$$

Letting $n \rightarrow \infty$ gives

$$x = \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j}.$$

Therefore

$$\mathcal{C} = \left\{ \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j} : \varepsilon_j \in \{0, 2\} \right\}.$$

EVERY CANTOR POINT IS A LIMIT POINT. Let

$$x = \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j} \in \mathcal{C}.$$

For every $N \in \mathbb{N}$, keep the first N digits fixed and replace the remaining digits by either all 0's or all 2's, choosing the one that does not reproduce x . This gives $y_N \in \mathcal{C}$ with

$$y_N \neq x, \quad |x - y_N| \leq \sum_{j=N+1}^{\infty} \frac{2}{3^j} = \frac{1}{3^N}.$$

Thus $y_N \rightarrow x$. Since $\mathcal{C}$ is closed and every point of $\mathcal{C}$ is a limit point of $\mathcal{C}$, the set $\mathcal{C}$ is perfect.

CARDINALITY. Define

$$\Phi : \{0, 2\}^{\mathbb{N}} \longrightarrow \mathcal{C}, \quad \Phi((\varepsilon_j)_{j \in \mathbb{N}}) = \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j}.$$

The ternary description proves that Φ is surjective. To prove injectivity, suppose that two sequences first differ at index k . Their k -th terms differ by $2/3^k$, while the total possible difference of the remaining tails is

$$\sum_{j=k+1}^{\infty} \frac{2}{3^j} = \frac{1}{3^k}.$$

Thus the two sums are distinct. Hence Φ is bijective.

Finally,

$$\{0, 2\}^{\mathbb{N}} \cong \mathcal{P}(\mathbb{N}),$$

by sending a sequence to the set of indices at which its digit is 2. Since

$$|\mathcal{P}(\mathbb{N})| = \mathfrak{c},$$

we obtain

$$|\mathcal{C}| = \mathfrak{c}.$$

□

6.2.6. Remark: Ternary expansions. Ternary expansions are not unique in general: for example,

$$\frac{1}{3} = 0.1_3 = 0.0222\dots_3.$$

The expansion of a point of $\mathcal{C}$ using only the digits 0 and 2 is unique. Consequently, the map $\varphi : \{0, 1\}^{\mathbb{N}} \rightarrow \mathcal{C}$ obtained by replacing each binary digit z_j with the ternary digit $2z_j$ is bijective.

6.2.7. Remark: Cantor tree. The component intervals of C_n are naturally indexed by finite binary strings of length n : appending 0 selects the left third and appending 1 the right third. An infinite branch therefore determines a point of $\mathcal{C}$, whose ternary expansion is obtained by replacing the binary digits by 0 and 2, respectively.

6.3 The Cantor–Lebesgue Function

6.3.1. Definition: Cantor–Lebesgue approximants. For $n \in \mathbb{N}$ and $x \in [0, 1]$, define

$$f_n(x) = \left(\frac{3}{2}\right)^n \lambda(C_n \cap [0, x]).$$

6.3.2. Remark: Shape of the approximants. The graph of f_0 is the straight line from $(0, 0)$ to $(1, 1)$. The function f_1 equals $1/2$ on $(1/3, 2/3)$; the function f_2 equals $1/4$ on $(1/9, 2/9)$ and $3/4$ on $(7/9, 8/9)$. In general, each f_n is polygonal, satisfies $f_n(0) = 0$ and $f_n(1) = 1$, is constant on each component of $[0, 1] \setminus C_n$, and has slope

$(3/2)^n$ on every component of C_n . If $[a_{n,j}, b_{n,j}]$ is the j -th component interval of C_n , then

$$f_n(a_{n,j}) = \frac{j-1}{2^n}, \quad f_n(b_{n,j}) = \frac{j}{2^n}.$$

Moreover,

$$\|f_{n+1} - f_n\|_\infty \leq \frac{1}{6 \cdot 2^n}.$$

Thus the uniform convergence of (f_n) follows from the geometric series $\sum_{n \geq 0} (6 \cdot 2^n)^{-1}$.

6.3.3. Theorem (Cantor–Lebesgue function). — *The sequence $(f_n)_{n \in \mathbb{N}}$ converges uniformly to a continuous nondecreasing function $f : [0, 1] \rightarrow [0, 1]$, called the Cantor–Lebesgue function or Devil’s staircase. The function f is constant on every interval removed in the construction of C , yet $f(0) = 0$ and $f(1) = 1$.*

Proof. Recall that f_n is the continuous nondecreasing piecewise-linear function obtained at the n -th stage of the Cantor construction: on every component interval of C_n , it rises by 2^{-n} , and it is constant on every interval removed by that stage. In particular,

$$0 \leq f_n \leq 1, \quad f_n(0) = 0, \quad f_n(1) = 1.$$

SUCCESSIVE APPROXIMATIONS DIFFER BY A SUMMABLE AMOUNT. Let I be a component interval of C_n . By construction, both f_n and f_{n+1} take values in the same interval of length 2^{-n} on I . Outside C_n , the functions f_n and f_{n+1} agree. Therefore

$$\|f_{n+1} - f_n\|_\infty \leq 2^{-n}.$$

Hence, if $m > n$,

$$\begin{aligned} \|f_m - f_n\|_\infty &\leq \sum_{k=n}^{m-1} \|f_{k+1} - f_k\|_\infty \\ &\leq \sum_{k=n}^{m-1} 2^{-k} \\ &\leq \sum_{k=n}^{\infty} 2^{-k} = 2^{1-n}. \end{aligned}$$

Thus (f_n) is uniformly Cauchy. Since $\mathbb{R}$ is complete, there is a function

$$f : [0, 1] \longrightarrow \mathbb{R}$$

such that

$$f_n \longrightarrow f$$

uniformly on $[0, 1]$.

CONTINUITY AND MONOTONICITY PASS TO THE LIMIT. Every f_n is continuous. By the Uniform Limit Theorem,

$$f \text{ is continuous on } [0, 1].$$

Let $x \leq y$. Since every f_n is nondecreasing,

$$f_n(x) \leq f_n(y) \quad (n \in \mathbb{N}).$$

Taking the limit gives

$$f(x) \leq f(y).$$

Thus f is nondecreasing. Since $0 \leq f_n \leq 1$, taking limits gives

$$0 \leq f \leq 1.$$

Therefore

$$f : [0, 1] \longrightarrow [0, 1].$$

THE REMOVED INTERVALS REMAIN PLATEAUS. Let I be an interval removed at stage N . By construction,

$$f_n|_I \text{ is constant} \quad (n \geq N).$$

Since $f_n \rightarrow f$ pointwise, the limit $f|_I$ is also constant. Hence f is constant on every interval removed in the construction of $\mathcal{C}$.

Finally, since

$$f_n(0) = 0, \quad f_n(1) = 1 \quad (n \in \mathbb{N}),$$

taking limits gives

$$f(0) = 0, \quad f(1) = 1.$$

Thus f is the Cantor–Lebesgue function. □

6.3.4. Remark. The entire increase of the Cantor–Lebesgue function from 0 to 1 occurs on the null set $\mathcal{C}$.

6.3.5. Proposition (Binary-to-ternary description). — *The map*

$$\varphi : \{0, 1\}^{\mathbb{N}} \longrightarrow \mathcal{C}, \quad \varphi((z_j)_{j \in \mathbb{N}}) = 2 \sum_{j=1}^{\infty} \frac{z_j}{3^j}$$

is a bijection. For $z = (z_j)_{j \in \mathbb{N}}$,

$$f(\varphi(z)) = \sum_{j=1}^{\infty} \frac{z_j}{2^j}.$$

In particular, $f[\mathcal{C}] = [0, 1]$.

Proof. REPLACE TERNARY DIGITS 0, 2 BY BINARY DIGITS 0, 1. By the ternary description of the Cantor set,

$$\mathcal{C} = \left\{ \sum_{j=1}^{\infty} \frac{\varepsilon_j}{3^j} : \varepsilon_j \in \{0, 2\} \right\}.$$

The correspondence

$$z_j \in \{0, 1\} \longmapsto \varepsilon_j = 2z_j \in \{0, 2\}$$

is bijective at every index j . Therefore

$$\varphi((z_j)_{j \in \mathbb{N}}) = 2 \sum_{j=1}^{\infty} \frac{z_j}{3^j}$$

is surjective onto $\mathcal{C}$.

To prove injectivity, suppose $z \neq w$, and let k be the first index at which they differ. After interchanging z and w if necessary, assume

$$z_k = 1, \quad w_k = 0.$$

Then

$$\begin{aligned}\varphi(z) - \varphi(w) &= \frac{2}{3^k} + 2 \sum_{j=k+1}^{\infty} \frac{z_j - w_j}{3^j} \\ &\geq \frac{2}{3^k} - 2 \sum_{j=k+1}^{\infty} \frac{1}{3^j} \\ &= \frac{2}{3^k} - \frac{1}{3^k} = \frac{1}{3^k} > 0.\end{aligned}$$

Hence $\varphi(z) \neq \varphi(w)$, so φ is injective.

THE n -TH CANTOR APPROXIMATION READS THE FIRST n BINARY DIGITS. Let

$$z = (z_j)_{j \in \mathbb{N}}, \quad x = \varphi(z).$$

At stage n , the point x belongs to the component interval of $\mathcal{C}_n$ labeled by

$$z_1, \dots, z_n.$$

By the construction of f_n , the values of f_n on this component interval lie between

$$\sum_{j=1}^n \frac{z_j}{2^j} \quad \text{and} \quad \sum_{j=1}^n \frac{z_j}{2^j} + \frac{1}{2^n}.$$

Therefore

$$\left| f_n(x) - \sum_{j=1}^n \frac{z_j}{2^j} \right| \leq \frac{1}{2^n}.$$

Since $f_n \rightarrow f$ uniformly and

$$\sum_{j=1}^n \frac{z_j}{2^j} \rightarrow \sum_{j=1}^{\infty} \frac{z_j}{2^j},$$

we obtain

$$f(\varphi(z)) = \sum_{j=1}^{\infty} \frac{z_j}{2^j}.$$

EVERY VALUE IN $[0, 1]$ OCCURS ON $\mathcal{C}$. Let $t \in [0, 1]$. By the binary-expansion theorem, there is $z = (z_j)_{j \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}}$ such that

$$t = \sum_{j=1}^{\infty} \frac{z_j}{2^j}.$$

By the formula just proved,

$$t = f(\varphi(z)).$$

Since $\varphi(z) \in \mathcal{C}$,

$$t \in f[\mathcal{C}].$$

Thus

$$[0, 1] \subseteq f[\mathcal{C}].$$

The reverse inclusion follows from $f : [0, 1] \rightarrow [0, 1]$. Therefore

$$f[\mathcal{C}] = [0, 1].$$

□

6.4 Choice and Nonmeasurable Sets

6.4.1. Definition: Axiom of choice. For every indexed family $(X_\alpha)_{\alpha \in A}$ of nonempty sets,

$$\prod_{\alpha \in A} X_\alpha \neq \emptyset.$$

6.4.2. Corollary (Selectors). — If $(X_\alpha)_{\alpha \in A}$ is a pairwise disjoint family of nonempty sets, then there is a selector $Y \subseteq \bigcup_{\alpha \in A} X_\alpha$ such that $Y \cap X_\alpha$ has exactly one element for every $\alpha \in A$.

Proof. AXIOM OF CHOICE. Since $(X_\alpha)_{\alpha \in A}$ is a family of nonempty sets, the Axiom of Choice gives a function

$$c : A \longrightarrow \bigcup_{\alpha \in A} X_\alpha$$

such that

$$c(\alpha) \in X_\alpha \quad (\alpha \in A).$$

Define

$$Y = c[A] = \{c(\alpha) : \alpha \in A\}.$$

Then

$$Y \subseteq \bigcup_{\alpha \in A} X_\alpha.$$

Fix $\alpha \in A$. By construction,

$$c(\alpha) \in Y \cap X_\alpha,$$

so $Y \cap X_\alpha \neq \emptyset$.

Let $y \in Y \cap X_\alpha$. By the definition of Y , there is $\beta \in A$ such that

$$y = c(\beta) \in X_\beta.$$

Since also $y \in X_\alpha$, the pairwise disjointness hypothesis gives

$$\alpha = \beta.$$

Therefore

$$y = c(\alpha).$$

Hence

$$Y \cap X_\alpha = \{c(\alpha)\},$$

so $Y \cap X_\alpha$ has exactly one element. □

6.4.3. Definition: Rational-translation equivalence. For $x, y \in [0, 1]$, define $x \sim y$ if $x - y \in \mathbb{Q}$. This is an equivalence relation on $[0, 1]$.

6.4.4. Remark. Its equivalence classes form a partition of $[0, 1]$: two classes are either equal or disjoint, and

$$[0, 1] = \bigcup_{\alpha \in A} E_\alpha.$$

Choosing one point from every class produces a selector for this family.

6.4.5. Theorem (Vitali). — *There exists a non-Lebesgue-measurable subset of $\mathbb{R}$.*

Proof. EQUIVALENCE MODULO RATIONAL TRANSLATION. Define an equivalence relation on $[0, 1]$ by

$$x \sim y \iff x - y \in \mathbb{Q}.$$

The equivalence classes form a pairwise disjoint family of nonempty subsets of $[0, 1]$. By the Selector Corollary, there is a set

$$V \subseteq [0, 1]$$

containing exactly one representative from every equivalence class.

Put

$$Q_0 = \mathbb{Q} \cap [-1, 1], \quad S = \bigcup_{q \in Q_0} (V + q).$$

THE RATIONAL TRANSLATES ARE PAIRWISE DISJOINT. Suppose

$$v + q = w + r, \quad v, w \in V, \quad q, r \in Q_0.$$

Then

$$v - w = r - q \in \mathbb{Q}.$$

Thus $v \sim w$. Since V contains exactly one representative from each equivalence class,

$$v = w.$$

Therefore

$$q = r.$$

Hence the sets

$$(V + q)_{q \in Q_0}$$

are pairwise disjoint.

THE UNION IS TOO LARGE FOR MEASURE ZERO AND TOO SMALL FOR POSITIVE MEASURE. We prove both inclusions.

($\subseteq$) Let $x \in S$. Then

$$x = v + q$$

for some $v \in V \subseteq [0, 1]$ and $q \in Q_0 \subseteq [-1, 1]$. Thus

$$-1 \leq x \leq 2.$$

Hence

$$S \subseteq [-1, 2].$$

($\supseteq$) Let $x \in [0, 1]$. By the defining property of V , there is $v \in V$ such that

$$x \sim v.$$

Thus

$$q := x - v \in \mathbb{Q}.$$

Since $x, v \in [0, 1]$,

$$q \in [-1, 1].$$

Hence $q \in Q_0$, and

$$x = v + q \in V + q \subseteq S.$$

Therefore

$$[0, 1] \subseteq S.$$

Suppose not. Then V is Lebesgue measurable. By translation invariance, every $V + q$ is measurable and

$$\lambda(V + q) = \lambda(V) \quad (q \in Q_0).$$

Since the translates are pairwise disjoint, countable additivity gives

$$\lambda(S) = \sum_{q \in Q_0} \lambda(V + q) = \sum_{q \in Q_0} \lambda(V).$$

If $\lambda(V) = 0$, then

$$\lambda(S) = 0,$$

which contradicts

$$1 = \lambda([0, 1]) \leq \lambda(S).$$

If $\lambda(V) > 0$, then, since Q_0 is infinite,

$$\lambda(S) = \sum_{q \in Q_0} \lambda(V) = \infty.$$

This contradicts

$$\lambda(S) \leq \lambda([-1, 2]) = 3.$$

Both possibilities are impossible. Therefore V is not Lebesgue measurable. $\square$

6.4.6. Remark. A Vitali set is obtained by choosing one representative from each equivalence class of $\sim$ on $[0, 1]$. Its distinct rational translates are pairwise disjoint, their union contains $[0, 1]$, and that union is contained in $[-1, 2]$.

6.5 Kakeya Sets

6.5.1. Definition: Kakeya set. A set $K \subseteq \mathbb{R}^d$ is a *Kakeya set* (or *Besicovitch set*) if it contains a unit line segment in every direction: for every $e \in \mathbb{R}^d$ with $|e| = 1$, there is $y \in \mathbb{R}^d$ such that

$$\{y + te : t \in [0, 1]\} \subseteq K.$$

6.5.2. Remark: Kakeya's needle problem. Kakeya asked for the least area of a planar region in which a unit needle can be turned through 180° . Besicovitch's 1919 zero-measure construction shows that no positive least area exists.

6.5.3. Theorem (Besicovitch). — *There exists a compact Kakeya set in $\mathbb{R}^2$ of Lebesgue measure zero.*

Proof. PERRON-TREE CONSTRUCTION. We use the following geometric lemma.

6.5.4. Lemma (Nested Perron trees). — *There is a decreasing sequence of compact sets*

$$K_1 \supseteq K_2 \supseteq \cdots \subseteq \mathbb{R}^2$$

such that:

(a) for every $n \in \mathbb{N}$ and every direction $\theta \in [0, \pi)$, the set K_n contains a line segment of length 1 parallel to

$$(\cos \theta, \sin \theta);$$

(b)

$$\lambda_2(K_n) \leq 2^{-n} \quad (n \in \mathbb{N}).$$

The lemma is obtained by the Perron-tree technique: at each stage, one slides and overlaps thin triangles carrying many directions, then refines the tree inside the preceding stage. The overlap makes the total area arbitrarily small while preserving one unit segment in every direction.

Define

$$K = \bigcap_{n \in \mathbb{N}} K_n.$$

Since K is a closed subset of the compact set K_1 , the set K is compact.

A UNIT SEGMENT SURVIVES IN EVERY DIRECTION. Fix $\theta \in [0, \pi)$. For every n , choose a unit segment

$$L_n \subseteq K_n$$

parallel to $(\cos \theta, \sin \theta)$. Write

$$L_n = [a_n, b_n], \quad b_n - a_n = (\cos \theta, \sin \theta).$$

Since $a_n, b_n \in K_1$ and K_1 is compact, pass to a subsequence such that

$$a_n \longrightarrow a, \quad b_n \longrightarrow b.$$

Then

$$b - a = (\cos \theta, \sin \theta),$$

so

$$L = [a, b]$$

is a unit segment parallel to $(\cos \theta, \sin \theta)$.

Fix $m \in \mathbb{N}$. Since $K_n \subseteq K_m$ for every $n \geq m$,

$$a_n, b_n \in K_m$$

for all sufficiently large n . Since K_m is closed,

$$a, b \in K_m.$$

Because K_m contains the limiting segments L_n and is closed,

$$L \subseteq K_m.$$

Since this holds for every m ,

$$L \subseteq \bigcap_{m \in \mathbb{N}} K_m = K.$$

Thus K contains a unit segment in every direction.

THE LIMITING SET HAS MEASURE ZERO. Since $K \subseteq K_n$ for every n , monotonicity gives

$$0 \leq \lambda_2(K) \leq \lambda_2(K_n) \leq 2^{-n} \quad (n \in \mathbb{N}).$$

Hence

$$\lambda_2(K) = 0.$$

Therefore K is a compact Kakeya set of Lebesgue measure zero. $\square$

6.5.5. Remark: Kakeya conjecture. The Kakeya conjecture asserts that every Kakeya set in $\mathbb{R}^d$ has Hausdorff dimension d . It is classical for $d = 2$; the slides record its recent proof for $d = 3$ by Wang and Zahl.

6.6 Problems on Cantor-Type Sets and Positive Measure

6.6.1. Construction: Generalized Cantor sets. Let $(\alpha_j)_{j \in \mathbb{N}} \subseteq (0, 1)$. Starting with $K_0 = [0, 1]$, obtain K_j from K_{j-1} by removing the open middle α_j -th of every component interval. The set

$$K = \bigcap_{j \in \mathbb{N}} K_j$$

is the *generalized Cantor set* determined by $(\alpha_j)_{j \in \mathbb{N}}$.

6.6.2. PROBLEM. Let K be a generalized Cantor set determined by $(\alpha_j)_{j \in \mathbb{N}} \subseteq (0, 1)$. Show that:

- (a) K is compact, perfect, and nowhere dense;
- (b) $|K| = c$;
- (c) if $\alpha_j = 1/3$ for every j , then $K = \mathcal{C}$ and $\lambda(K) = 0$;
- (d) $\lambda(K) = \prod_{j=1}^{\infty} (1 - \alpha_j)$, so $\lambda(K) > 0$ if and only if $\sum_{j=1}^{\infty} \alpha_j < \infty$. Show that every $\beta \in (0, 1)$ occurs as $\lambda(K)$ for a suitable choice of (α_j) .

Solution. Let

$$K_0 = [0, 1].$$

Inductively, at stage n , remove from the middle of every component interval of K_{n-1} an open interval whose length is α_n times the length of that component. The remaining set is denoted by K_n , and

$$K = \bigcap_{n \in \mathbb{N}} K_n.$$

Put

$$\ell_n = \frac{1}{2^n} \prod_{j=1}^n (1 - \alpha_j).$$

Then K_n is a union of 2^n disjoint closed intervals, each of length ℓ_n .

(a) **NESTED COMPACT CONSTRUCTION.** Since

$$K_0 \supseteq K_1 \supseteq K_2 \supseteq \cdots,$$

and every K_n is a nonempty compact subset of $[0, 1]$, the Nested Compact Set Theorem gives

$$K = \bigcap_{n \in \mathbb{N}} K_n \neq \emptyset.$$

Since K is closed in $[0, 1]$, it is compact.

BINARY ADDRESSES GIVE NEARBY DISTINCT POINTS. Every $x \in K$ determines a unique sequence

$$\varepsilon(x) = (\varepsilon_j(x))_{j \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}},$$

where $\varepsilon_j(x)$ records whether x lies in the left or right surviving interval at stage j .

Let $x \in K$, and fix $n \in \mathbb{N}$. Keep the first n entries of $\varepsilon(x)$, change the $(n+1)$ -st entry, and choose any remaining entries. The resulting address determines a point $y_n \in K$ such that

$$y_n \neq x.$$

Both x and y_n lie in the same component interval of K_n . Therefore

$$|x - y_n| \leq \ell_n \leq 2^{-n}.$$

Hence $y_n \rightarrow x$. Thus every point of K is a limit point of K , so K is perfect.

NO INTERVAL SURVIVES EVERY STAGE. Suppose that $I \subseteq K$ is a nontrivial interval. Since $I \subseteq K_n$, the connected set I lies in one component interval of K_n . Hence

$$\text{length}(I) \leq \ell_n \leq 2^{-n} \quad (n \in \mathbb{N}).$$

Letting $n \rightarrow \infty$ gives

$$\text{length}(I) = 0,$$

a contradiction. Thus K contains no nontrivial interval.

Since K is closed and has empty interior, it is nowhere dense.

(b) BINARY ADDRESSES PARAMETRIZE K . For every

$$\varepsilon = (\varepsilon_j)_{j \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}},$$

let $I_n(\varepsilon)$ be the component interval of K_n selected by the first n digits

$$\varepsilon_1, \dots, \varepsilon_n.$$

Then

$$I_1(\varepsilon) \supseteq I_2(\varepsilon) \supseteq \dots, \quad \text{length}(I_n(\varepsilon)) = \ell_n \rightarrow 0.$$

Hence

$$\bigcap_{n \in \mathbb{N}} I_n(\varepsilon)$$

consists of exactly one point of K . This defines a bijection

$$\{0, 1\}^{\mathbb{N}} \rightarrow K.$$

Since

$$\{0, 1\}^{\mathbb{N}} \cong \mathcal{P}(\mathbb{N}),$$

we obtain

$$|K| = |\mathcal{P}(\mathbb{N})| = \mathfrak{c}.$$

(c) THE CONSTANT CHOICE $\alpha_j = 1/3$. If

$$\alpha_j = \frac{1}{3} \quad (j \in \mathbb{N}),$$

then, at every stage, the removed interval is the middle third of every component interval. Thus

$$K = \mathcal{C}.$$

Moreover,

$$\lambda(K) = \prod_{j=1}^{\infty} \left(1 - \frac{1}{3}\right) = \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0.$$

- (d) THE SURVIVING TOTAL LENGTH. Since K_n has 2^n component intervals of length ℓ_n , finite additivity gives

$$\begin{aligned}\lambda(K_n) &= 2^n \ell_n \\ &= 2^n \left(\frac{1}{2^n} \prod_{j=1}^n (1 - \alpha_j) \right) \\ &= \prod_{j=1}^n (1 - \alpha_j).\end{aligned}$$

Since $K_n \downarrow K$ and $\lambda(K_0) = 1 < \infty$, continuity from above gives

$$\lambda(K) = \lim_{n \rightarrow \infty} \lambda(K_n) = \prod_{j=1}^{\infty} (1 - \alpha_j).$$

We prove both implications:

$$\lambda(K) > 0 \iff \sum_{j=1}^{\infty} \alpha_j < \infty.$$

($\Rightarrow$) Suppose

$$\prod_{j=1}^{\infty} (1 - \alpha_j) > 0.$$

Then

$$\sum_{j=1}^{\infty} -\log(1 - \alpha_j) < \infty.$$

Since

$$-\log(1 - t) \geq t \quad (0 < t < 1),$$

we obtain

$$\sum_{j=1}^{\infty} \alpha_j < \infty.$$

($\Leftarrow$) Suppose

$$\sum_{j=1}^{\infty} \alpha_j < \infty.$$

Then $\alpha_j \rightarrow 0$, so there is N such that

$$0 < \alpha_j \leq \frac{1}{2} \quad (j \geq N).$$

For $0 < t \leq 1/2$,

$$-\log(1 - t) \leq 2t.$$

Hence

$$\sum_{j=N}^{\infty} -\log(1 - \alpha_j) \leq 2 \sum_{j=N}^{\infty} \alpha_j < \infty.$$

Therefore

$$\prod_{j=1}^{\infty} (1 - \alpha_j) > 0.$$

Finally, let $\beta \in (0, 1)$. Define

$$\alpha_j = 1 - \beta^{2^{-j}} \quad (j \in \mathbb{N}).$$

Then

$$0 < \alpha_j < 1,$$

and

$$\begin{aligned} \lambda(K) &= \prod_{j=1}^{\infty} (1 - \alpha_j) \\ &= \prod_{j=1}^{\infty} \beta^{2^{-j}} \\ &= \beta^{\sum_{j=1}^{\infty} 2^{-j}} \\ &= \beta. \end{aligned}$$

□

6.6.3. PROBLEM. Assuming $|\text{Bor}(\mathbb{R})| = \mathfrak{c}$, use the preceding problem to show that not every Lebesgue measurable subset of $\mathbb{R}$ is Borel.

Solution. A NULL CANTOR SET HAS CONTINUUM MANY POINTS. Choose

$$\alpha_j = \frac{1}{3} \quad (j \in \mathbb{N}).$$

By the preceding problem, the corresponding generalized Cantor set is the usual Cantor set $\mathcal{C}$, and

$$\mathcal{C} \text{ is compact,} \quad \lambda(\mathcal{C}) = 0, \quad |\mathcal{C}| = \mathfrak{c}.$$

Since $\mathcal{C}$ is compact,

$$\mathcal{C} \in \text{Bor}(\mathbb{R}).$$

COMPLETENESS MAKES EVERY SUBSET MEASURABLE. Let $A \subseteq \mathcal{C}$. Since $\mathcal{C}$ is a Borel null set, the completeness of Lebesgue measure gives

$$A \in \mathcal{L}(\mathbb{R}).$$

Thus

$$\mathcal{P}(\mathcal{C}) \subseteq \mathcal{L}(\mathbb{R}).$$

THERE ARE TOO MANY SUBSETS TO BE BOREL. Since $|\mathcal{C}| = \mathfrak{c}$,

$$|\mathcal{P}(\mathcal{C})| = 2^{\mathfrak{c}}.$$

By Cantor's Theorem,

$$2^{\mathfrak{c}} > \mathfrak{c}.$$

On the other hand, by hypothesis,

$$|\text{Bor}(\mathbb{R})| = \mathfrak{c}.$$

Therefore

$$\mathcal{P}(\mathcal{C}) \not\subseteq \text{Bor}(\mathbb{R}).$$

Hence there is

$$A \subseteq \mathcal{C}$$

such that

$$A \in \mathcal{L}(\mathbb{R}), \quad A \notin \text{Bor}(\mathbb{R}).$$

Thus not every Lebesgue measurable subset of $\mathbb{R}$ is Borel. $\square$

6.6.4. PROBLEM. Construct a set $X \subseteq \mathbb{R}$ such that every nonempty open $V \subseteq \mathbb{R}$ satisfies

$$\lambda(V \cap X) > 0, \quad \lambda(V \cap X^c) > 0.$$

Hint: Use Problem 1.

Solution. A POSITIVE-MEASURE NOWHERE-DENSE SET INSIDE EVERY INTERVAL. By the preceding generalized Cantor-set problem, choose a compact nowhere-dense set

$$K \subseteq [0, 1]$$

such that

$$\lambda(K) > 0.$$

By translation and homogeneity, every nonempty open interval contains a compact nowhere-dense set of strictly positive Lebesgue measure.

Let

$$(I_n)_{n \in \mathbb{N}}$$

be an enumeration of all bounded open intervals with rational endpoints. Every nonempty open subset of $\mathbb{R}$ contains some I_n .

INDUCTIVELY RESERVE TWO POSITIVE PIECES IN EACH RATIONAL INTERVAL. We construct compact nowhere-dense sets $C_n, D_n \subseteq I_n$ such that

$$\lambda(C_n) > 0, \quad \lambda(D_n) > 0,$$

and such that all the sets

$$C_1, D_1, C_2, D_2, \dots$$

are pairwise disjoint.

Suppose that C_j, D_j have been constructed for $j < n$. Put

$$H_{n-1} = \bigcup_{j=1}^{n-1} (C_j \cup D_j).$$

Since H_{n-1} is a finite union of compact nowhere-dense sets, it is closed and nowhere dense. Hence

$$I_n \setminus H_{n-1}$$

contains a nonempty open interval J_n . Choose two disjoint nonempty open intervals

$$J_n^0, J_n^1 \subseteq J_n.$$

Inside J_n^0 , choose an affine copy C_n of K ; inside J_n^1 , choose an affine copy D_n of K . Then

$$C_n, D_n \subseteq I_n \setminus H_{n-1},$$

they are compact and nowhere dense, and

$$\lambda(C_n) > 0, \quad \lambda(D_n) > 0.$$

This completes the induction.

Define

$$X = \bigcup_{n \in \mathbb{N}} C_n.$$

EVERY OPEN SET MEETS BOTH X AND ITS COMPLEMENT IN POSITIVE MEASURE. Let $V \subseteq \mathbb{R}$ be nonempty and open. Choose $n \in \mathbb{N}$ such that

$$I_n \subseteq V.$$

Since $C_n \subseteq I_n \subseteq V$ and $C_n \subseteq X$,

$$\lambda(V \cap X) \geq \lambda(C_n) > 0.$$

Since the family $(C_n, D_n)_{n \in \mathbb{N}}$ is pairwise disjoint,

$$D_n \cap X = \emptyset.$$

Thus

$$D_n \subseteq V \cap X^c,$$

and therefore

$$\lambda(V \cap X^c) \geq \lambda(D_n) > 0.$$

Hence every nonempty open set has positive-measure intersection with both X and X^c . $\square$

6.6.5. PROBLEM (Density of measurable sets). Let $E \subseteq \mathbb{R}$ be Lebesgue measurable with $\lambda(E) > 0$. Show that for every $\alpha \in (0, 1)$, there is an open interval I such that

$$\lambda(E \cap I) > \alpha \lambda(I).$$

Solution. LOCALIZE TO A FINITE POSITIVE-MEASURE PIECE. Since

$$E = \bigcup_{n \in \mathbb{N}} (E \cap [-n, n]),$$

continuity from below gives

$$\lambda(E) = \lim_{n \rightarrow \infty} \lambda(E \cap [-n, n]).$$

Since $\lambda(E) > 0$, there is $n \in \mathbb{N}$ such that

$$0 < \lambda(A) < \infty, \quad A := E \cap [-n, n].$$

CHOOSE AN OPEN ENLARGEMENT WHOSE RELATIVE ERROR IS SMALL. Let

$$\varepsilon = \frac{1}{2} \left(\frac{1}{\alpha} - 1 \right) \lambda(A) > 0.$$

By outer regularity, choose an open set $O \supseteq A$ such that

$$\lambda(O \setminus A) < \varepsilon.$$

Since $A \subseteq O$,

$$\begin{aligned}\lambda(O) &= \lambda(A) + \lambda(O \setminus A) \\ &< \lambda(A) + \frac{1}{2} \left(\frac{1}{\alpha} - 1 \right) \lambda(A) \\ &= \frac{1 + \alpha}{2\alpha} \lambda(A) \\ &< \frac{1}{\alpha} \lambda(A).\end{aligned}$$

Thus

$$\lambda(A) > \alpha \lambda(O).$$

ONE COMPONENT INTERVAL HAS THE REQUIRED DENSITY. Write the open set O as a pairwise disjoint countable union of open intervals:

$$O = \bigsqcup_{j \in \mathbb{N}} I_j.$$

Since $\lambda(O) < \infty$, every I_j has finite length. Suppose not. Then

$$\lambda(A \cap I_j) \leq \alpha \lambda(I_j) \quad (j \in \mathbb{N}).$$

Since $A \subseteq O$ and the I_j 's are pairwise disjoint, countable additivity gives

$$\begin{aligned}\lambda(A) &= \sum_{j \in \mathbb{N}} \lambda(A \cap I_j) \\ &\leq \alpha \sum_{j \in \mathbb{N}} \lambda(I_j) \\ &= \alpha \lambda(O),\end{aligned}$$

contradicting

$$\lambda(A) > \alpha \lambda(O).$$

Therefore, for some $j \in \mathbb{N}$,

$$\lambda(A \cap I_j) > \alpha \lambda(I_j).$$

Since $A \subseteq E$,

$$\lambda(E \cap I_j) \geq \lambda(A \cap I_j) > \alpha \lambda(I_j).$$

Taking $I = I_j$ proves the claim. $\square$

6.6.6. PROBLEM (Steinhaus' theorem). Let $E \subseteq \mathbb{R}$ be Lebesgue measurable with $\lambda(E) > 0$. Show that

$$E - E = \{x - y : x, y \in E\}$$

contains an interval centered at 0. *Hint:* If I is the interval furnished by Problem 4 with $\alpha > 3/4$, then $E - E$ contains $(-\lambda(I)/2, \lambda(I)/2)$.

Solution. CHOOSE AN INTERVAL ON WHICH E OCCUPIES MORE THAN THREE QUARTERS. Apply the preceding density result with

$$\alpha > \frac{3}{4}.$$

There is an open interval I such that

$$\lambda(E \cap I) > \alpha \lambda(I) > \frac{3}{4} \lambda(I).$$

Put

$$A = E \cap I.$$

A SMALL TRANSLATE OF A MUST OVERLAP A . Let

$$|t| < \frac{\lambda(I)}{2}.$$

Suppose not. Then

$$A \cap (A + t) = \emptyset.$$

Since translation preserves Lebesgue measure,

$$\lambda(A + t) = \lambda(A).$$

Thus, by additivity on disjoint sets,

$$\lambda(A \cup (A + t)) = 2\lambda(A) > \frac{3}{2} \lambda(I).$$

On the other hand,

$$A \cup (A + t) \subseteq I \cup (I + t).$$

Since I is an interval,

$$\lambda(I \cup (I + t)) = \lambda(I) + |t| < \frac{3}{2} \lambda(I).$$

By monotonicity,

$$\lambda(A \cup (A + t)) \leq \lambda(I \cup (I + t)) < \frac{3}{2} \lambda(I),$$

which is a contradiction. Therefore

$$A \cap (A + t) \neq \emptyset.$$

Choose $z \in A \cap (A + t)$. By the definition of $A + t$, there is $y \in A$ such that

$$z = y + t.$$

Since $z, y \in A \subseteq E$,

$$t = z - y \in E - E.$$

Thus

$$\left(-\frac{\lambda(I)}{2}, \frac{\lambda(I)}{2} \right) \subseteq E - E.$$

Hence $E - E$ contains an interval centered at 0. □

6.6.7. PROBLEM (Cantor–Bendixson theorem). Show that every closed set $A \subseteq \mathbb{R}$ is the union of a perfect set P and a countable set C . The set P may be empty.

Solution. CONDENSATION POINTS. Define

$$P = \{x \in \mathbb{R} : \text{every open interval } I \ni x \text{ satisfies } |A \cap I| > \aleph_0\}.$$

Thus P is the set of condensation points of A .

Since every neighborhood of a point of P meets A , we have

$$P \subseteq \overline{A}.$$

By hypothesis, A is closed, so

$$P \subseteq A.$$

THE SET OF CONDENSATION POINTS IS CLOSED. Let $x \notin P$. By the definition of P , there is an open interval $I \ni x$ such that

$$A \cap I$$

is countable. For every $y \in I$, the same interval I is a neighborhood of y whose intersection with A is countable. Thus

$$y \notin P.$$

Hence

$$I \subseteq P^c.$$

Therefore P^c is open, so P is closed.

THE DISCARDED SET IS COUNTABLE. Let $\mathcal{B}$ be the countable basis of open intervals with rational endpoints. Define

$$\mathcal{B}_0 = \{I \in \mathcal{B} : A \cap I \text{ is countable}\}.$$

If $x \in A \setminus P$, then, by the definition of P , there is an open interval $U \ni x$ such that $A \cap U$ is countable. Since $\mathcal{B}$ is a basis, choose

$$I \in \mathcal{B}$$

such that

$$x \in I \subseteq U.$$

Then

$$A \cap I \subseteq A \cap U,$$

so $A \cap I$ is countable. Hence $I \in \mathcal{B}_0$. Therefore

$$A \setminus P \subseteq \bigcup_{I \in \mathcal{B}_0} (A \cap I).$$

Since $\mathcal{B}_0$ is countable and every $A \cap I$ is countable,

$$A \setminus P$$

is countable. Put

$$C = A \setminus P.$$

THE CONDENSATION SET HAS NO ISOLATED POINTS. Suppose not. Then there are $p \in P$ and an open interval $J \ni p$ such that

$$P \cap J = \{p\}.$$

Thus

$$J \setminus \{p\} \subseteq P^c.$$

By the preceding countable-basis argument,

$$A \cap (J \setminus \{p\})$$

is countable. Since $\{p\}$ is finite,

$$A \cap J = (A \cap (J \setminus \{p\})) \cup \{p\}$$

is countable. This contradicts $p \in P$, because every open interval containing p must meet A in an uncountable set.

Therefore P has no isolated points. Since P is closed, P is perfect. Finally,

$$A = P \cup C,$$

where P is perfect and C is countable. The possibility

$$P = \emptyset$$

is allowed. □

6.6.8. PROBLEM. Let $E \subseteq \mathbb{R}$ be Lebesgue measurable with $0 < \lambda(E) \leq \infty$. Show that for every $q \in (0, \lambda(E))$, there is a perfect set $B \subseteq E$ satisfying $\lambda(B) = q$. *Hint:* Use inner regularity of λ and Problem 6.

Solution. CHOOSE A COMPACT SUBSET OF E WITH MORE THAN q MEASURE. Since

$$q < \lambda(E),$$

the inner regularity of Lebesgue measure gives a compact set

$$K \subseteq E$$

such that

$$\lambda(K) > q.$$

In particular,

$$\lambda(K) < \infty.$$

By the Cantor–Bendixson Theorem, there are a perfect set P and a countable set D such that

$$K = P \cup D.$$

Since every countable set has Lebesgue measure zero,

$$\lambda(P) = \lambda(K) > q.$$

CUT THE PERFECT SET AT EXACTLY THE DESIRED MEASURE. Define

$$F(t) = \lambda(P \cap (-\infty, t]) \quad (t \in \mathbb{R}).$$

Since $\lambda(P) < \infty$ and Lebesgue measure has no atoms,

$$\lambda(\{t\}) = 0 \quad (t \in \mathbb{R}),$$

the function F is continuous. Indeed, continuity from above gives right-continuity, while continuity from below gives

$$\lim_{s \uparrow t} F(s) = \lambda(P \cap (-\infty, t)) = \lambda(P \cap (-\infty, t]) = F(t).$$

Since P is compact, choose

$$a < \min P, \quad b > \max P.$$

Then

$$F(a) = 0, \quad F(b) = \lambda(P) > q.$$

By the Intermediate Value Theorem, there is $x \in [a, b]$ such that

$$F(x) = q.$$

Set

$$A = P \cap (-\infty, x].$$

Then A is compact, $A \subseteq E$, and

$$\lambda(A) = q.$$

REMOVE THE COUNTABLE PART ONCE MORE. Apply the Cantor–Bendixson Theorem to the closed set A . There are a perfect set B and a countable set C such that

$$A = B \cup C.$$

Since $\lambda(C) = 0$,

$$\lambda(B) = \lambda(A) = q.$$

Finally,

$$B \subseteq A \subseteq P \subseteq K \subseteq E.$$

Thus B is a perfect subset of E satisfying

$$\lambda(B) = q.$$

□

6.6.9. PROBLEM. Show that every Lebesgue measurable $A \subseteq \mathbb{R}$ with $\lambda(A) > 0$ has cardinality $\mathfrak{c}$. *Hint:* Use Problem 7.

Solution. USE THE PRECEDING PERFECT-SUBSET CONSTRUCTION. Choose

$$q = \begin{cases} \lambda(A)/2, & \lambda(A) < \infty, \\ 1, & \lambda(A) = \infty. \end{cases}$$

Then

$$0 < q < \lambda(A).$$

By the preceding problem, there is a perfect set

$$B \subseteq A$$

such that

$$\lambda(B) = q.$$

In particular,

$$B \neq \emptyset.$$

A NONEMPTY PERFECT SUBSET OF $\mathbb{R}$ CONTAINS A BINARY TREE. We construct an injection

$$\{0, 1\}^{\mathbb{N}} \longrightarrow B.$$

Choose a closed interval $I_\emptyset$ whose interior meets B . Inductively, suppose that I_s has been chosen for a finite binary word s , with

$$B \cap \text{int}(I_s) \neq \emptyset.$$

Since B is perfect, every point of B is an accumulation point of B . Hence choose two distinct points of

$$B \cap \text{int}(I_s),$$

and choose disjoint closed intervals I_{s0}, I_{s1} such that

$$I_{s0}, I_{s1} \subseteq \text{int}(I_s), \quad B \cap \text{int}(I_{s\epsilon}) \neq \emptyset, \quad \text{length}(I_{s\epsilon}) \leq 2^{-|s|-1}.$$

Let

$$z = (z_1, z_2, \dots) \in \{0, 1\}^{\mathbb{N}}.$$

The intervals

$$I_{z_1} \supseteq I_{z_1 z_2} \supseteq \dots$$

are nested and have lengths converging to 0. Hence their intersection consists of exactly one point, denoted by x_z . Since B is closed and every interval meets B ,

$$x_z \in B.$$

If $z \neq w$, let n be the first index at which they differ. Then x_z and x_w lie in the disjoint intervals

$$I_{z_1 \dots z_n}, \quad I_{w_1 \dots w_n},$$

so

$$x_z \neq x_w.$$

Thus

$$z \mapsto x_z$$

is injective. Therefore

$$|B| \geq |\{0, 1\}^{\mathbb{N}}| = \mathfrak{c}.$$

Since

$$B \subseteq A \subseteq \mathbb{R},$$

we also have

$$|A| \leq |\mathbb{R}| = \mathfrak{c}.$$

Consequently

$$|A| = \mathfrak{c}.$$

□

6.6.10. PROBLEM. Let $A \subseteq [a, b]$ be Lebesgue measurable with $\lambda(A) > 0$. Show that there are $x, y \in A$ such that $|x - y|$ is irrational. *Hint:* Use Problem 8; this also follows immediately from Steinhaus' theorem.

Solution. A POSITIVE-MEASURE SET CANNOT LIE IN ONE RATIONAL TRANSLATE. Suppose not. Then

$$|x - y| \in \mathbb{Q} \quad (x, y \in A).$$

Since $\lambda(A) > 0$, the set A is nonempty. Choose

$$x_0 \in A.$$

For every $x \in A$,

$$|x - x_0| \in \mathbb{Q}.$$

Hence

$$x - x_0 \in \mathbb{Q},$$

because a real number whose absolute value is rational is rational. Therefore

$$A \subseteq x_0 + \mathbb{Q}.$$

Since $\mathbb{Q}$ is countable,

$$x_0 + \mathbb{Q}$$

is countable. Thus A is countable. By the preceding problem, every Lebesgue measurable set of positive measure has cardinality $\mathfrak{c}$, a contradiction.

Therefore there are $x, y \in A$ such that

$$|x - y| \notin \mathbb{Q}.$$

□

6.6.11. Remark: *Immediate consequence of Steinhaus' theorem.* By Steinhaus' theorem,

$$A - A$$

contains an open interval centered at 0. Every nonempty open interval contains an irrational number, so choose

$$t \in (A - A) \setminus \mathbb{Q}.$$

By the definition of $A - A$, there are $x, y \in A$ such that

$$t = x - y.$$

Hence

$$|x - y| = |t|$$

is irrational.